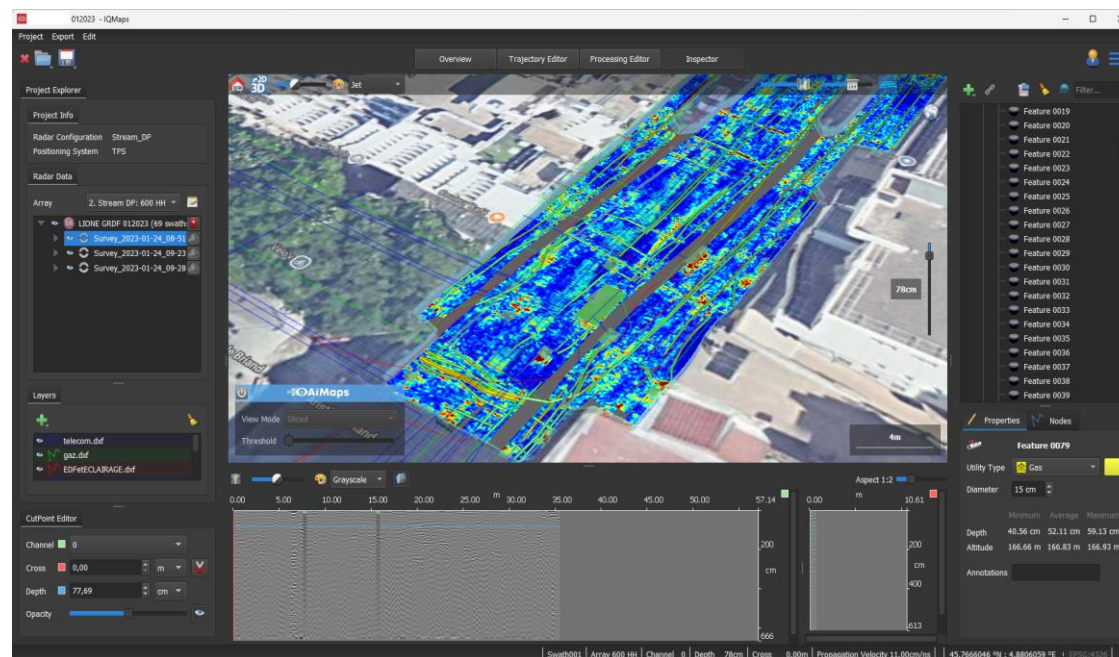


IQMaps

Version 1.013.001



ORIGINAL INSTRUCTION IN ENGLISH LANGUAGE



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1 INTRODUCTION

IQMaps is the software by IDS GeoRadar s.r.l. designed to post-process IDS GeoRadar data of both entry level and single antenna systems (**Opera Duo/DS2000, Ris Hi-Mod 1A, DSX, DS4000/Opera XR, Chaser XR...**), as well as massive array systems (e.g.: **Stream EM, Stream X, Stream C, Stream UP, Stream DP...**).

IQMaps is designed to implement advanced processing and to optimise interpretation of Georadar data, in 2D and 3D environments, in a simplified workflow.

The aim of this User's Manual is to give the User a complete set of procedures to finalize the complete workflow. The User shall be led, step by step, from the advanced GPR data processing to the visualization and export in several formats, including CAD environment-compatible formats.

1.1 Application field

This manual is addressed to the Users of an IDS GeoRadar GPR that uses **IQMaps** to post-process georadar data.

1.2 Intended readership

The intended reader of this manual is the technician in charge of using the system that has attended the IDS GeoRadar training for **IQMaps**.



NOTE: The product is intended for professional use only; not for Consumers use.

2 HOW TO READ THE MANUAL

2.1 Manual Layout

It is strongly recommended to go through this manual thoroughly, and to read it carefully.

However, this document is divided into:

- a first part (par. **4 HOW TO INSTALL IQMAPS AND PLUGINS**, and par. **5 HOW TO USE IQMAPS: THE WORKFLOW**) including specific procedures that can be read sparsely, and
- a second part (**APPENDIX B** to **APPENDIX H**), with insight on specific topics.

Hence it may not be mandatory to read this document chapter after chapter, in case of need.

Please, note that images and pictures shown in the manual aims at supporting the Users in understanding the operation described. However, pictures and images portrayed herein may, from time to time, slightly differ from the actual Hardware and Software they describe.

2.2 Symbols

Warning messages are an essential part of the Safety Concept of the instrument. They appear wherever hazards or hazardous situations can occur.

Type	Description
 DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.



Indicates a potentially hazardous situation or an unintended use which, if not avoided, could result in death or serious injury.



Indicates a potentially hazardous situation or an unintended use which, if not avoided, may result in minor or moderate injury.

Warning messages:

- make the user alert about direct and indirect hazards concerning the use of the product.
- contain general rules of behaviour.

For the user's safety, all safety instructions and safety messages shall be strictly observed and followed. Therefore, the manual must always be available to all people performing any tasks described herein.

DANGER, WARNING, CAUTION and **NOTICE** are standardized signal words for identifying levels of hazards and risks related to personal injury and property damage.

For your safety it is important to read and fully understand the table below with the different signal words and their definitions. Supplementary safety information symbols may be placed within a warning message as well as supplementary text.



Note text/to keep in mind



Tip information

2.3 Glossary & Acronyms

The following acronyms and naming are used in this manual:

- **AI** Artificial Intelligence
- **APS**: Accurate Positioning System
- **CDS**: Cloud Data Sharing.
- **Controller**: PC / Tablet used by the user for data acquisition. The data acquisition SW (client) is installed on the controller.
- **CLM**: Client License Manager is the Leica geosystems software used to activate software licenses.
- **Free Running**: acquisition mode triggered by the “time” not by encoder (wheel or any other type).
- **GNSS**: Global Navigation Satellite System.
- **GPR**: Ground Penetrating Radar or Georadar or Ground Probing Radar.
- **GUI**: Graphical User Interface. A graphical user interface, or GUI, is a form of user interface that allows users to interact with electronic devices.
- **IDS GCAD**: IDS Georadar Free plugin for AutoCAD and BricsCAD.
- **LLM**: Local License Manager is the software by NovAtel to activate Inertial Explorer license for **APS**.
- **NMEA format**: National Marine Electronics Association format.
- **Re.D.:** Remote Diagnostics.
- **RTK**: Real Time Kinematic. This is a positioning correction service that is subscription-based and real time.
- **SNR**: Signal to Noise Ratio.
- **SFTP**: Secure File Transfer Protocol.
- **UBLOX**: GNSS device within the antenna case.
- **HD**: High Density.
- **TS**: Total Station.
- **WKT**: Well Known Text.

3 SYSTEMS REQUIREMENTS

3.1 Minimum PC System Requirements

- 1.6GHz quad core processor
- 8 GB RAM
- Graphic Display Adapter Open GL 3.3 compatible
- Internet Connection (for license authentication and use of maps)
- Windows 10, 64 bit.

3.2 Recommended PC System Requirements

- 10th Generation Intel® Core™ i7 Processors equivalent or better.
- 16 or 32 GB RAM
- NVIDIA Graphic Display Adapter OpenGL 3.3 compatible. NVIDIA GeForce RTX 20 series GPU or better
- Fast disk access (NVMe or SSD)
- Full HD screen (1920 x 1080) or higher
- Broadband Internet Connection (for license authentication and use of maps).
- Windows 11

3.3 Operating System

- Windows 10, 64 bit only
- Windows 11

4 HOW TO INSTALL IQMAPS AND PLUGINS

IQMaps and its plugins (**AIMaps**, **Accurate Positioning System (APS)**, **IDS GCAD**) can all be downloaded from the [MyWorld](#) website.

IQMaps and plugins (except for IDS GCAD) must be activated before the first use by entering a 25-digit code in the software named **Client License Manager (CLM)**.

The **CLM** must be installed into the very same PC where **IQMaps** and plugins are installed. To do this refer to **Leica Geosystems Software Licensing Introduction & Installation User Manual**.

The full software installation flow is illustrated hereinafter, and it can be summarized as follows:

1. Download and Install **CLM**, “run as administrator”,
2. Download and install **IQMaps**,
 - a. Activate **IQMaps** through **CLM**,
3. Download and install **AIMaps** (optional),
 - a. Activate **AIMaps** through **CLM**,
4. Download and install **APS** (optional),
 - a. Activate **APS** through **CLM** and **LLM**,
5. Download and install **IDS GCAD** (NO NEED activation).

4.1 How to activate an EID using CLM

To activate any SW EID using **CLM**, the steps to be implemented are the following:

1. Start CLM through the icon , the window of Fig. 1 shall appear,

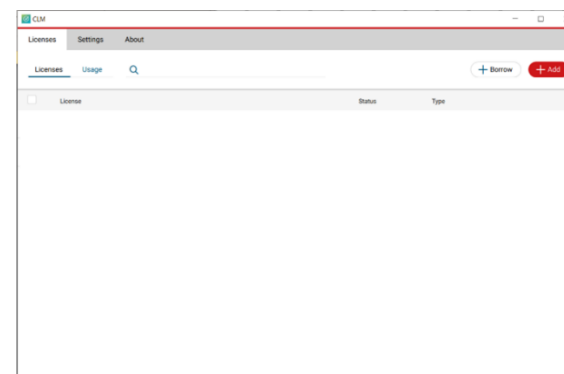


Fig. 1 CLM user interface

2. Press the  icon on the right side of the CLM,
3. Add the 25-digit code relating to the software (or plugin) to be activated, in the space indicated in Fig. 2

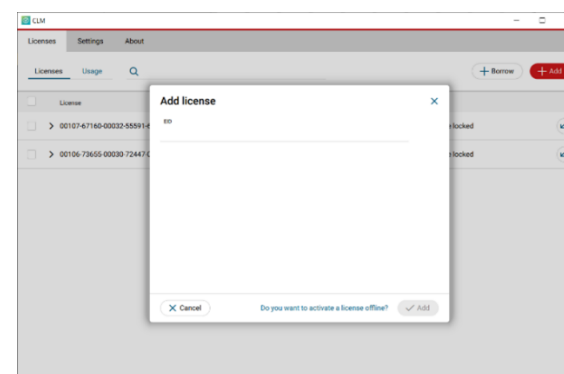


Fig. 2 License activation window

4. Press  as shown add in Fig. 3, and wait for the CLM to activate the license,

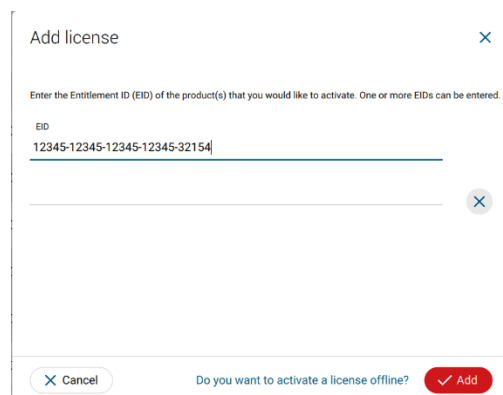


Fig. 3 Adding license EID to CLM

5. As the EID is correctly activated, the **CLM** interface shall show all the active licenses in the PC being used (Fig. 4).

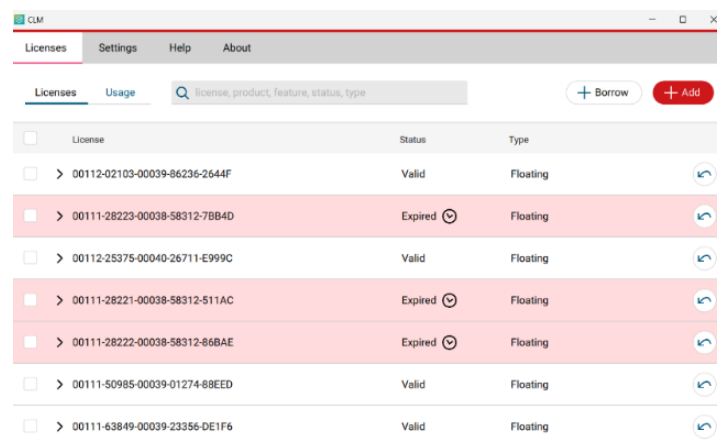


Fig. 4 CLM interface with list of activated EID

4.2 How to Install and activate IQMaps

To install **IQMaps** after downloading, follow the steps below:

1. Right click on the **IQMaps.exe** and select “run as administrator”,
2. Press the **NEXT** button of Fig. 5

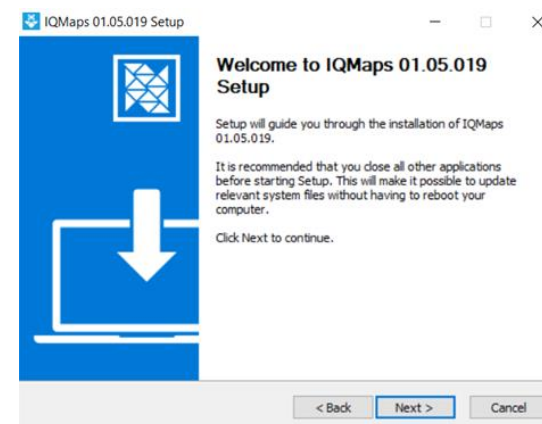


Fig. 5 IQMaps install step 1

3. Select the installation folder in the window of Fig. 6 and then click on **Install**,

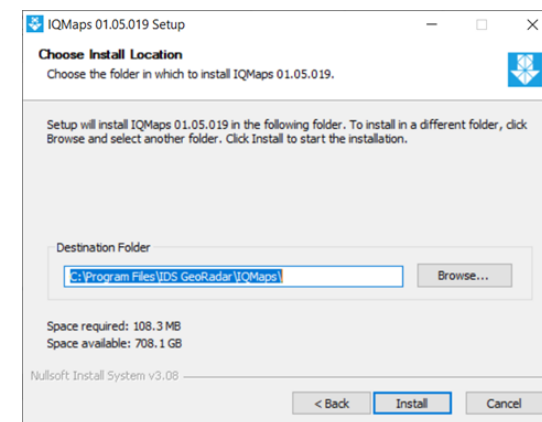


Fig. 6 IQMaps install Step 2

4. Press **Finish** (Fig. 7), for the icon  to be visualized on the PC screen desktop,

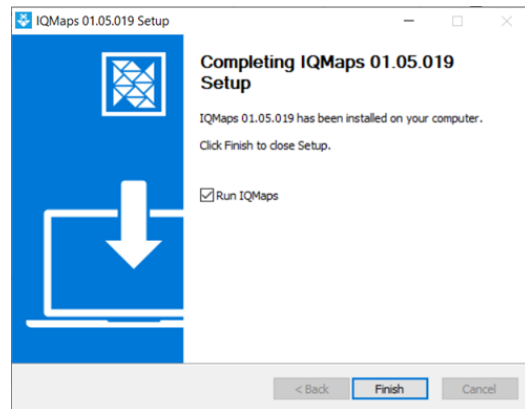


Fig. 7 IQMaps install step 3

5. Activate ***IQMaps*** by entering a valid 25-digit code in CLM following the procedure described in par. 4.1,
6. Start ***IQMaps*** by pressing the icon  to activate (FIRST START ONLY) the Server License window (Fig. 8),

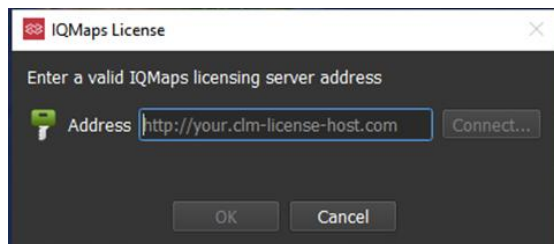


Fig. 8 IQMaps server license window

7. If the license has been installed locally type ***“localhost”*** to the address box and press ***Connect...***, then press ***OK***,
8. If the license has been installed on a PC-server, type the server IP address and press ***Connect...***, then press ***OK***.

4.3 How to install and activate AiMaps

To install ***AiMaps*** after downloading, follow the steps below:

1. Right click on the ***AiMaps for IQMaps.exe*** and select “run as administrator”,
2. Press the ***NEXT*** button of Fig. 9,

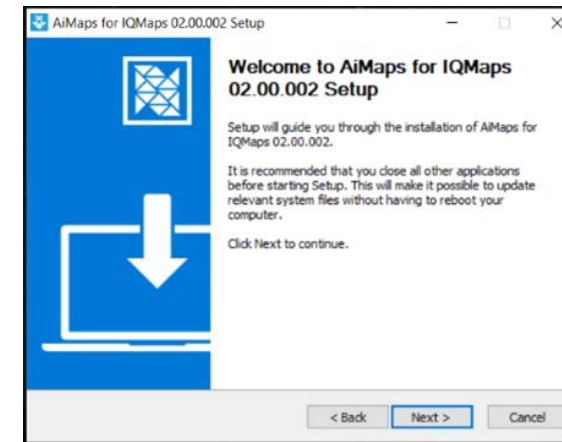


Fig. 9 AiMaps install step 1

3. Select ***I Agree*** button to accept the License Agreement (Fig. 10),

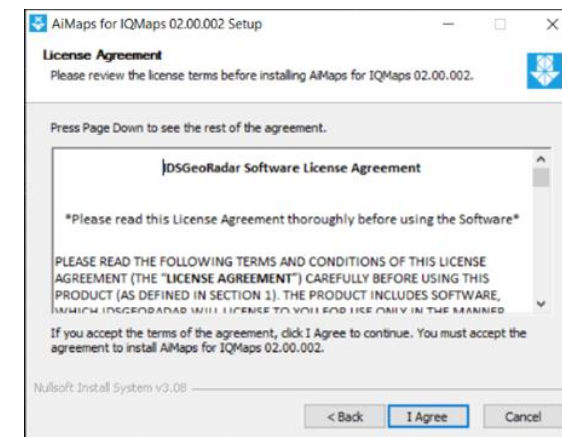


Fig. 10 AiMaps install step 2

4. Wait until installation process is completed (Fig. 11) and press **NEXT**,

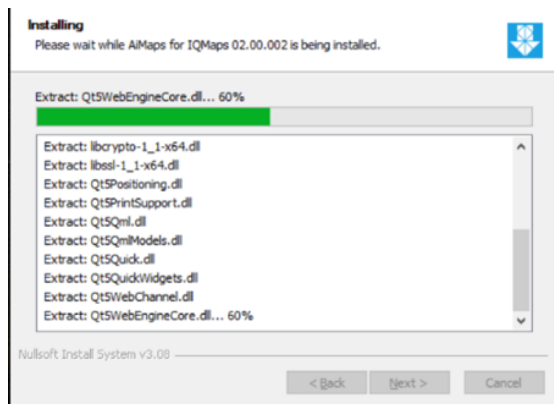


Fig. 11 AiMaps install step 3

5. Complete the installation process by clicking on **Finish** button of Fig. 12.

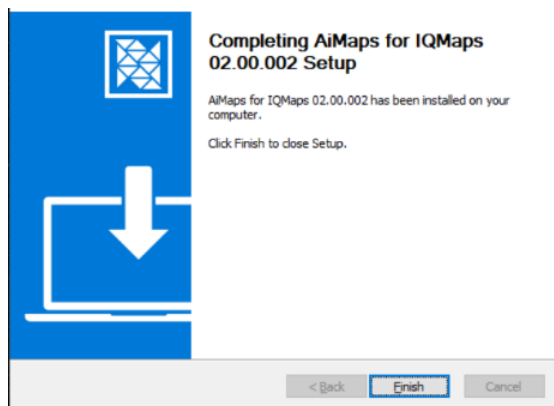


Fig. 12 AiMaps install step 4

6. Activate **AiMaps** by entering a valid 25-digit code in CLM following the procedure described in par. 4.1,
7. Send an e-mail to the address AiMaps.geo@idsgeoradar.com requesting the registration link to HxDR platform,



NOTE: DO NOT CREATE AN ACCOUNT WITHOUT INVITATION FROM IDS

8. After receiving back the e-mail with the activation link, enter the personal details and press **NEXT** (Fig. 13),

Fig. 13 AiMaps activation step 1

9. Enter the account details and press **NEXT** (Fig. 14),

Fig. 14 AiMaps activation step 2

10. Accept terms and condition and press **SIGN UP** (Fig. 15),

Create an HxDR account
Follow the simple 3 steps to complete your registration

Personal details

Account details

HxDR Terms of Use

HxDR Terms of Use

☒ I have the authority to enter into the Agreement on behalf of the Customer*

☒ The Customer is a business customer and it and its Users will access and use HxDR for commercial use only*

☒ I have read, understood and agree to:

- the HxDR Terms of Use, including the arbitration clause contained therein;
- the terms of the Subscription Plan I chose; (for freemium)
- the Lelica Geosystems Privacy Policy;
- the Acceptable Use Policy;
- the HxDR Documentation;

☐ I consent to receive electronic marketing communications on products, services, publications, and events from Hexagon

Lelica Geosystems Privacy Policy describes how we process your personal data, as well as information on how you may exercise your rights to update access, manage, export and delete your personal data. If you subsequently change your mind, you may withdraw consent at any time by clicking on the "unsubscribe" link within the marketing emails sent to you.

*Required Field

Sign Up

TERMS OF USE | IMPRINT | COOKIE POLICY | PRIVACY POLICY | SUPPORT | COOKIE SETTINGS

Fig. 15 AiMaps activation step 3

11. verify the link received at the e-mail address entered, and then press **login** to confirm (Fig. 16),

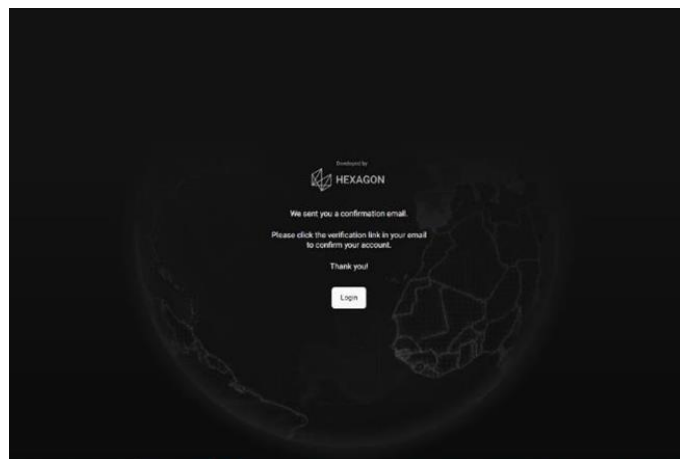


Fig. 16 AiMaps activation step 4

12. login **HxDR** platform by using the credentials just verified (Fig. 17).

Log in

Email

This field is required

Password

This field is required

Forgot your password?

Log in

TERMS OF USE | IMPRINT | COOKIE POLICY | PRIVACY POLICY

Fig. 17 AiMaps activation step 5



NOTE: AiMaps is a cloud-based data processing solution relying on AI algorithms "Artificial Neural Network". User must be aware that all AiMaps outputs simply represents reality to some extent. User assumes all risks associated with using AiMaps outputs for any purposes. AiMaps output is not meant to be the only basis for any business decision. The user must always verify the accuracy of any AiMaps output with other independent means and sources.



NOTE: When running AiMaps on Stream UP dataset REDUCED array configuration are NOT supported for processing.

4.4 How to install and activate APS (for Stream UP ONLY)

To install and activate **APS** after downloading, follow the steps below:

1. Right click on the **Accurate Positioning System for IQMaps.exe** and select “run as administrator”,
2. Click on **Next** button in the window shown in Fig. 18,

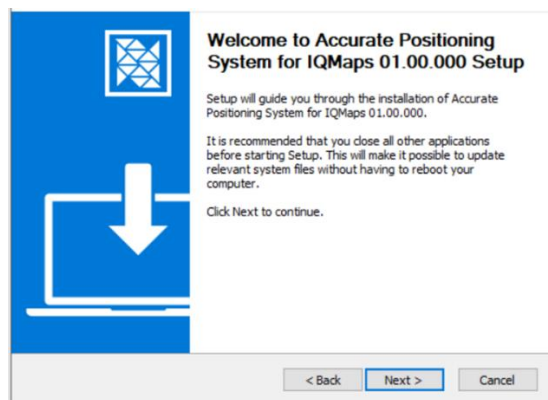


Fig. 18 APS install step 1

3. Select **I Agree** button to accept the License Agreement (Fig. 19),

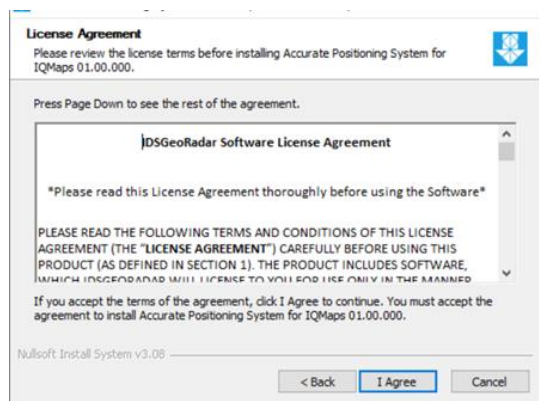


Fig. 19 APS install step 2

4. Wait until installation process is complete and press NEXT (Fig. 20),

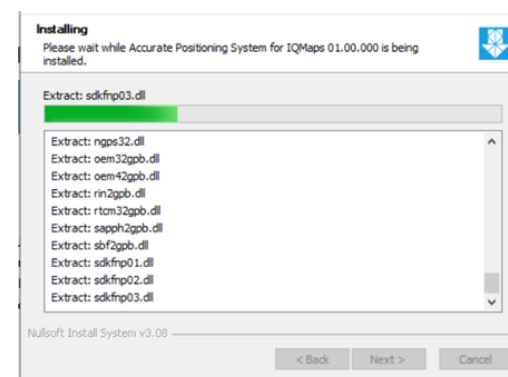


Fig. 20 APS install step 3

5. Press **FINISH** to complete installation (Fig. 21),

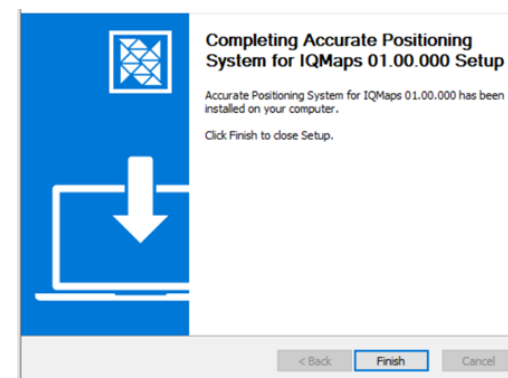


Fig. 21 APS install step 4

6. Activate **NavPrep** for **APS** by entering a valid 25-digit code in CLM following the procedure described in par. 4.1,
7. Activate **Inertial Explorer** for **APS** by starting the **LLM** software at the following path: `C:\Program Files\IDS GeoRadar\IQMaps\plugins\org.idsgeoradar.iqmaps.aps\LeicaPegasusNavPrep\WayptSDK890\bin\LLMForm.exe`,
8. Enter the activation code into the Activate License box of Fig. 22, and press **ACTIVATE**,

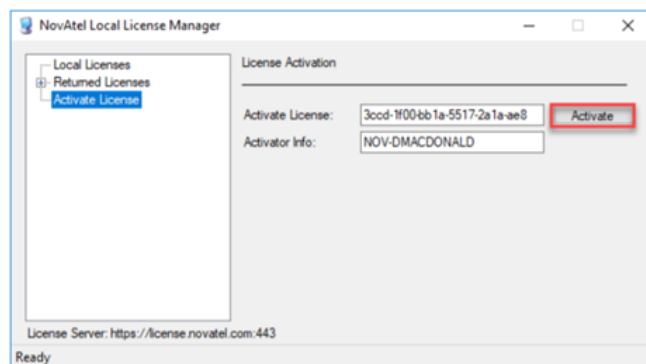


Fig. 22 APS activation step 5



NOTE: SDK license must be installed on device where IQMaps APS plugin runs.



NOTE: If the activation fails, please contact IDS GeoRadar Customer Support with your activation ID and a screenshot of the error.



NOTE: The User can return and re-host the Inertial Explorer SDK licenses limitless, provided that this has been previously returned by using LLM "Return License" function.



NOTE: To activate a license on a virtual machine, please contact the local sales representative to make sure that the activation of the entitlement is possible.



NOTE: Every time that IQMaps version is updated, also plugins installed must be updated. I.E.: the latest version of IQMaps, cannot run with older version of APS or AIMaps

4.5 How to install IDS GCAD

To install and activate **IDS GCAD** after downloading, follow the steps below:

1. click on the **idsGcad.exe** which can be found inside the folder **IDS_GCAD_XYZ**. and select the "run as administrator" option,

2. Press  on Fig. 23,



Fig. 23 IDS GCAD install step 1



3. The icon  shall be created on the PC desktop
4. To start **IdsGcad_xyz** there are two options:
 - a. clicking directly on the **IdsGcad_xyz** icon: in this case the program will automatically open AutoCAD with IdsGcad_xyz toolbar, (for first activation use "run as administrator" from the desktop icon) OR,
 - b. if **AutoCAD** is already open, the User may type **IdsGcad_xyz** in the command window, to recall the **IdsGcad_xyz** toolbar (Fig. 24).

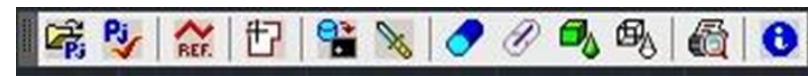


Fig. 24 IDS GCAD toolbar

5 HOW TO USE IQMAPS: THE WORKFLOW

IQMaps allows the full post-processing of IDS GeoRadar data by implementing a complete and simplified workflow (Fig. 25), including the following steps:

1. **Import/upload GPR data** (IMPORT SURVEY or IMPORT ACQUISITION PROJECT function par. 5.1) including:
 - a. Single antenna GPR,
 - b. Massive array GPR,
 - c. Multiple Systems (e.g.: Stream UP + Stream DP + Chaser XR + DS4000)
 - d. Multiple Positioning Systems
2. **QUALITY CONTROL** (in the **Overview** tab par. 5.2),
3. **EDITING OF TRAJECTORY** (in the **Trajectory Editor** tab par. 5.3),
4. **EDITING OF STANDARD DATA PROCESSING** (in the **Processing Editor** tab par. 5.4),
5. **TARGETS INSERT AND EDITING** (in the **Inspector** Tab par. 5.5),
6. **Export of targets, data, tomography, and images** in several formats (par. 5.6).

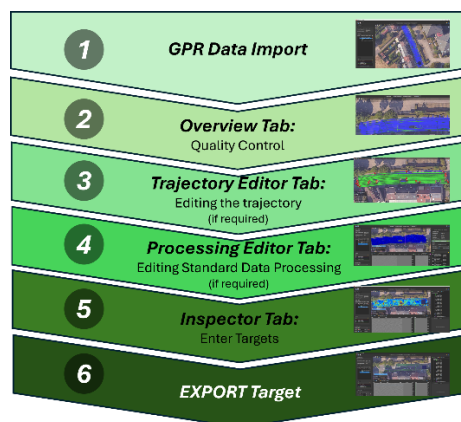


Fig. 25 IQMaps Workflow

5.1 How to import GPR data

When **IQMaps** is started the software's main page pops up (Fig. 26).



Fig. 26 IQMaps main page

5.1.1 How to import an acquisition project

1. Press on the **Project** icon (top left corner of Fig. 26) and select **Import Acquisition Project...** option (Fig. 27),

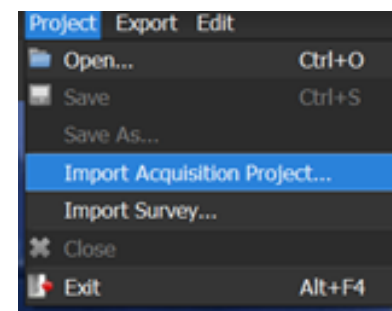


Fig. 27 Import acquisition project step 1

2. Follow the path where project data were saved and select the *.uproj file inside the project folder (Fig. 28), and press OPEN,

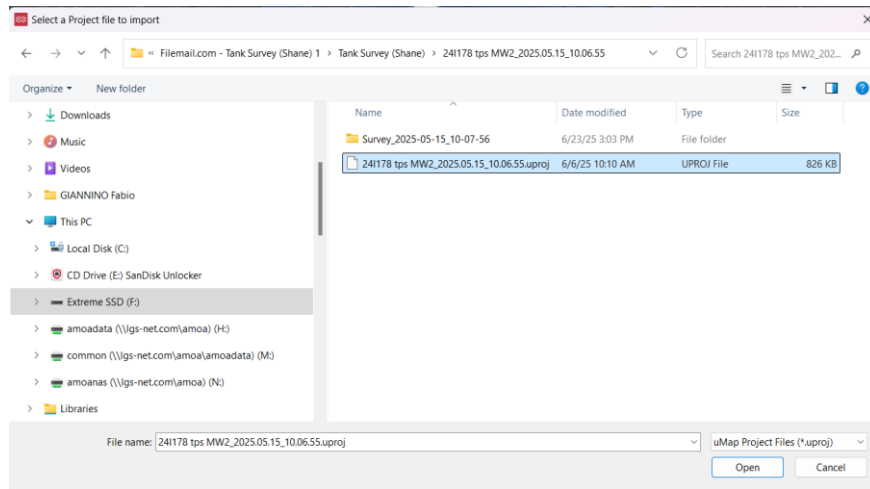


Fig. 28 Import acquisition project step 2

7. Project is now visualized as in Fig. 29,

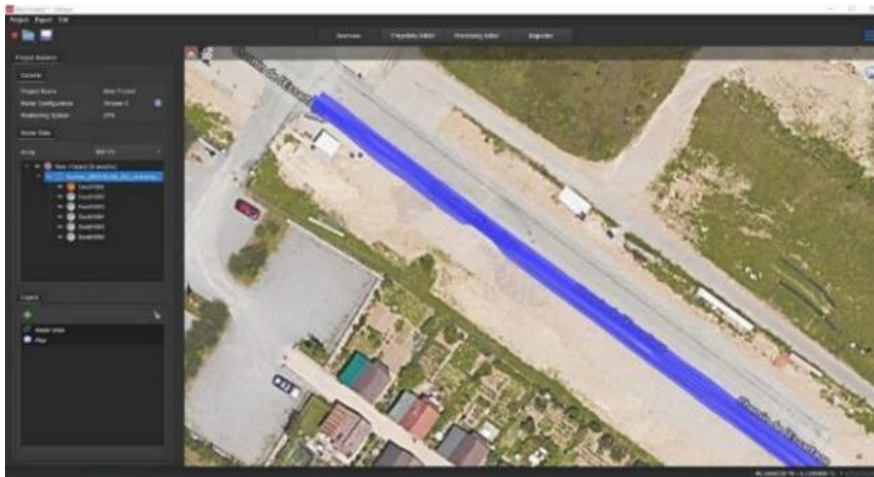


Fig. 29 data uploaded

8. Press Save, and this will create the project format *.iproj Fig. 30.

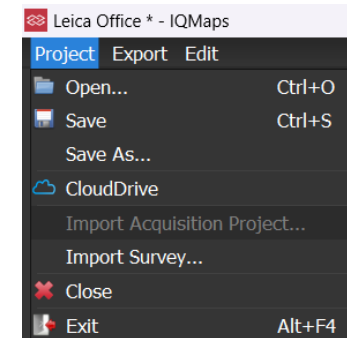


Fig. 30 Save a project



NOTE: After importing an Acquisition Project, ONLY single surveys can be uploaded in the same IQMaps Project.

5.1.2 How to import survey

One or more surveys can be included in the same project, following the steps below:

1. Press on the **Project** icon (top left corner of Fig. 26) and select **Import Survey...** option (Fig. 31),

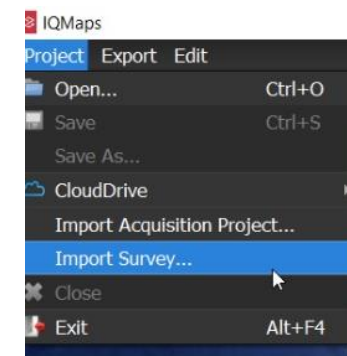


Fig. 31 Import survey step 1

- Follow the path where survey data folder was saved and select the survey folder to be uploaded (Fig. 32).

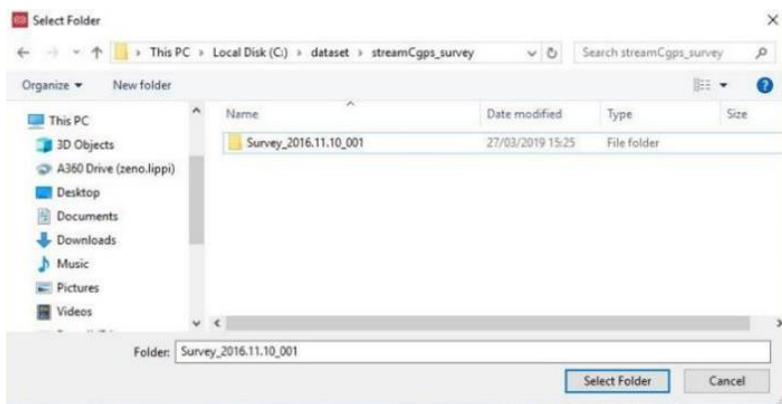
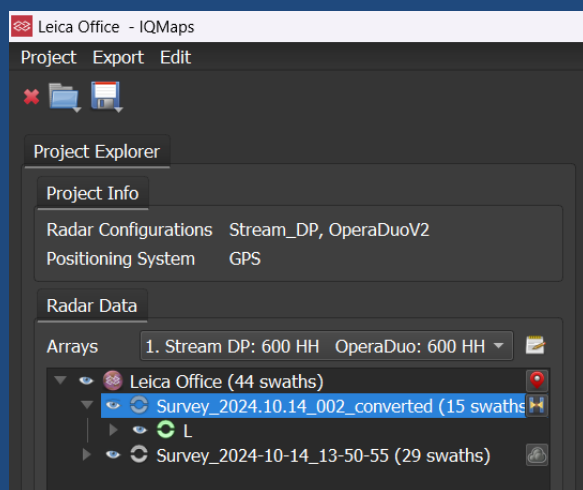


Fig. 32 Import survey step 2

- Repeat step 1 and step 2 as many times depending on how many surveys to be uploaded in the same Project.



NOTE: GPR surveys collected on site using different IDS GeoRadar system and different positioning systems can be uploaded into the same IQMaps project.



- Project is visualized as in Fig. 33,

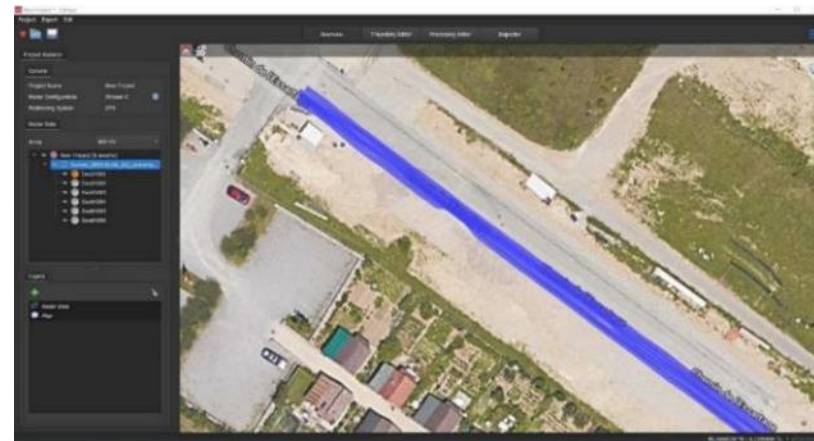


Fig. 33 data uploaded

- Press Save, and this will create the project format *.iproj Fig. 34.

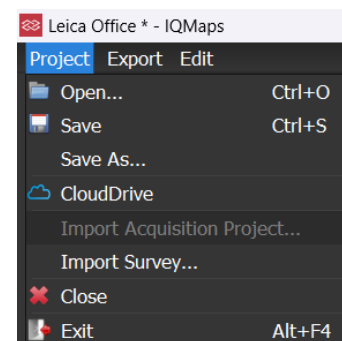


Fig. 34 Import survey step 3

5.1.3 How to open a project

To Open Projects previously created on IQMaps:

- Press on the **Project** icon (top left corner of Fig. 26) and select **Open...** option (Fig. 35), or directly press on  icon,

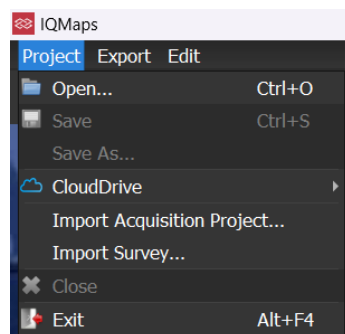


Fig. 35 Open project

- Follow the path where project was saved and select the *.iqproj file (Fig. 36) to be open.

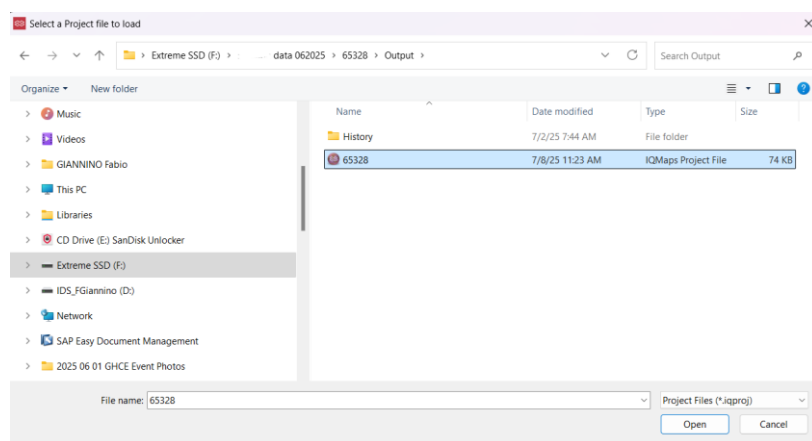


Fig. 36 Open a Project

- Alternatively, by keeping the left mouse pressed on  icon, would pop up a short list of recent projects open that can quickly be recalled Fig. 37.

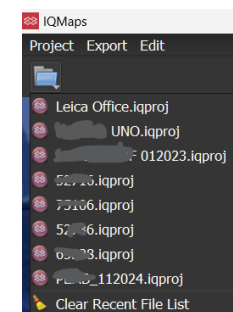


Fig. 37 Open a recent project

5.1.4 How to relocate a Project

If a project has been moved from the original directory the following window shall pop up Fig. 38, when attempting to Open a Project,

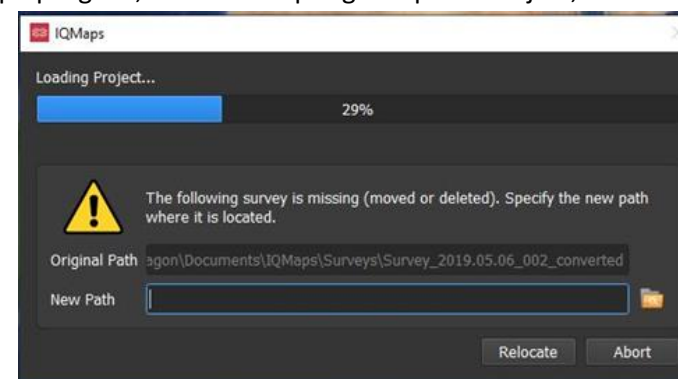


Fig. 38 The Relocate Project window

- Press on ,
- Select the path where the project has been moved to,
- Select each survey of the Project individually,
- Press  for all surveys belonging to the original project,
- Repeat as many times as many surveys are part of the Project, and the Project is now open.

5.2 Overview: How to...

As IDS GeoRadar data set is uploaded on **IQMaps** and a project is created, IQMaps **OVERVIEW** window page shall pop up (Fig. 39 and Fig. 40) and the data set NORTH direction is always oriented as indicated by the green arrow.

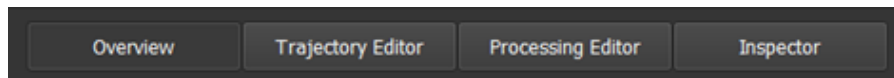


Fig. 39 Project tabs

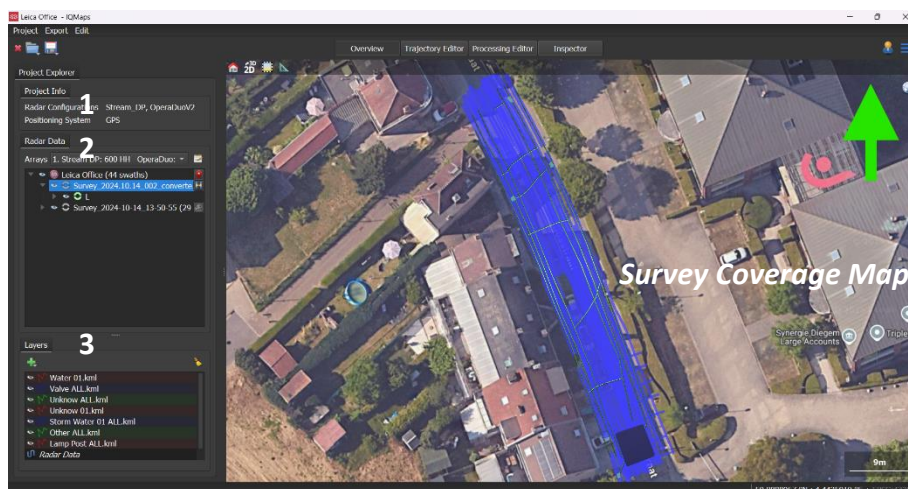


Fig. 40 IQMaps Project Overview page with indication of North direction (green arrow)

- **ON THE LEFT** side panel of Fig. 40, it is located the Project Explorer tab, including three tabs:
 1. **Project Info** (includes information or GPR systems Configurations and Positioning system),
 2. **Radar Data**, and
 3. **Layers**.
- **ON THE RIGHT** side panel of Fig. 40, it is located the survey coverage (the blue lines) overlapped to a base map (selectable by the User

among Google Hybrid, Google Maps, Google Satellite, OpenStreetMap, Bing, or ArcGis).

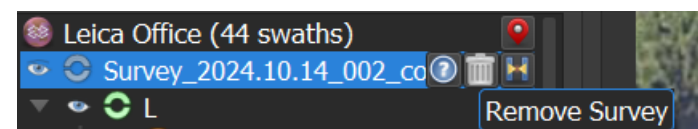
The following actions described hereinafter, including a general quality control of the project, can be performed in the Overview tab.

5.2.1 ...hide or show survey

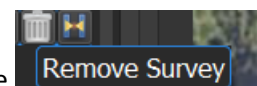
1. Point the mouse to the Radar data tab, then,
2. Click on the  icon beside the Project name to Hide/Show all swath from the Survey Coverage Map.
3. Click on the  icon beside the Survey name to Hide/Show all swath of that Survey, from the Survey Coverage Map.
4. Click on the  icon beside the Swath name to Hide/Show that swath from the Survey Coverage Map.

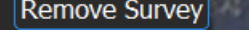
5.2.2 ...remove survey from a Project

1. Point the mouse to the Radar data tab over a Survey, then,



2. A  icon shall pop up,



3. Press on the  to Remove that Survey from the Project.

5.2.3 ...show survey information

1. Point the mouse to the Radar data tab over a Survey, then,

2. A  icon shall pop up,
3. Press on the  icon to show information on that Survey, as shown on Fig. 41.

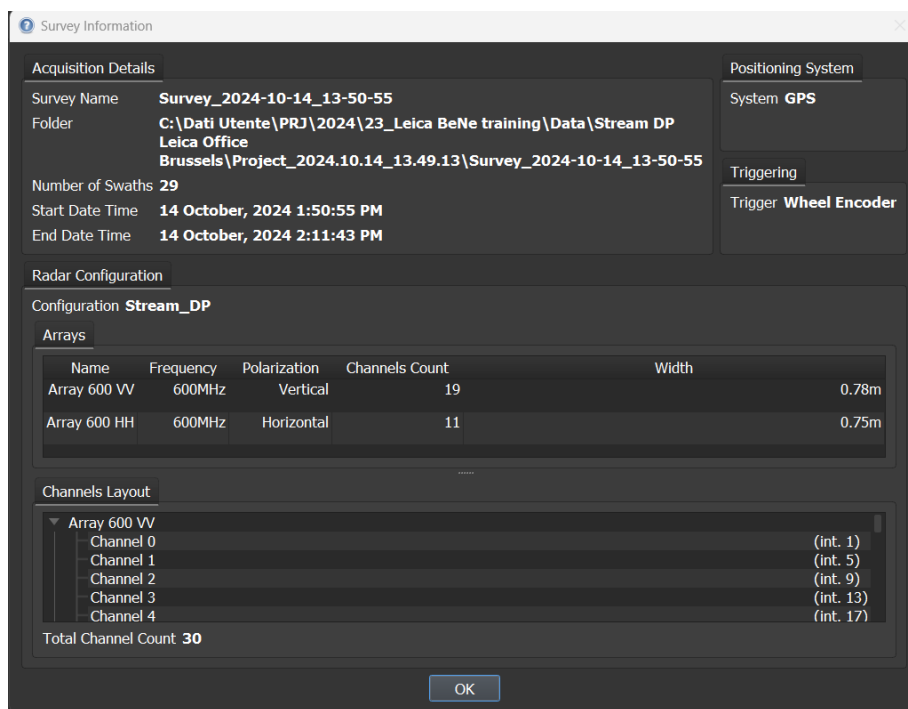


Fig. 41 Survey Information window

5.2.4 ...merge swath on C Scan view

1. Point the mouse to the Radar data tab over a Survey, then,

2. A  icon shall pop up,
3. Press on the  icon to merge all the swath of that Survey.

5.2.5 ...add vector layers

1. Point the mouse to the Layers tab over a Survey, then,
2. A  icon shall pop up,
3. Press on the  and select the Layer to be added to the project (supported formats are *.kml, *.dxf, *.gml, *.shp, *.geojson, *.json, *.ogpr) Fig. 42,

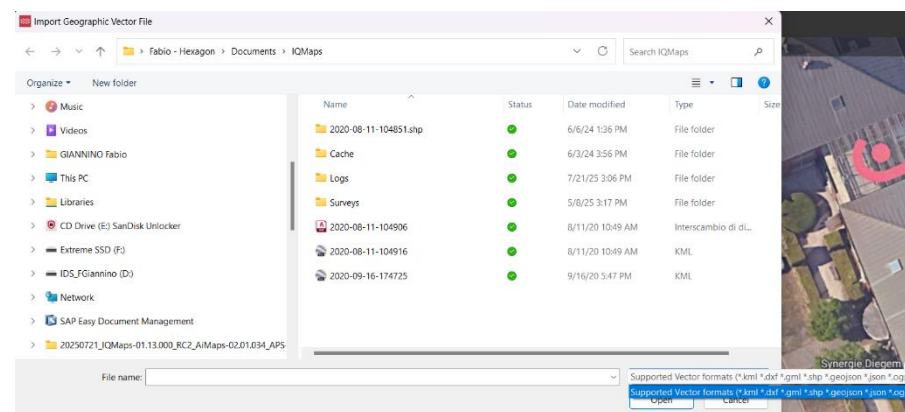


Fig. 42 Import vector layer

4. Specify the Spatial Reference System (that must be the same as the Project) Fig. 43, then press Import.

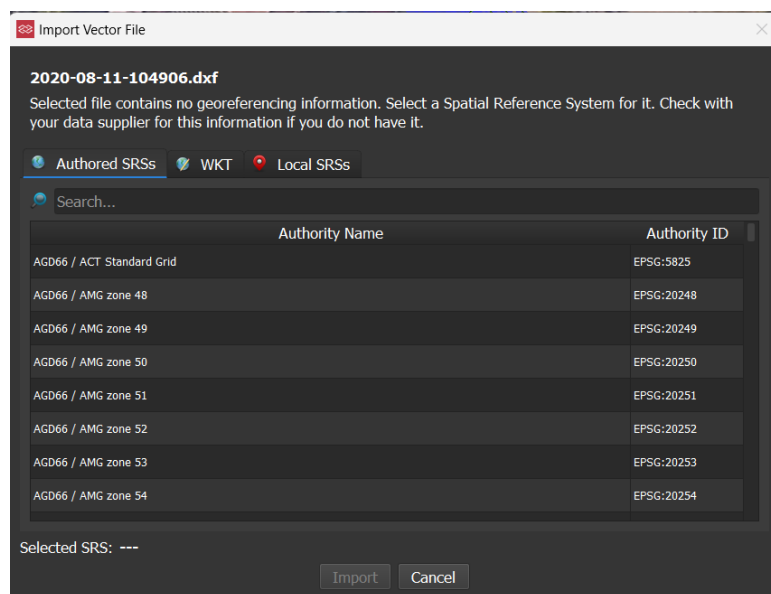


Fig. 43 Spatial Reference System selection of the Layers to import

- Layers added shall be listed as shown in Fig. 44 and they can be hidden or shown by using the  icon.

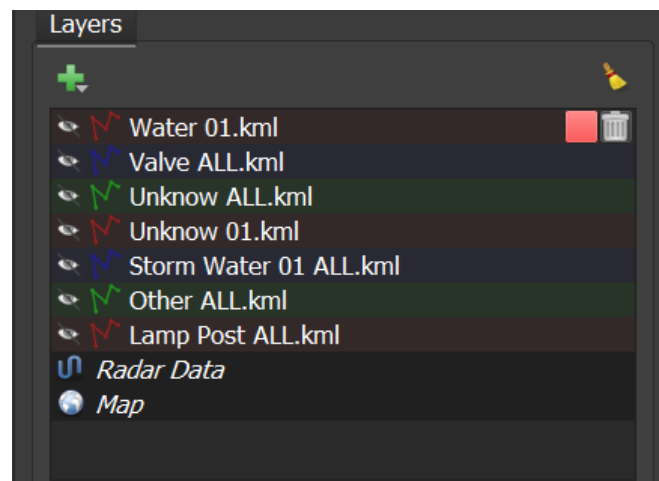


Fig. 44 Layers list

- By toggling the icon  beside the layers name, the color can be changed () or the layer can be removed ().
- By pressing the  icon, User shall clear the whole layers list.

5.2.6 ...change from 2D to 3D view

- Positioned in the top left corner of the survey coverage map menu



bar,

- Select the  icon to swap the project view from 2D to 3D.
- On 2D view: pan mouse, while keeping right mouse button pressed on C Scan to change plan orientation of the 2D map on screen.
- On 3D view: pan mouse, while keeping the right mouse button pressed on C Scan to change orientation in space of the 3D map on screen.
- Use the  icon to go back to the original position North-oriented.

5.2.7 ...select areas on the coverage map

- Aim at the top left corner of the survey coverage map,



- Select the  icon,
- Then press "alt" on keyboard, and select an area with Left Mouse Button pressed as shown in Fig. 45,

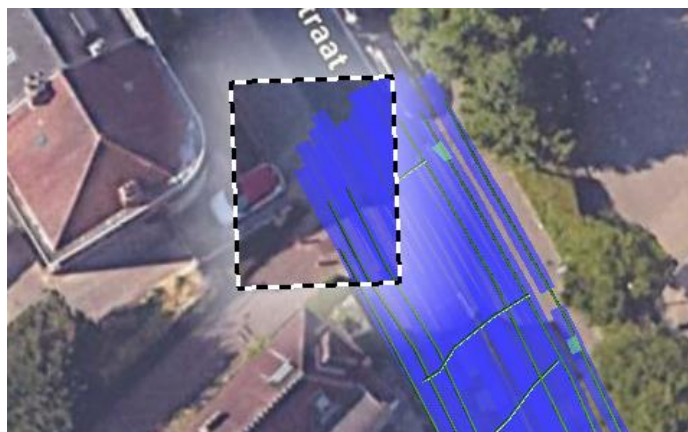


Fig. 45 Area selection

- Right click on the selected area and select one of the options to implement on the selected area Fig. 46.

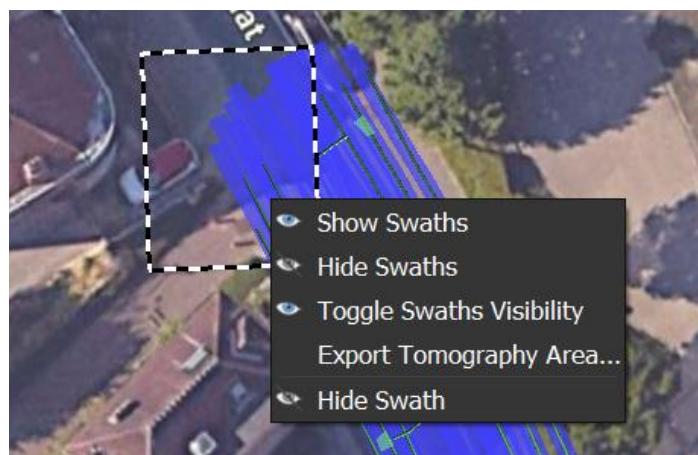


Fig. 46 Action that can be implemented in the selected area

- Select the  icon again to close the area selection option.

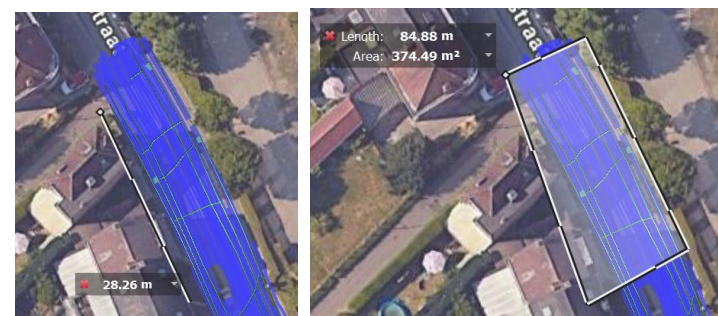
5.2.8 ...measure distances and areas

- Aim at the top left corner of the survey coverage map,



- Select the  icon,

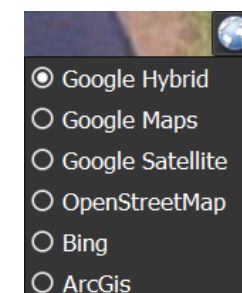
- Then press “alt” on keyboard and select (at least) two points to measure a distance, or at least 4 points (and close against the first point) to measure an area.



- Press again on  icon, to exit the measuring tool.

5.2.9 ...change the base map viewer

- Aim at the top right corner and press the  icon,
- Then, select one of the base maps among the Providers listed below.



5.3 Trajectory editor: How to...

As the **Trajectory editor tab** is selected (Fig. 47) the **Trajectory editor window** pop us (Fig. 48).

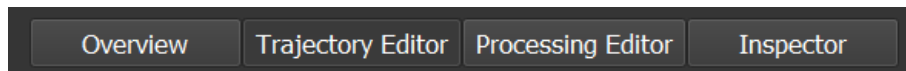


Fig. 47 Trajectory editor tab selected

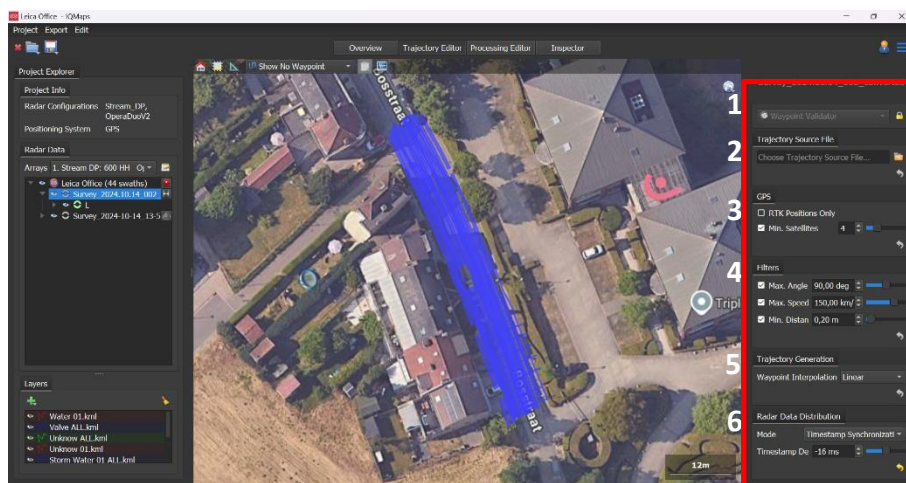


Fig. 48 Trajectory editor window

The left panel and the panel with the map of the survey coverage area remain unchanged with respect to the **Overview tab**, whereas the right panel (red rectangle in Fig. 48), includes:

1. *Trajectory Source file tab,*
2. *GPS tab,*
3. *Filters tab,*
4. *Trajectory generation tab, and*
5. *Radar data distribution tab.*

In this tab of IQMaps, it is possible to adjust the trajectory of the survey, if some sort of glitch occurred to the positioning datum paired to the GPR data.

5.3.1 ...hide/show trajectory waypoints

1. At the top left corner of the coverage area panel select **"Show No Waypoint"** from the drop-down menu, to Hide the waypoint of the trajectory for the whole project (Fig. 49),



Fig. 49 Hide trajectory waypoints

2. Select **"Show Swath Waypoints"** to visualize one single swath trajectory waypoints (Fig. 50),

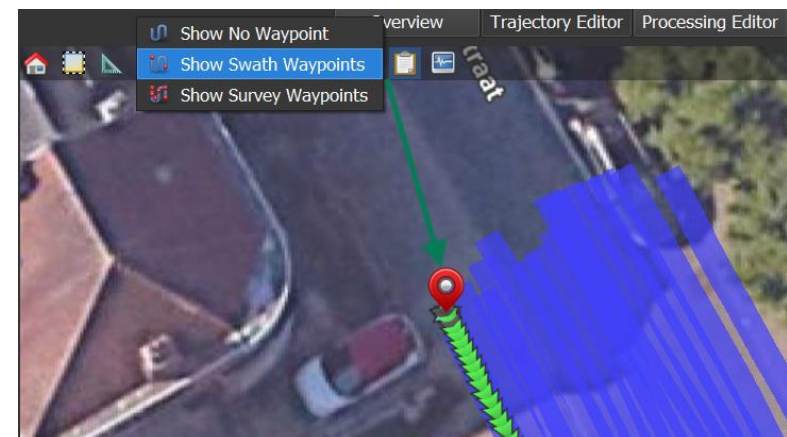


Fig. 50 Show Swath waypoints

3. Select “**Show Survey Waypoints**” to visualize the waypoints of the whole survey (Fig. 51).

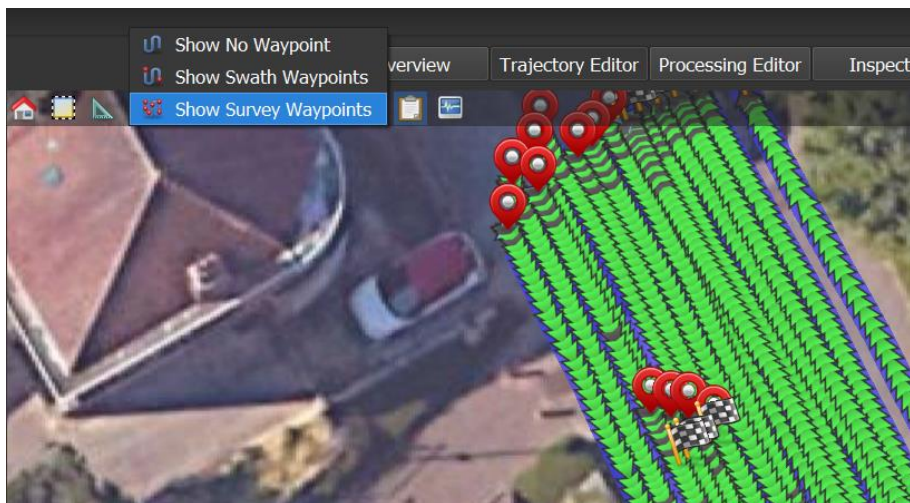


Fig. 51 Show Survey trajectory Waypoints

5.3.2 ...hide/show tomography

1. Press the  icon at the top left corner of the coverage area map (Fig. 52), to show the tomography map in the trajectory tab.

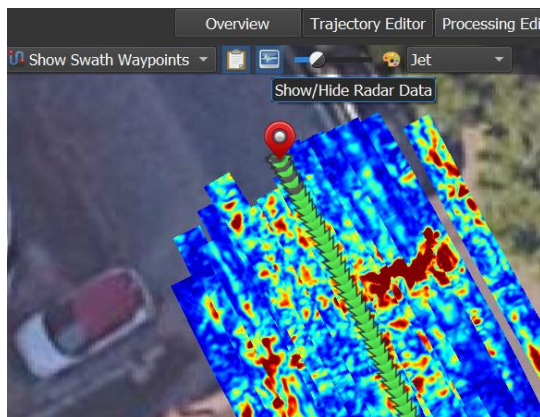


Fig. 52 How to show the tomography on the Trajectory editor

2. Press again the same icon to hide the tomography view (Fig. 53), and only show the swaths coverage.

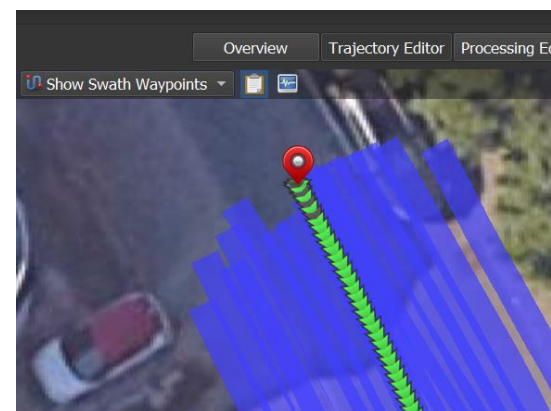


Fig. 53 How to hide tomography on the trajectory editor

5.3.3 ...import external trajectory (APS only)

In collecting data with Stream UP only, the **Accurate Positioning System** (APS) can be used as a positioning system. In this case, a trajectory file is created and associated with each survey of the project:

1. Press on the  icon of Fig. 54,

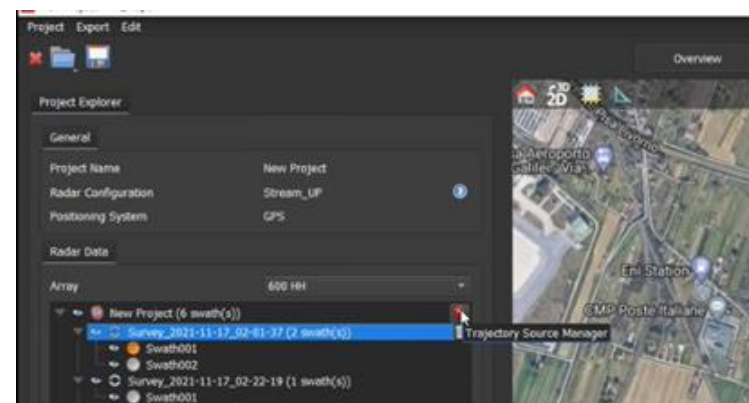


Fig. 54 APS Source Trajectory Manager

2. Select “Open File” from the Source Trajectory Manager that pops up Fig. 55,

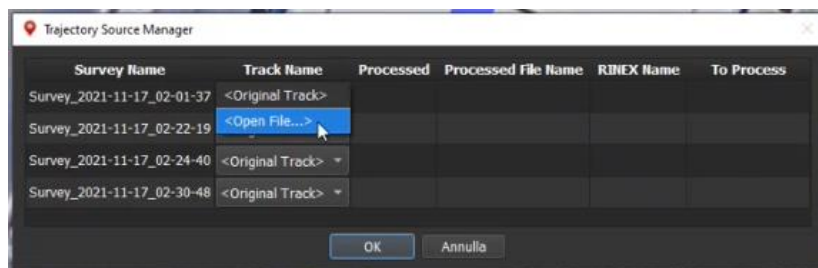


Fig. 55 APS Source Trajectory Manager step 2

3. Associate to each Survey of the project the specific .gps file Fig. 56

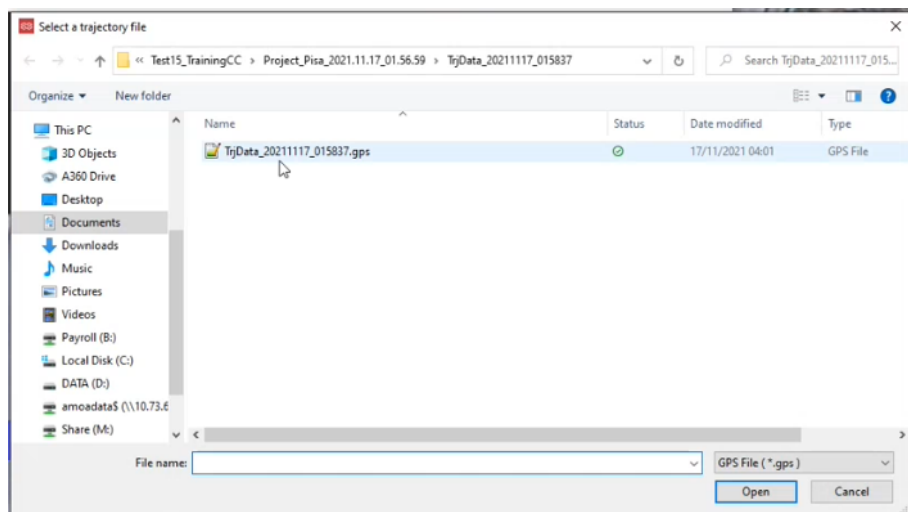


Fig. 56 APS Select trajectory file

Native APS Trajectory data for Stream UP can also be post-processed and imported after downloading the corrected trajectory as RINEX. How to download and process RINEX data is illustrated in Appendix B.10 and Appendix B.11.

5.3.4 ...edit non-georeferenced GRID

If a GPR survey using non-georeferenced grid has been gathered on site, this can be georeferenced on IQMaps, provided that the grid origin point coordinates are available. The procedure is the following (Fig. 57):

1. Import the dataset collected in grid mode,
2. Enter the trajectory editor tab,
3. Enter the Latitude (in decimal degree) of the grid origin point,
4. Enter the Longitude (in decimal degree) of the grid origin point,
5. Enter the altitude in meters,
6. Adjust the heading towards North.

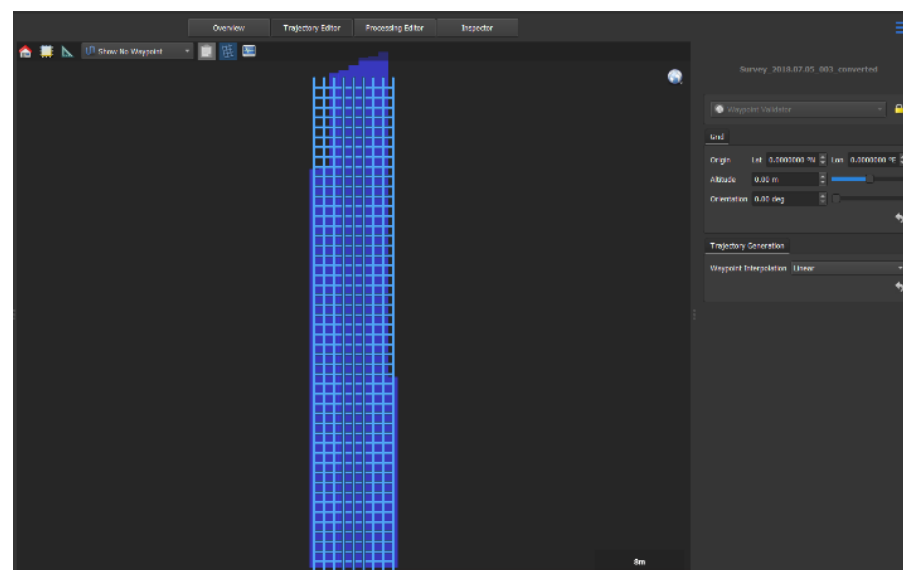


Fig. 57 Grid GeoReferencing

5.3.5 ...edit TPS positioning data

5.3.5.1 NON-Georeferenced TPS

If a GPR survey has been conducted on site by pairing a TPS, this can be georeferenced on IQMaps, provided that the TPS base station origin point coordinates are available, through the following procedure (Fig. 58):

1. Import the dataset collected using the TPS, as positioning system,
2. Enter the trajectory editor tab,
3. Enter the Latitude (in decimal degree) of the TPS base origin point,
4. Enter the Longitude (in decimal degree) of the TPS base origin point,
5. Enter the altitude in meters,
6. Adjust the heading (i.e.: orientation with respect to North).

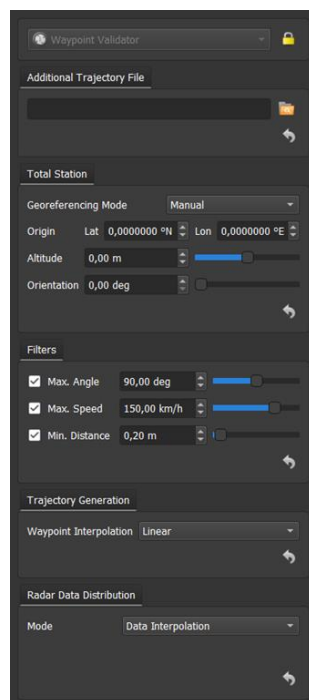


Fig. 58 TPS edit trajectory

5.3.5.2 Georeferenced TPS

If TPS data are georeferenced during data acquisition, then:

1. Import the project,
2. Enter the Trajectory editor tab,
3. Instead of the GPS tab, the *Total Station* tab shall appear on the right panel. Here, press on **Spatial Reference System** icon of Fig. 59,

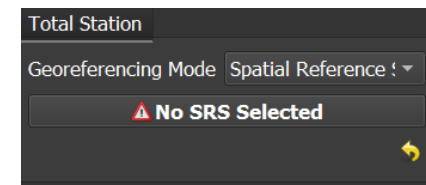


Fig. 59 SRS selection

4. Select the Spatial Reference System (SRS) in which TPS data were collected on site (Fig. 60),

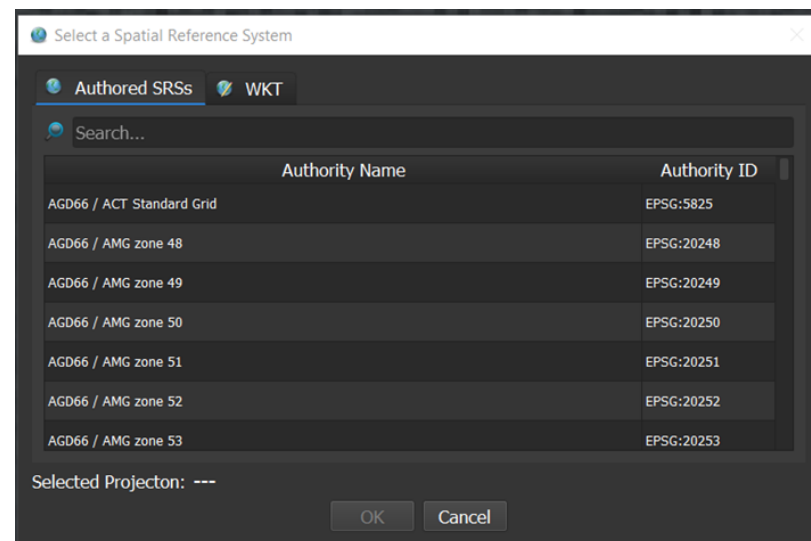
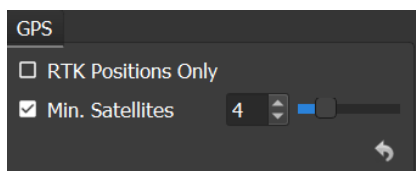


Fig. 60 SRS selection step 2

5. Press **OK** to confirm.

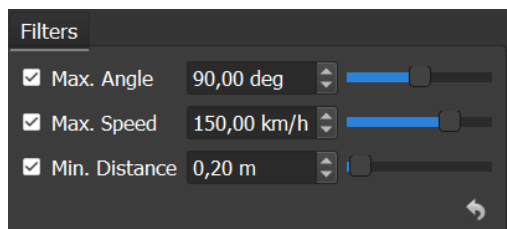
5.3.6 ...hide/show waypoints based on satellites or RTK coverage

1. In collecting GPR data paired with a GNSS, to reject positioning datum not covered by RTK correction, select the option “**RTK Position Only**” (by default this is un-selected), OR
2. To reject positioning datum based on a minimum number of satellites, select “**Min. Satellites**” (by default this is selected) and state number of satellites (by default is 4).



5.3.7 ...filter GNSS data by angle, speed, or distance filters

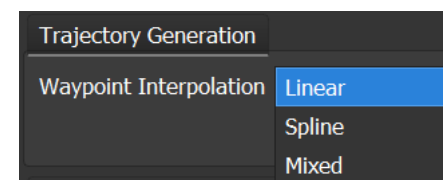
1. In collecting GPR data paired with a GNSS, to reject (filter out) positioning datum depending on the angle between two next datum points, select “**Max. Angle**” (selected by default) and enter the angle in degree between two next points (90° by default).
2. To reject (filter out) positioning datum depending on the speed between two next datum points, select “**Max. Speed**” (selected by default) and enter the velocity value in Km/h between two next points (150 Km/h by default).
3. To reject (filter out) positioning datum depending on the minimum distance between two next datum points, select “**Min. Distance**” (selected by default) and enter the distance in meter between two next points (0,20 m by default).



5.3.8 ...edit the Trajectory Generation method

Trajectory visualized on screen results from the Waypoints interpolation. This interpolation can be obtained according to three methods:

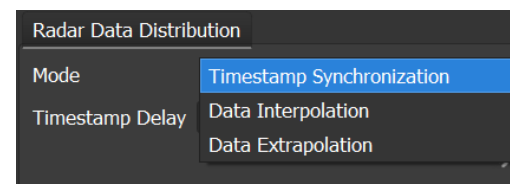
1. **Linear**: (Default option) this option is applied when Operator maintained a straight-line during data acquisition,
2. **Spline**: Select this option if during data acquisition Operator kept a non-linear trajectory (curves),
3. **Mixed**: select this option when the mixed condition of the two previous is met.



5.3.9 ...edit Radar Data Distribution method

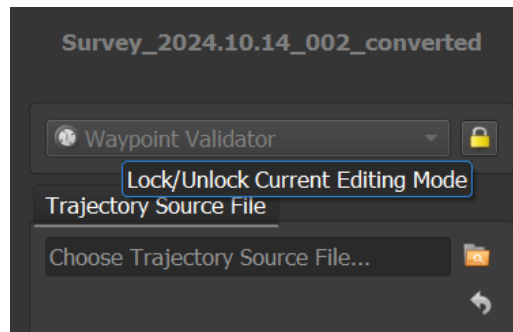
GPR data are associated (synchronized) with positioning datum in three possible modes:

1. **Timestamp Synchronization**: (Default option) this option allows to associate GPR sweep with positions taking into consideration the timestamp of the GNSS.
2. **Data Interpolation**: Select this option when/if clear misalignments between parallel lines are visible because of the poor GNSS coverage between beginning and end of the acquisition line.
3. **Data extrapolation**: Select this option when/if clear misalignments between parallel lines are visible because of the poor GNSS coverage by the end of the acquisition line.

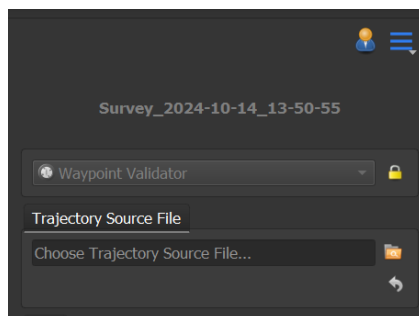


5.3.10 ...edit the trajectory manually

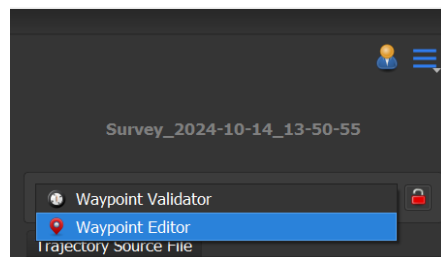
1. Aim at the top right corner of the trajectory editor tab to unlock the current editing mode,



2. Press on the  icon to activate the drop down-menu,



3. Select Waypoint Editor,



3. Press the  icon,

4. Drag-and-drop the waypoint to be moved by using the Left Mouse button,



5. Add missing points to fill the gaps between positioning datum by pressing "CTRL" on key-board and Left Mouse Button in the very point where the datum must be inserted.



5.4 Processing Editor: How to...

As the **Processing editor tab** is selected (Fig. 61) the **Processing editor window** pops up Fig. 62.

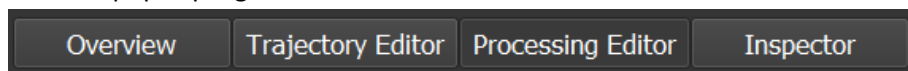


Fig. 61 Processing editor tab

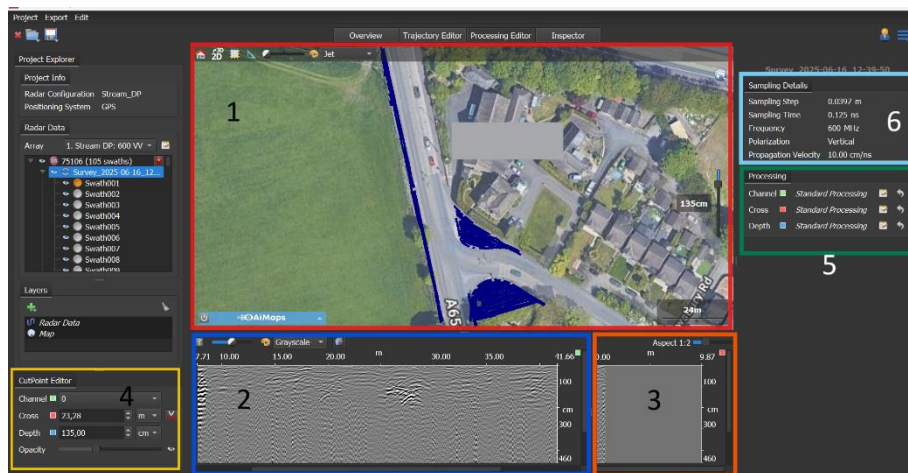


Fig. 62 Processing Editor Window

The upper left panel of the **Processing editor window** (Fig. 62) remain unchanged, whereas the rest is optimized for processing editing, as it follows:

1. (red rectangle): **C Scan**, shows the radar data as tomography view,
2. (blue rectangle): **B Scan**, shows each single radargram (i.e.: each channel),
3. (orange rectangle): **T Scan**, shows the cross-cut between parallel channels. This can also be customized.
4. (Yellow rectangle): **Cut Point editor**, allows to customize the view of B Scan and T Scan, by channel, and also to generate a Custom Cut for the T Scan view,

5. (green rectangle): **Processing tab**, where it is possible to edit the standard macro filters (add or remove filters, modify single filters parameters),
6. (light blue rectangle): **Sampling details tab**, showing the acquisition parameters including:
 - Sampling step,
 - Sampling time,
 - Frequency,
 - Polarization, and
 - Propagation Velocity.

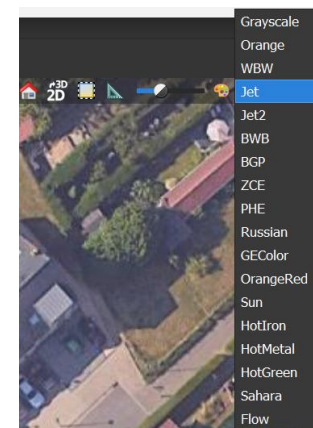
Inside the Processing Editor window, Users can perform several operations, as described hereinafter.

5.4.1 ...change color palette and C Scan view contrast

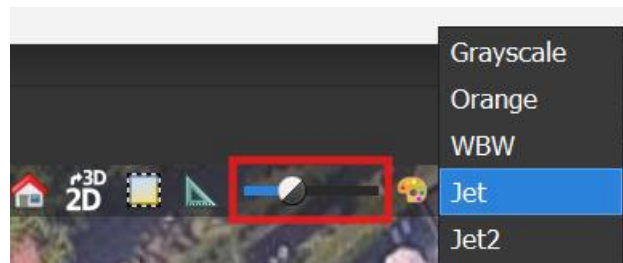
1. At the top left corner of the Processing editor window, using Left Mouse Button on the “Choose Colormap” drop-down menu,



2. Select the preferred color palette among those available (default is “Jet”)

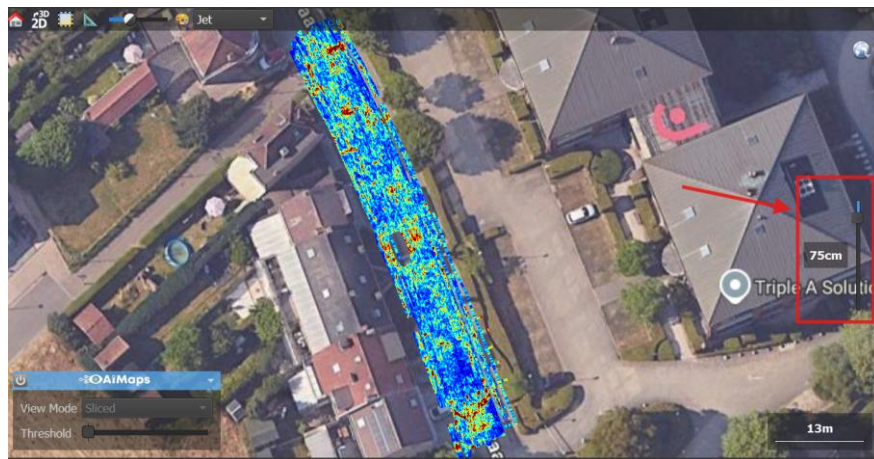


3. The color map of the C Scan shall update automatically.
4. By acting on the slider (red rectangle) at the left side of the color palette drop-down, the saturation of the color map can be changed.



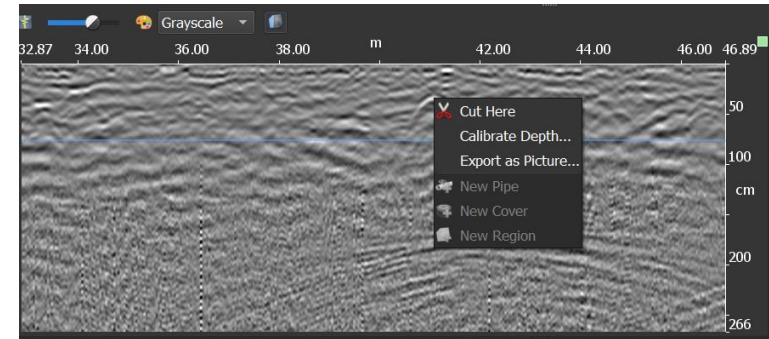
5.4.2 ...change C Scan depth view

1. With the left mouse button pressed act on the slider at the right side of the Tomography map view (red rectangle), or alternatively
2. Use the scroll of the mouse after placing the cursor on top of the slider.

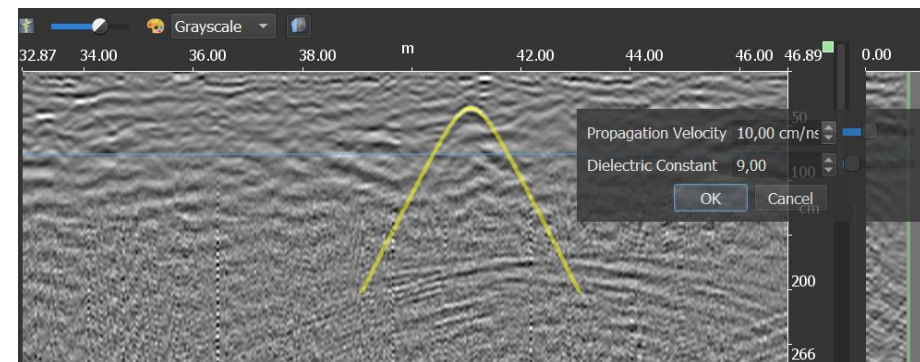


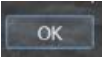
5.4.3 ...calibrate depth (Hyperbola fitting) on B Scan

1. Aim at the top of a hyperbola well defined on B Scan,
2. Right click with the mouse,



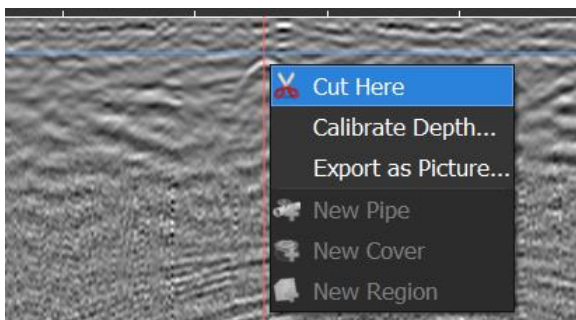
3. Select “Calibrate Depth”,
4. By using the  arrows icon or the slider icon  besides, adjust the yellow hyperbola so that it perfectly fits the actual hyperbola on B Scan,



5. Press  to apply the new Propagation Velocity value.

5.4.4 ...visualize A Scan

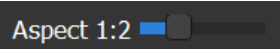
1. Select the  icon at the top left corner of the B Scan Panel,
2. Then, right click on a specific point of the B Scan and press “Cut Here”,

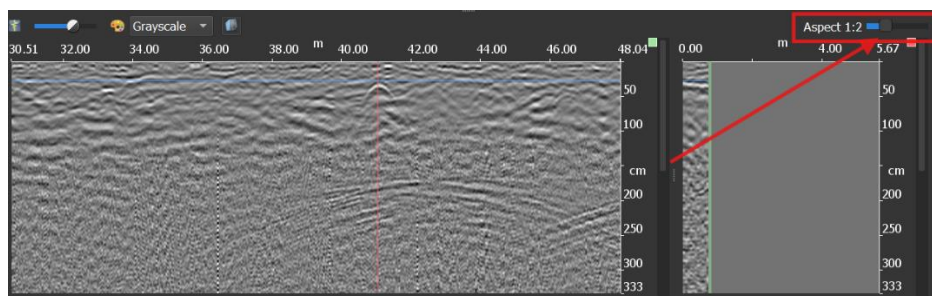


3. The A Scan shall be visualized at the very point where the cut has been specified.

4. Press again on the  icon to deactivate the A Scan view.

5.4.5 ...change the Aspect ratio (Horizontal distance respect to depth)

1. Aim at the “Aspect” slider  at the top right corner of the T Scan Panel,



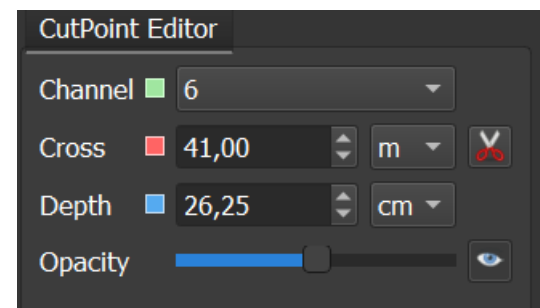
2. Vary the aspect ratio (default is 1:2, that means depth is twice horizontal distance) between horizontal-to-vertical. Admitted are:

- 1:1
- 1:2
- 1:3
- 1:4
- 1:5

3. Aspect ratio changes apply to both the B Scan and the T Scan, at the same time.

5.4.6 ...change the channel visualized in B Scan

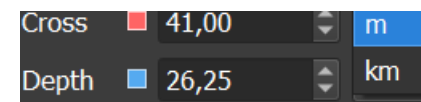
1. Aiming at the Cut Point Editor with the Left Mouse Button,
2. Press on the Channel Drop-down,



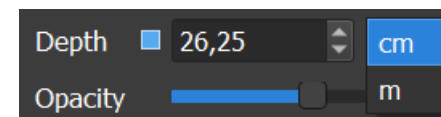
3. Select the channel to be visualized on B Scan panel,
4. Alternatively, use the scroll directly on the drop-down.

5.4.7 ...change unit measure for distance and depth in B Scan

1. Aiming at the Cut Point Editor with the Left Mouse Button,
2. Press on the Cross Drop-down, to change from “m” to “Km”, and the depth slice automatically update,

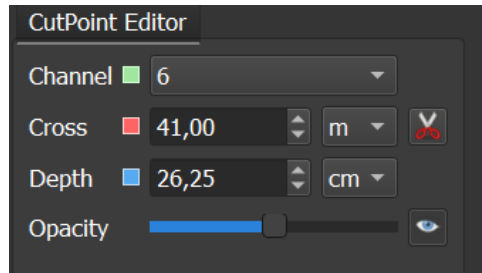


3. Press on the Depth Drop-down, to change from “cm” to “m”, and the B Scan and T Scan shall automatically update.

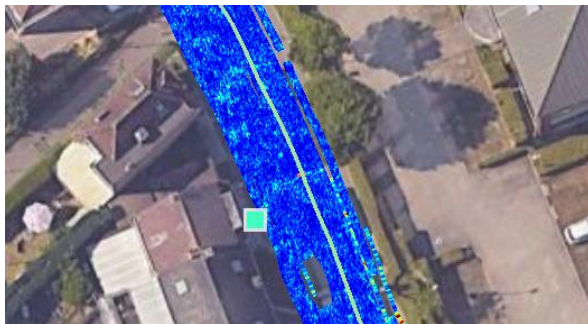


5.4.8 ...generate a Custom Cut

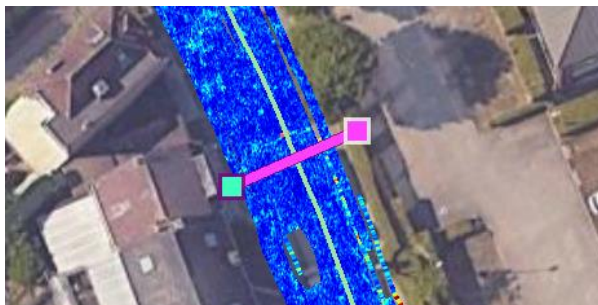
1. Aiming at the Cut Point Editor with the cursor,
2. Press on the  icon,



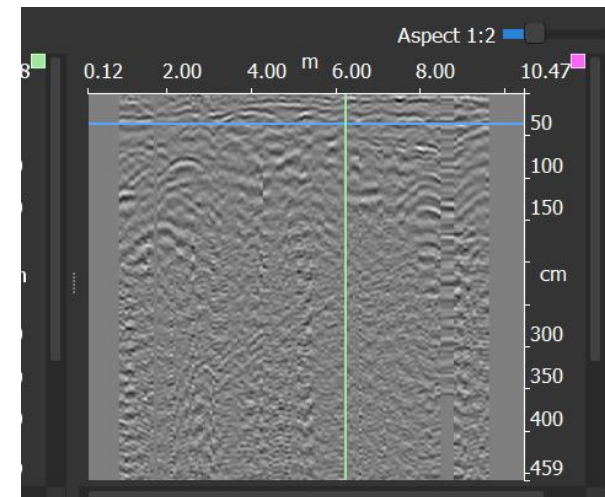
3. With the “alt” key pressed on keyboard, select the start point (green) of the Customized cross section, using the left mouse button,



4. Then press again, with the left mouse button, at a second point (purple) where the Customized cross section should end, and release the “alt” button



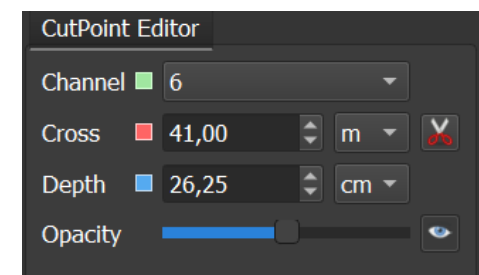
5. T Scan is generated with the beginning of the Custom Section on the left side (Green dot ) and the end of the section on the right side (purple dot ) ,



6. Drag and drop with left mouse button pressed on begin () or at the end () shall update the section.

5.4.9 ...view or hide the synchronization cut line

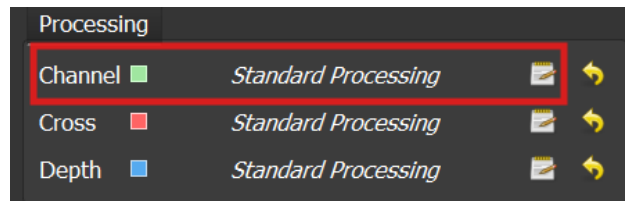
1. B Scan, C Scan, and T Scan view is synchronized, and the alignment of the synchronization can be shown or hidden activating the  icon in the Cut Point editor,



2. Act on the Opacity slider , to increase or decrease the visibility on screen of the cut line.

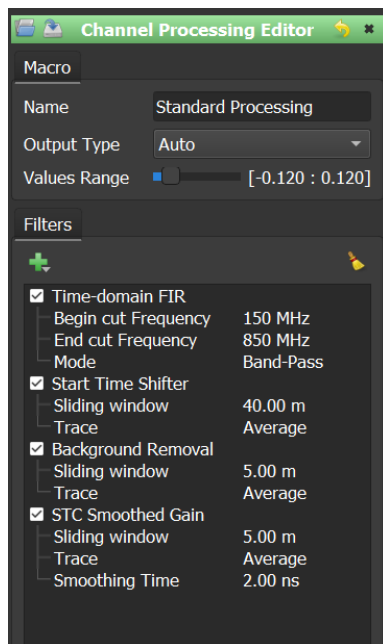
5.4.10 ...edit the Channel (B Scan) standard processing

1. Press the  icon, besides *Channel Standard Processing*,



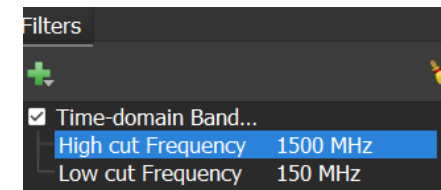
2. The *Channel Processing Editor* Window shall expand, showing two tabs:

- **Macro**, and
- **Filters**

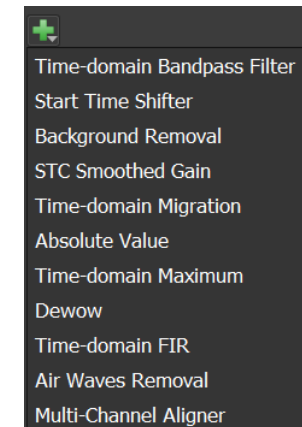


3. The **Filters** tab includes the list of filters applied by default (the so-called “Standard Macro”) for the specific Hardware.
4. Press on the  icon to activate or deactivate a specific filter.

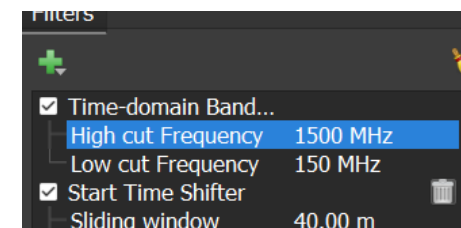
5. Select (by double-click) a specific filter parameter to edit manually the value for that “parameter” of that “filter” (**High cut Frequency**, in the example below).



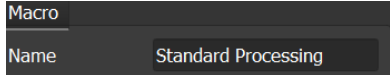
6. Press the  icon to add a filter from the list that pops up.

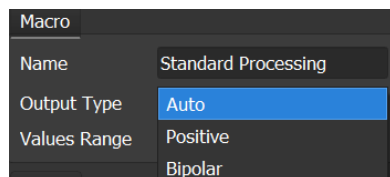


7. Press the  icon beside a specific filter to delete a filter from the list.

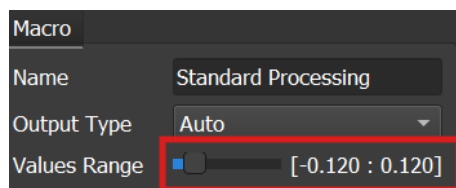


8. Press the  icon to clear the whole filter list.

9. Press the  icon to go back (undo) from the change made in the standard processing.
10. Press the  icon to save, in a **Macro** file, the change made in the standard processing.
11. Edit the name of the **Macro** saved in the text box
 (default name is “Standard Processing”).
12. Press the  icon to upload a **Macro** previously saved.
13. Select the type of Color Palette from the “Output Type” drop down,

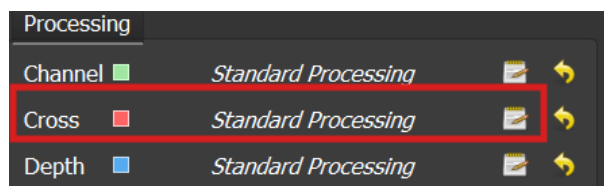


14. Choose the “Values Range” of the minimum and maximum value of the color palette by acting on the slider below (red rectangle),

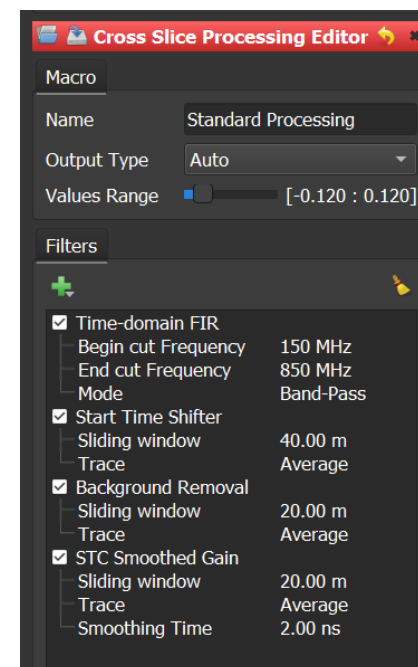


5.4.11 ...edit the Cross (T Scan) standard processing

1. Press the  icon, besides Cross Standard Processing,



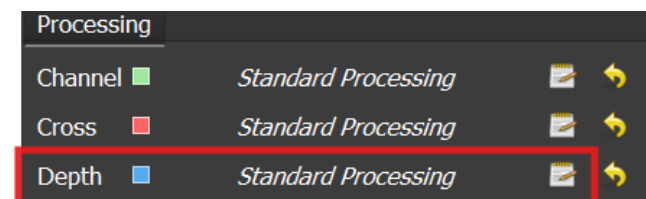
2. The Cross Slice Processing Editor Window shall expand, showing two tabs:
 - **Macro**, and
 - **Filters**



3. To edit the standard processing macro, the same procedure described at points 3 to 14 of par. 5.4.10, applies.

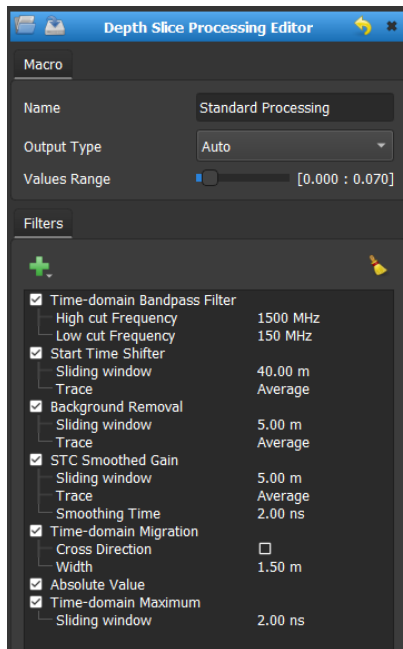
5.4.12 ...edit the Depth (C Scan) standard processing

1. Press the  icon, besides Depth Standard Processing,



2. The Depth Slice Processing Editor Window shall expand, showing two tabs:

- **Macro**, and
- **Filters**



3. To edit the standard processing macro, the same procedure described at points 3 to 14 of par. 5.4.10, applies.

5.5 Inspector tab: How to...

As the **Inspector** tab is selected (Fig. 63) the **Inspector editor** window pops up (Fig. 64).

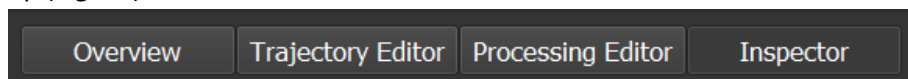


Fig. 63 Inspector tab

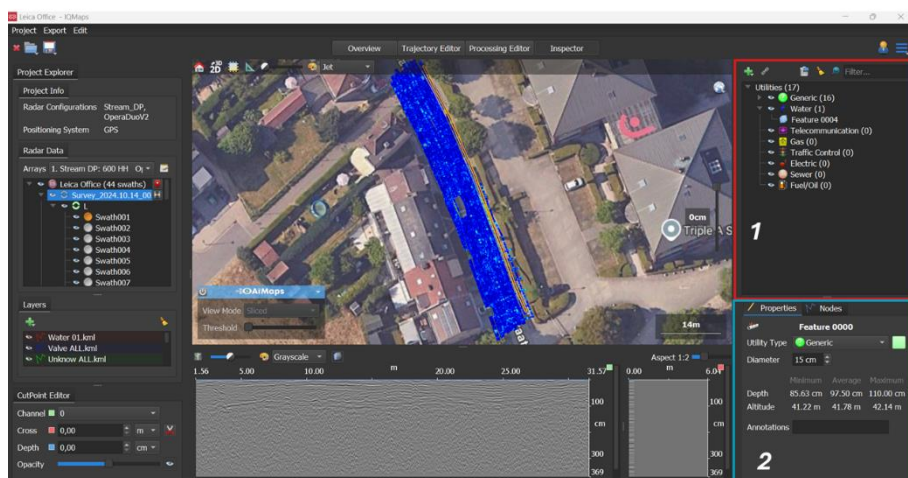


Fig. 64 Inspector editor window

Apart from the upper (1, red rectangle in Fig. 64) and lower (2, blue rectangle in Fig. 64) right panel, the **Inspector editor window** and functions remain unchanged with respect to the **Processing editor window** (Fig. 62). In Particular:

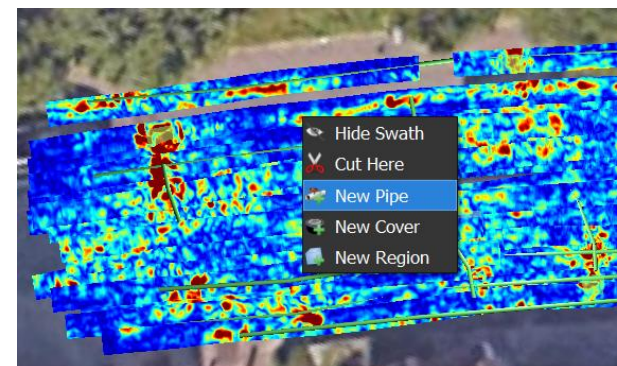
1. **Red rectangle** area: it includes all Utilities (Pipes, Covers and Regions) drawn and entered by the User.
2. **Blue rectangle** area: includes "Properties" and "Nodes" details of each of the "Utilities" entered by the User.

Inside the Inspector Editor window, User can perform several operations, as described hereinafter.

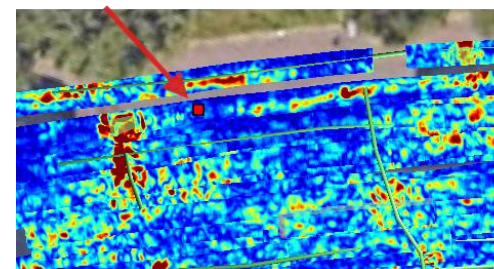
5.5.1 ...enter New Pipe

Pipes can be entered on C Scan, B Scan, as well as on T Scans.

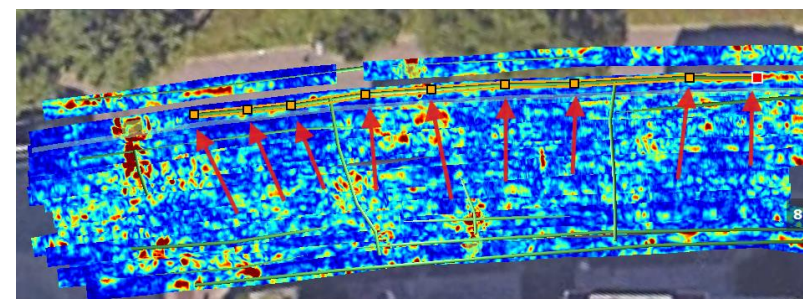
1. Right mouse click where the first point of the Pipes should be entered and select New Pipe,



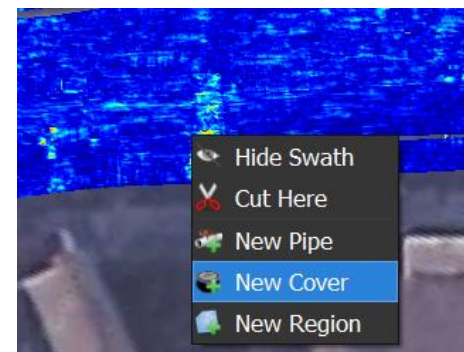
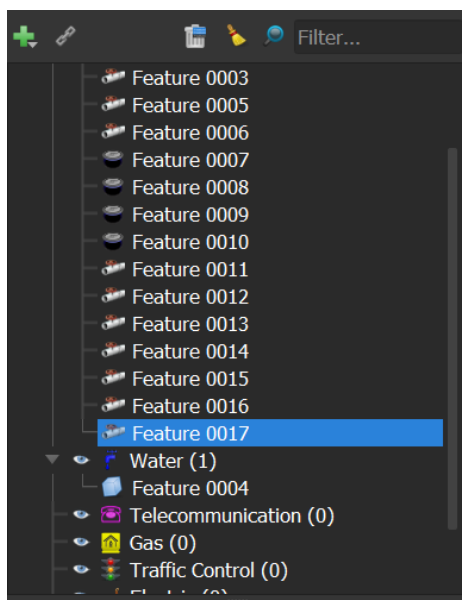
2. A red dot shall appear where the selection of the previous point was made,



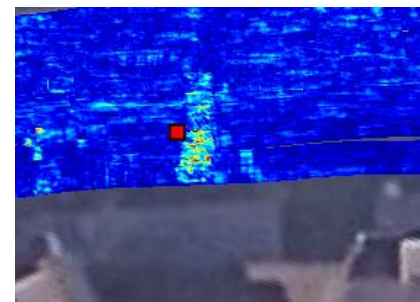
3. By holding "ctrl" key on keyboard enter as many points as required, to define the pipe identified,



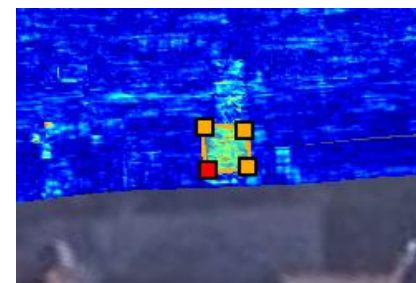
- Each of the “nodes” just entered to define a pipe, can be moved by drag-and-drop with left mouse button pressed on it,
- Release “ctrl” key and press anywhere with mouse to close the Pipe entering.
- The Pipe just entered is now listed on the right-side panel. Double-click on this to edit its name.



- A red dot shall appear where the selection of the previous point was made,



- By holding “ctrl” key on keyboard enter 4 points, to define the edge cover identified,

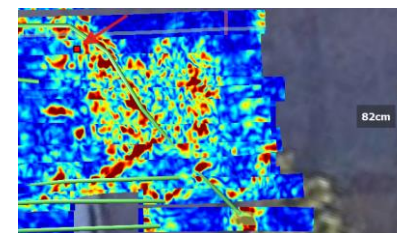
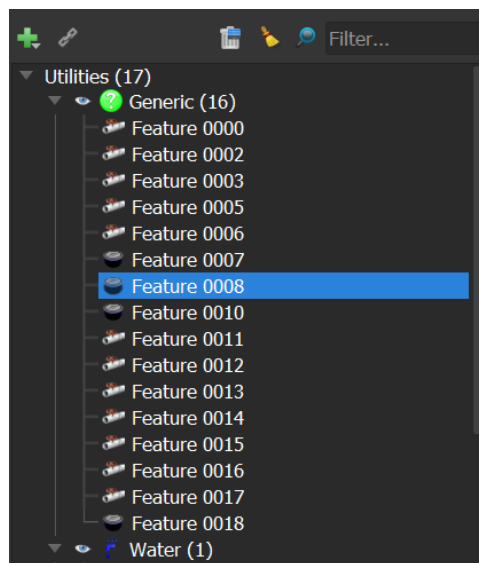


5.5.2 ...enter New Cover

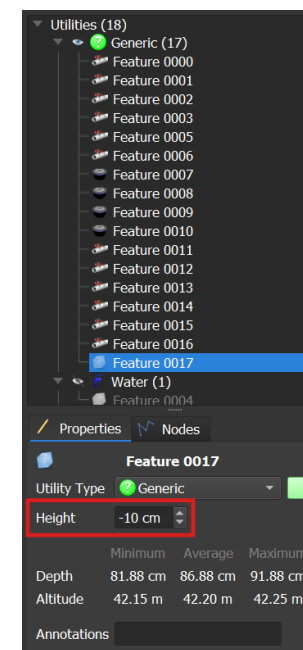
- Right mouse click where the first point of the cover should be entered and select New Cover,

- Each of the “nodes” just entered to define a cover, can be moved by drag-and-drop with left mouse button pressed on it,
- Release “ctrl” key and press anywhere with mouse to close the Cover entering.

- The Cover just entered is now listed on the right-side panel. Double-click on this to edit its name.

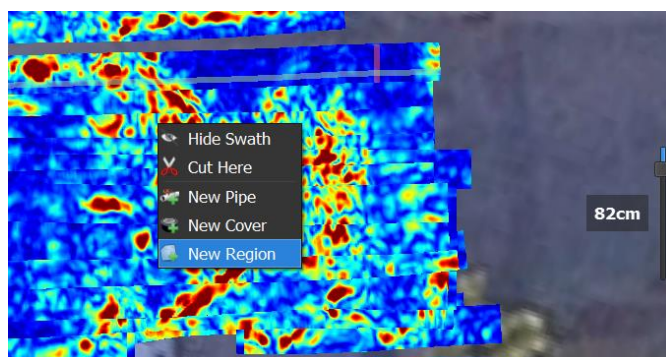


- By holding “ctrl” key on keyboard enter as many points as required, to define edge of the Region identified,
- Each of the “nodes” just entered to define a region, can be moved by drag-and-drop with left mouse button pressed on it,
- Release “ctrl” key and press anywhere with mouse to close the Cover entering.
- The Region just entered is now listed on the right-side panel. Double-click on this to edit its name.



5.5.3 ...enter New Region

- Right mouse click where the first point of the region should be entered and select New Region,

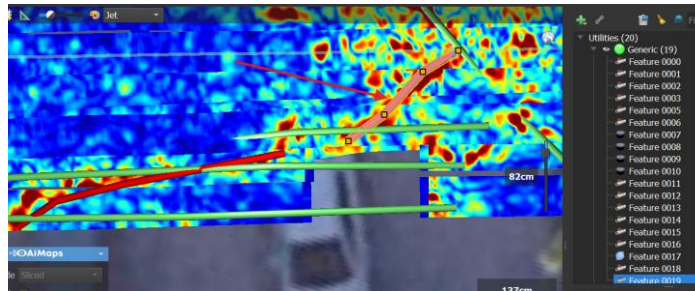


- A red dot shall appear where the selection of the previous point was made,

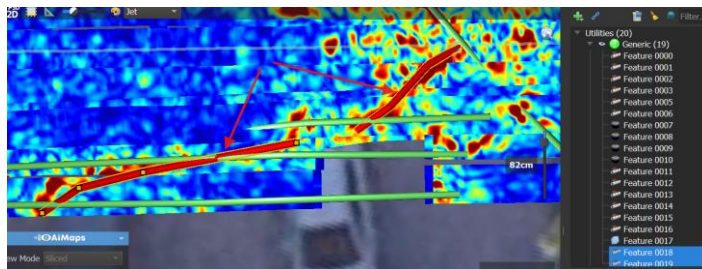
- Edit the thickness of the region from the “Height” tab (red rectangle).

5.5.4 ...merge two pipes

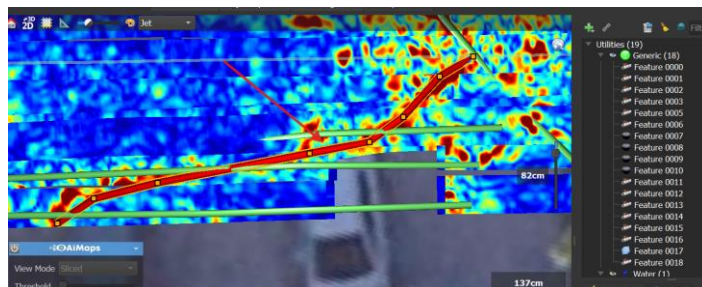
1. Hold “ctrl” key keyboard,
2. Select one pipe by clicking with left mouse button,



3. Select the second pipe to be merged with the first one,

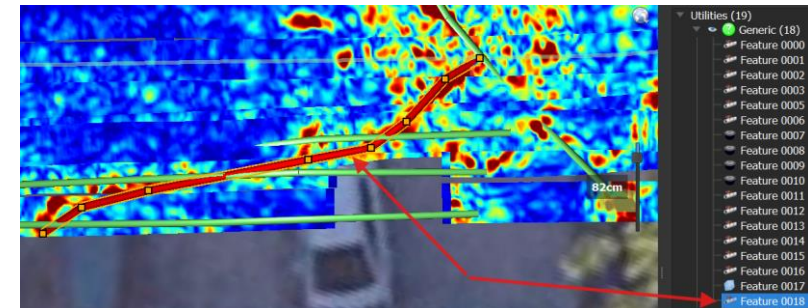


4. Press the Link Pipes icon ,
5. The two selected pipes are now linked in one single pipe.

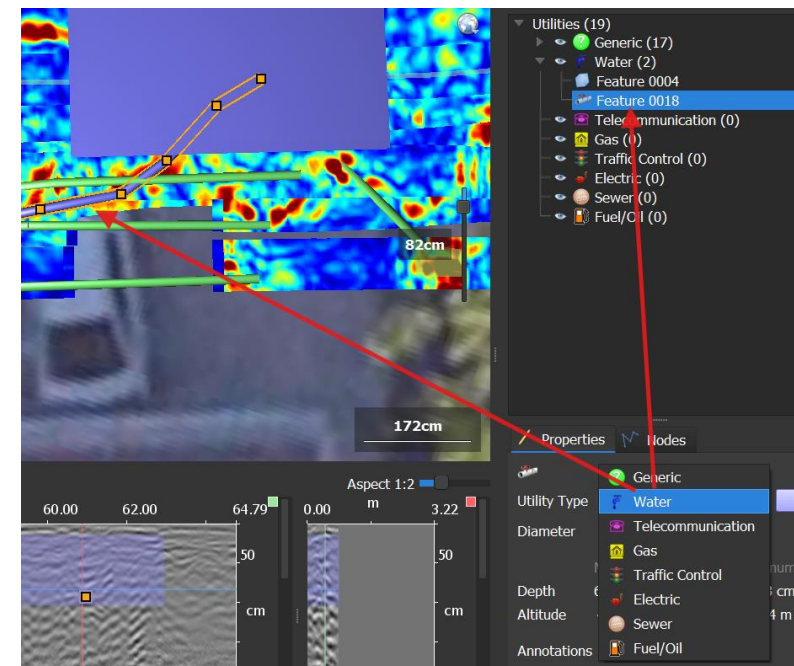


5.5.5 ...categorize features by type

1. Select a feature from the Utilities panel list (e.g.: “Feature 18” in the image below),

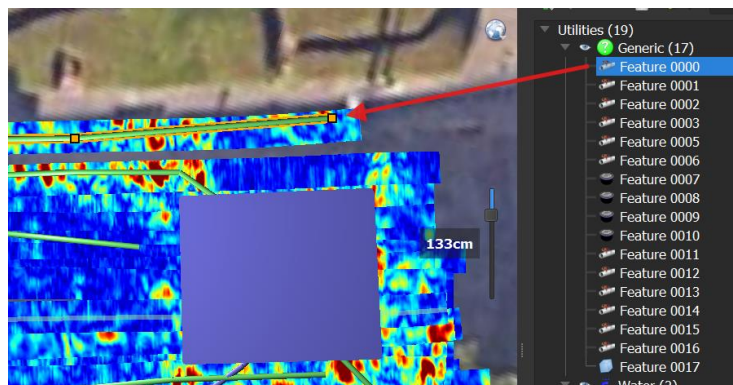


2. Press on the Utility Type drop-down and select the type (Water in the figure below). The feature selected shall automatically be listed in the “Water” group and change color.

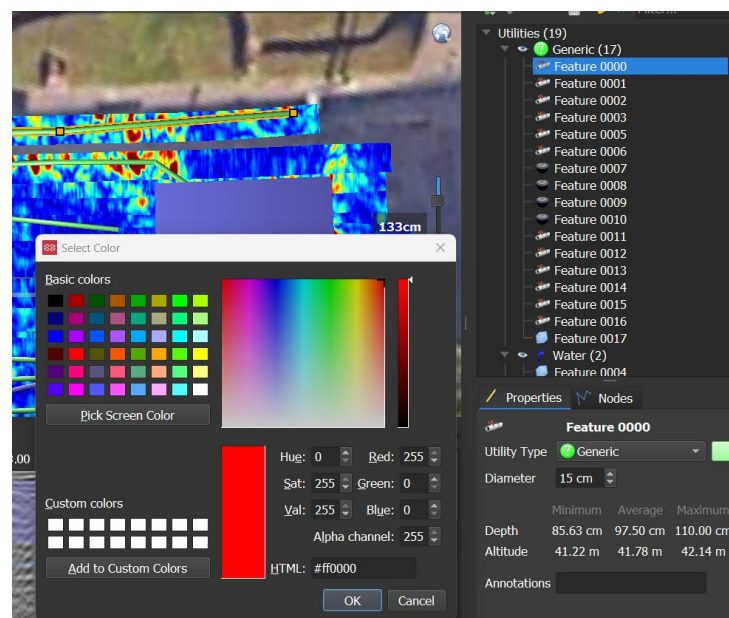


5.5.6 ...change features color

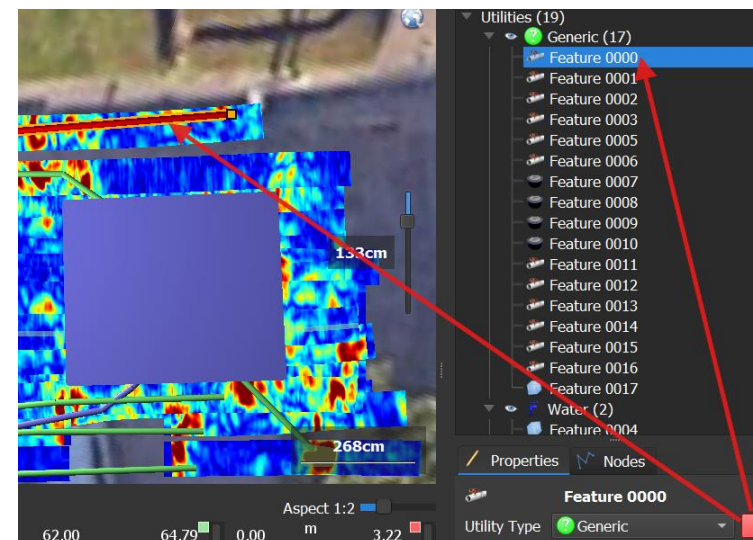
1. Select a feature from the Utilities panel list (e.g.: "Feature 0000" in the image below),



2. Press on the  icon and the Select Color window pops up. Choose a color from the palette and press .

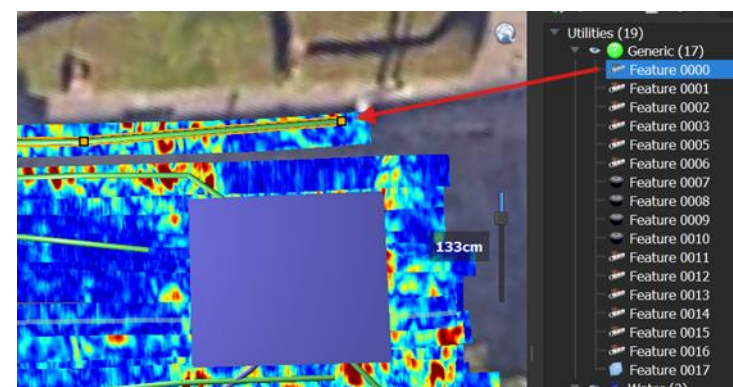


3. The color of the selected feature shall automatically update (from green to red in the image below).

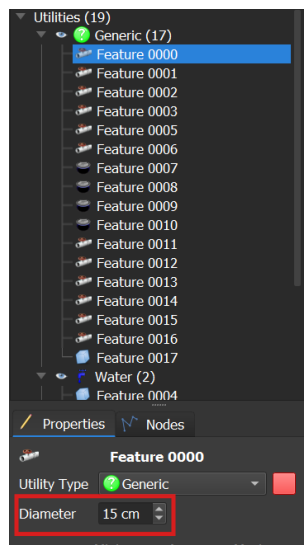


5.5.7 ...change diameter

1. Select a feature from the Utilities panel list (e.g.: "Feature 0000" in the image below),

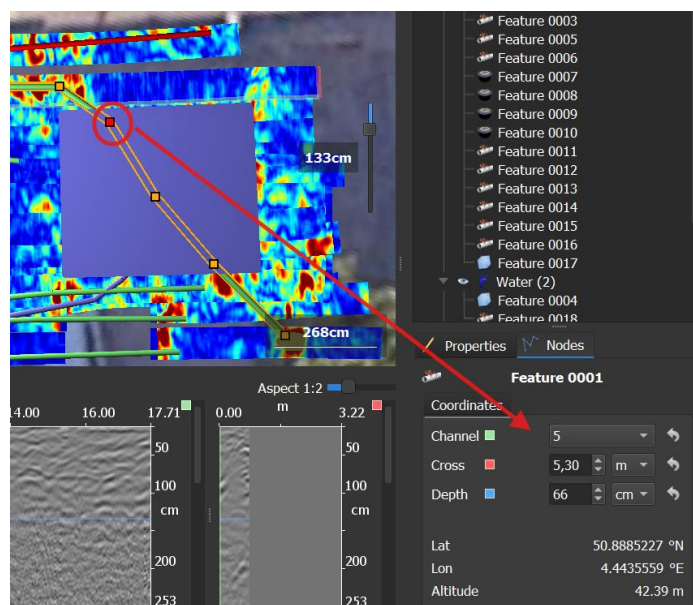


2. Edit the diameter value (15 cm is the default value set) from the Diameter spin-box (red rectangle in the image below),



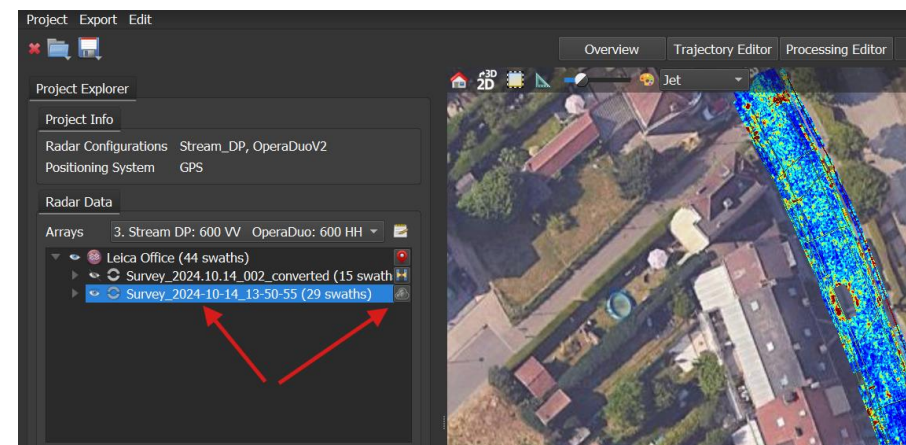
5.5.8 ...visualize features nodes coordinates

1. Select the node of a feature in the Utilities List, and the coordinates details of that Node are visualized in the Nodes tab.

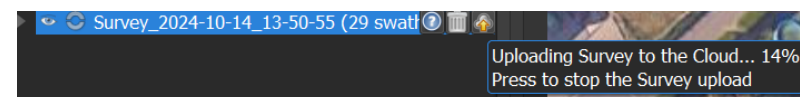


5.5.9 ...launch a AImaps data processing session (optional, for supported IDS GeoRadar system)

1. Verify that a valid AImaps subscription is active on the PC the IQMaps project is running,
2. Verify that data to be processed on the cloud are collected on site with an IDS GeoRadar supported system (that is: Stream UP, Stream DP, Stream C),
3. Verify that the IQMaps project is saved on PC disk,
4. Verify that Project total length is up to 10 Km,
5. Press on the  beside the Survey to be processed located in the Project tree,



6. Press confirm, to start the upload of HxDR cloud,
7. The  shall turn into the  confirming that the Project is uploading on the Cloud,



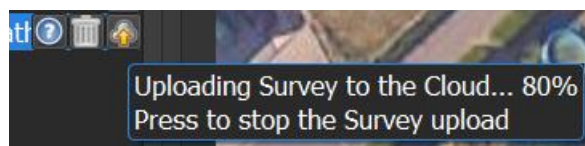


NOTE: For the Project to be correctly uploaded, an active internet connection must be active during the process.

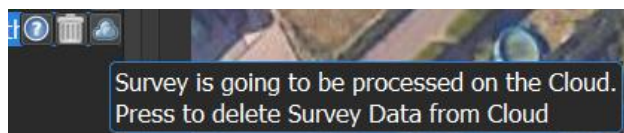


NOTE: Do not turn off the PC during the project upload as the process will fail.

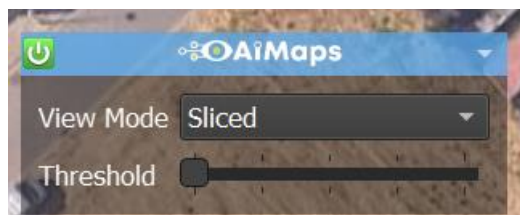
8. The upload can be interrupted by pressing the same icon again,



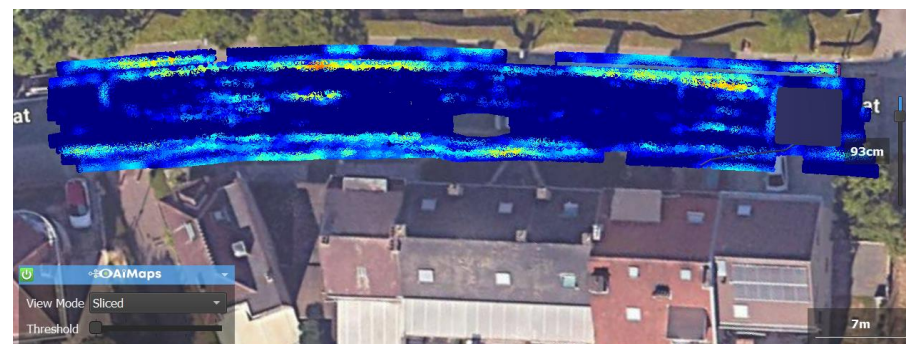
9. As the project is correctly uploaded on cloud the  icon shall turn into  icon indicating that the Processing on cloud is ongoing,



10. As the processing is in progress, the  shall turn in to the .
11. Processing is complete as the  icon appears, and results can then be visualized.
12. The panel below drives adjustments of the AiMaps point cloud,



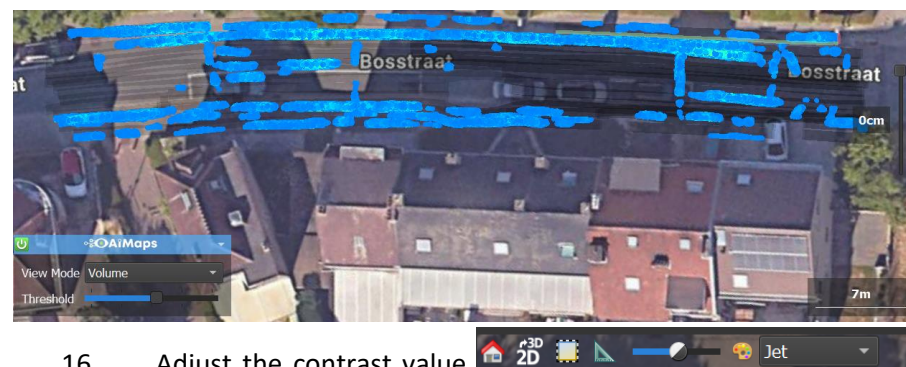
13. Where the **Sliced** "View Mode" allows for the point cloud to be visualized by depth slice (at 93cm in the image below),



14. Adjust the amplitude threshold value using the slider  to cut off amplitude values below the selected threshold,



15. Change the "View Mode" to **Volume** to visualize the entire volume rather than single depth slices,



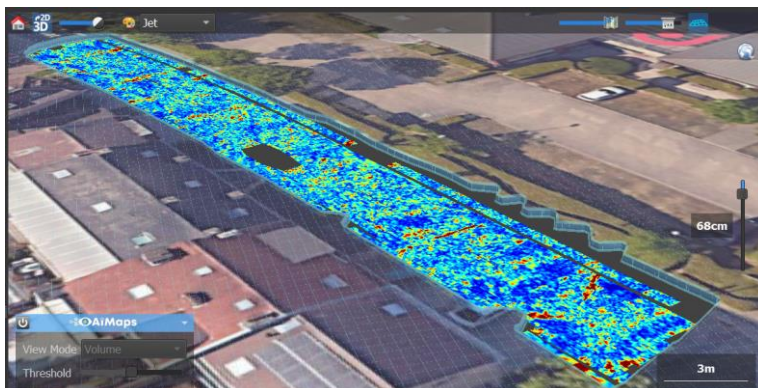
16. Adjust the contrast value  to enhance visibility of anomalies.



17. Toggle the  icon to change the visibility of the area from 2D to 3D.



18. Press the  icon again, to disable the visibility of AIMaps results and view the C Scan tomography view.



5.6 How to export features, tomography, data and images

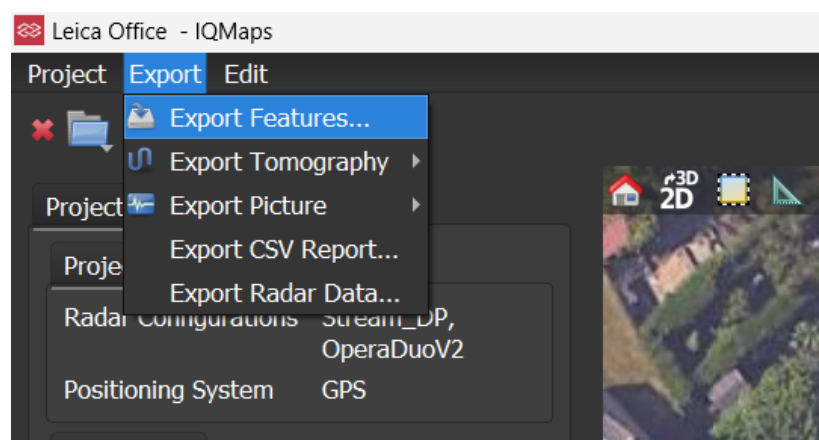
At the end of the workflow in IQMaps the projects, depending on IDS GeoRadar system used and type of processing applied, includes the following:

1. **Channels trajectories**
2. **Targets entered by the User**
 - a. Pipes
 - b. Covers
 - c. Regions
3. **Tomography data/view** (the Depth Slice known as C Scan)
4. **Channels data** (the B Scans)
5. **Cross section data** (the T Scan)

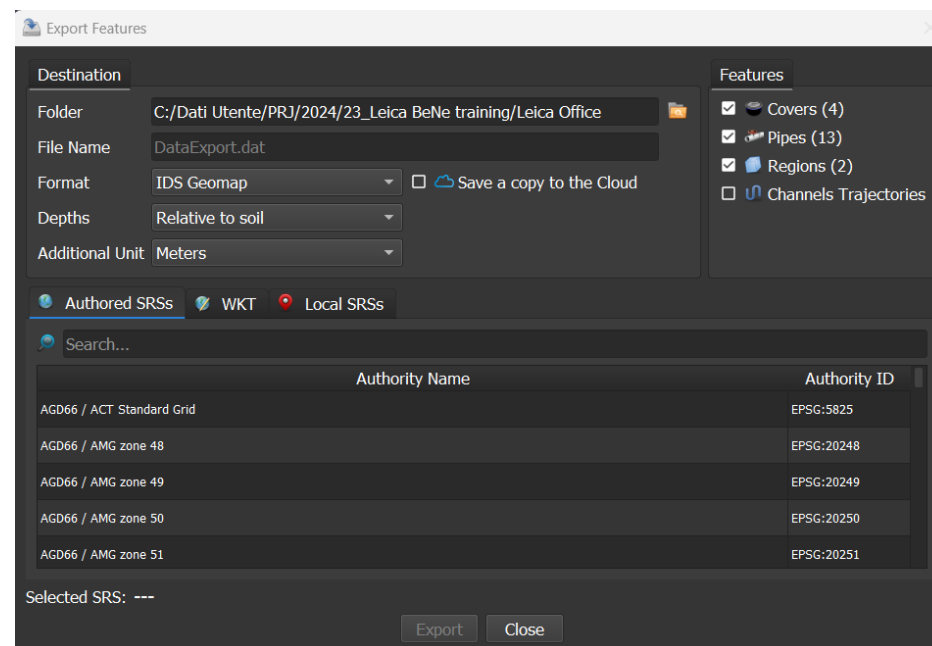
All the above, can be exported into various industry formats and visualized through external software.

5.6.1 How to export Features

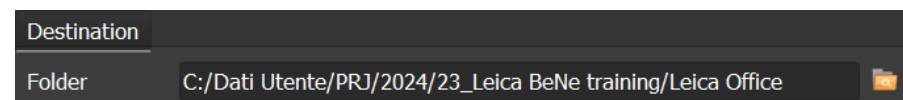
1. Press the **Export** menu at the top left corner, and select **Export Features**,



2. The **Export Features** window pops up,



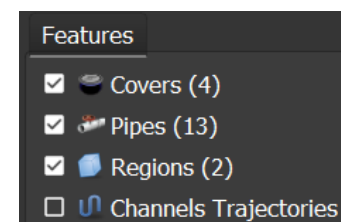
3. Here, defines the folder location where to save the export in the **Folder** box,



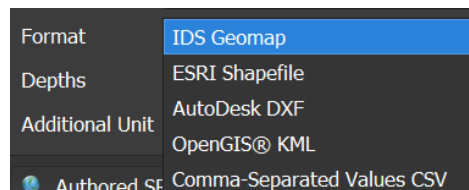
4. Enter the file name of the export in the **File Name** box,



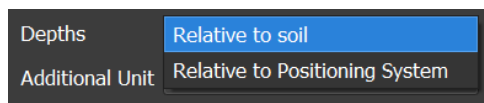
5. Select the features to be saved using the “tick box”, from the **Features** tab,



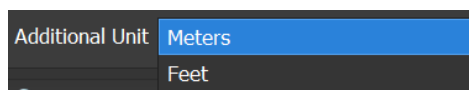
6. Select the format in which to save the features (it can be IDS Geomap, *.shp, *.dxf, *.kml, *.csv), from the **Format** drop-down,



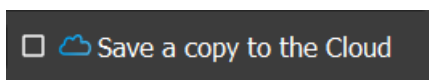
7. Select if features are exported with respect to depth (**Relative to soil**) or in altitude (**Relative to Positioning System**), from the **Depth** drop-down,



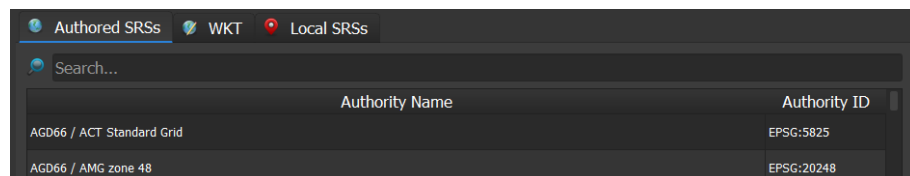
8. Select if the export is in **Metric** or **Imperial**, from the **Additional Unit** drop-down,



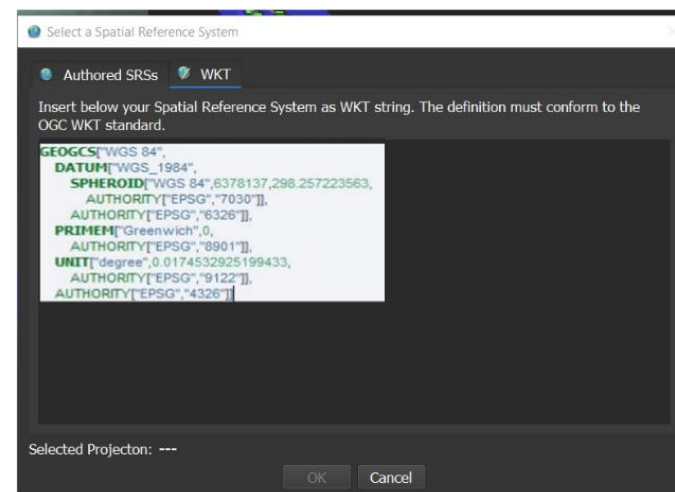
9. Tick if a copy of the features should be saved into the cloud (either, Google Drive, Microsoft OneDrive, HxGN GeoCloud Drive, or Leica ConX Drive),



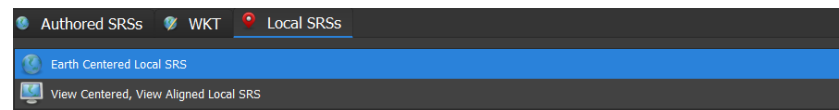
10. Select the correct **Authoried Spatial Reference System** (SRS) from the available library list,



11. Alternatively, enter the customized parameters of the Coordinate System into the **WKT (Well Known Text)** space,



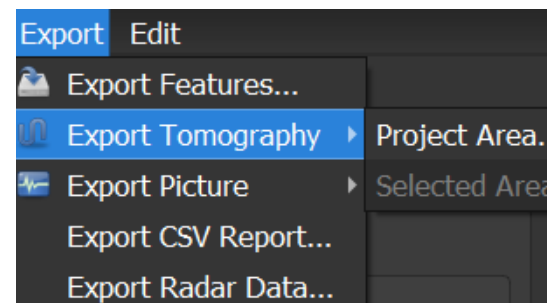
12. Further to the above, it is possible to select **Local SRSs**, when none of the previous two options apply,



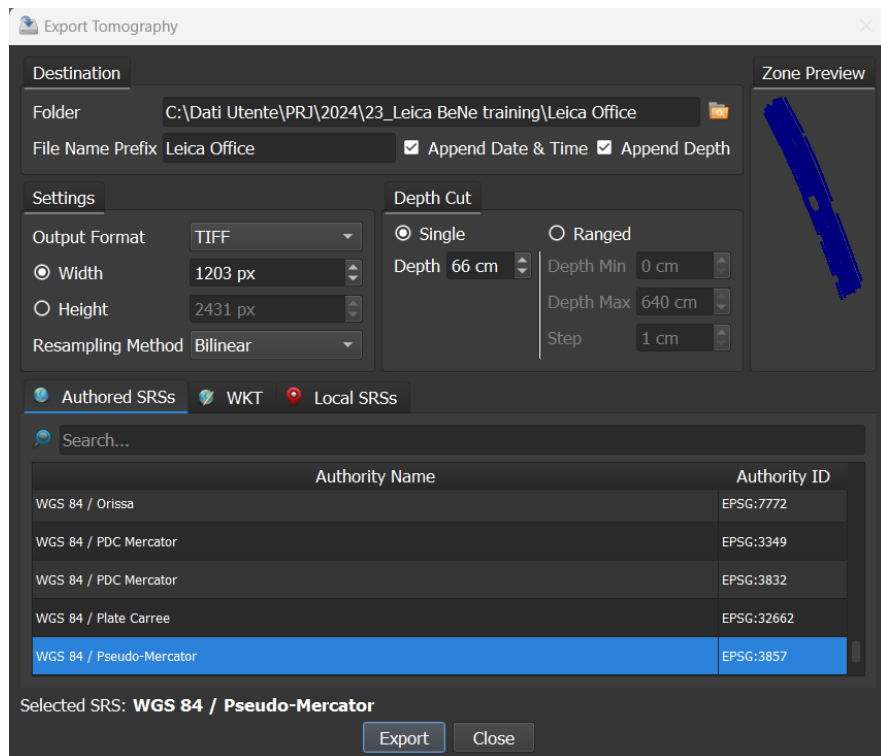
13. Eventually, press **Export** to finalize the procedure.

5.6.2 How to export tomography

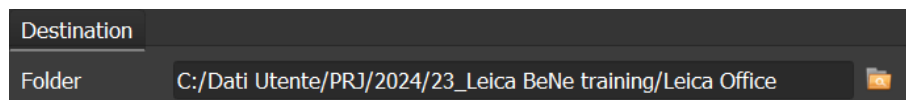
1. Press the **Export** menu at the top left corner, and select **Export Tomography**, (it can be either the whole project area or an area previously selected with the  tool),



2. and the **Export Tomography** Window pops up,



3. Defines the folder location where to save the export in the Folder box,



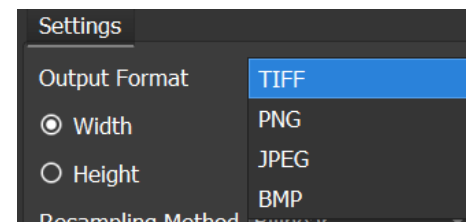
4. Define the File Name Prefix,

File Name Prefix Leica Office

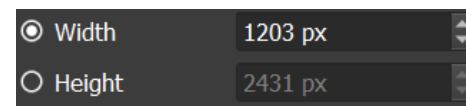
5. Select whether to append Date, Time, and Depth from the specific tick boxes,

☒ Append Date & Time ☒ Append Depth

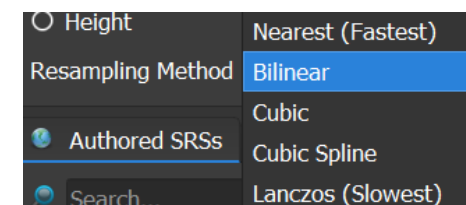
6. Select the format of the tomography export,



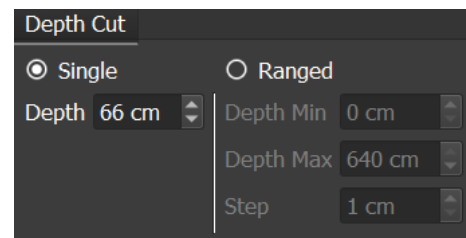
7. Define the dimension of the image,



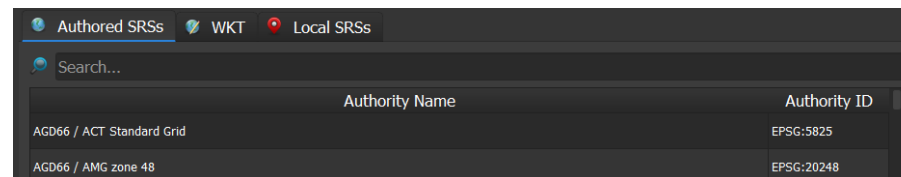
8. Define the method to resample the image of the tomography while exporting,



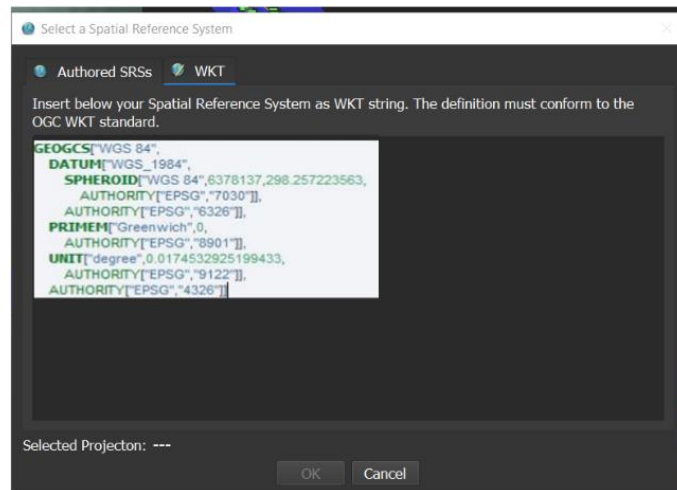
9. Define whether to export a single depth slice (**Single**), or a defined number of tomography depth slices (**Ranged**),



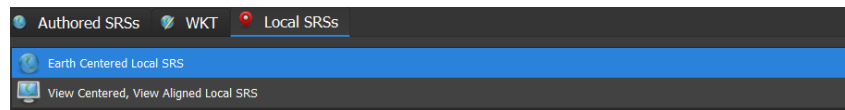
14. Select the correct **Authored Spatial Reference System** (SRS) from the available library list,



15. Alternatively, enter the customized parameters of the Coordinate System into the **WKT (Well Known Text)** space,



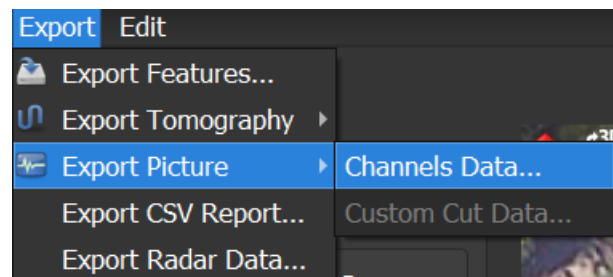
16. Further to the above, it is possible select **Local SRSs**, when none of the previous two options apply,



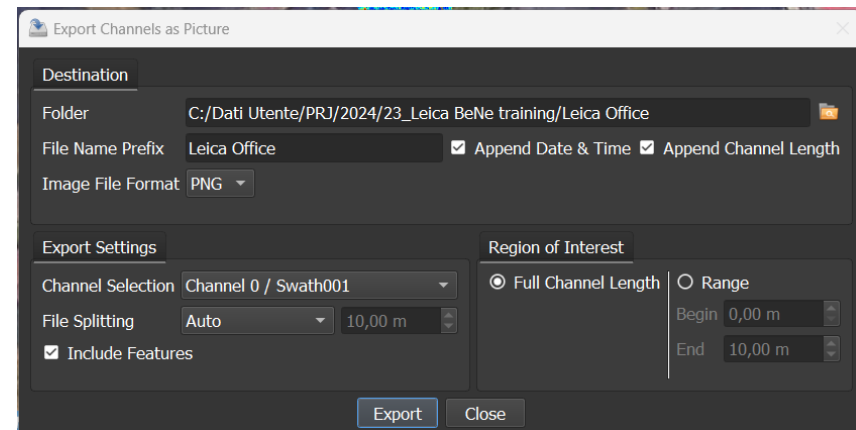
17. Eventually, press **Export** to finalize the procedure.

5.6.3 How to export pictures of B Scan, T Scan and Custom Cut

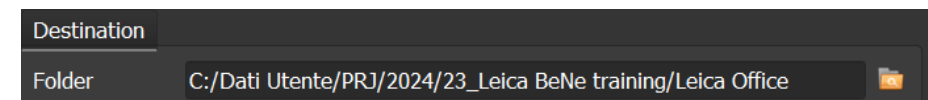
1. Press the **Export** menu at the top left corner, and select **Export picture**, (it can be either **Channels** or **Custom Cut**),



2. and the **Export Channels as Picture** Window pops up,



3. defines the folder location where to save the export in the **Folder** box,



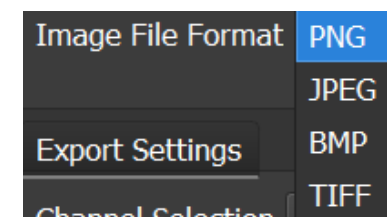
4. define the **File Name Prefix**,

File Name Prefix Leica Office

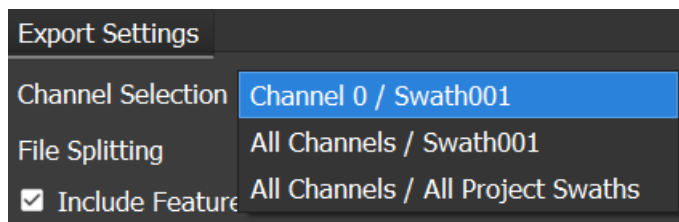
5. select whether to append **Date**, **Time**, and **Channel length** from the specific tick boxes,

☒ Append Date & Time ☒ Append Channel Length

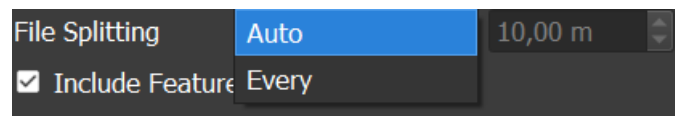
6. Choose the **Image File Format**,



7. Choose the Channel(s) to export,



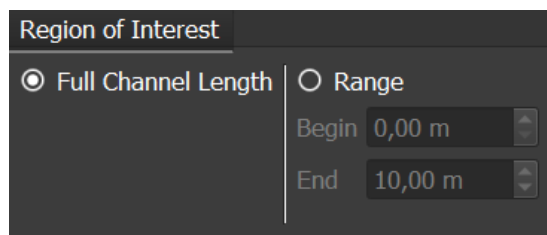
8. Select whether to split the files and the length of each,



9. Choose whether to include the Features entered



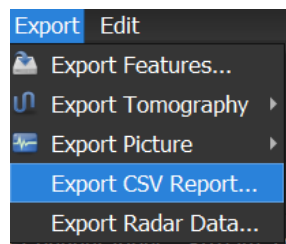
10. Select whether to export the Full Channel Length or only part of it, in a User-defined range



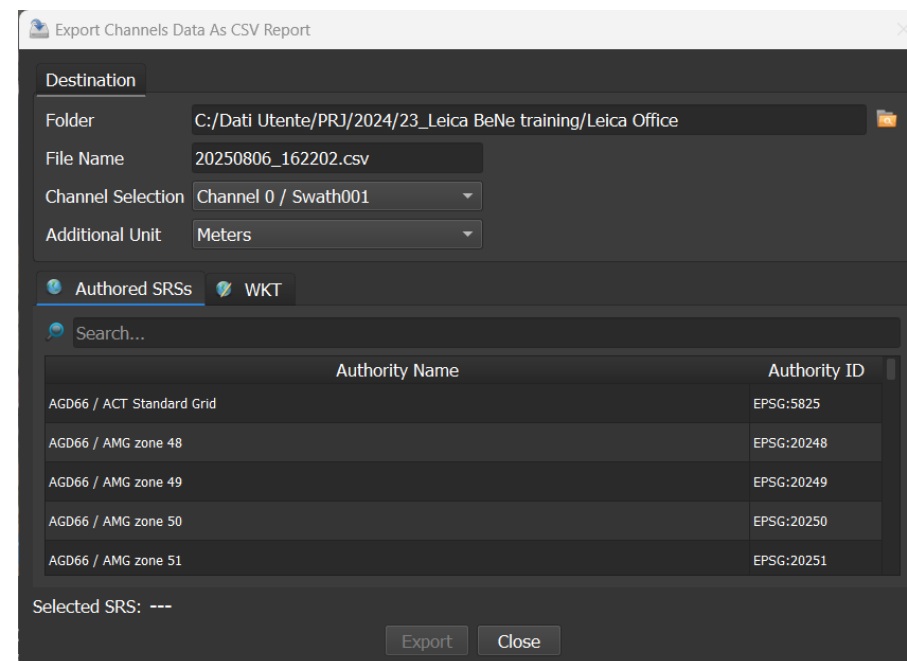
18. Eventually, press **Export** to finalize the procedure.

5.6.4 How to export Features as .csv report

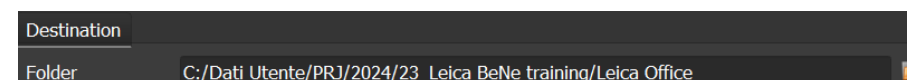
1. Press the **Export** menu at the top left corner, and select **Export CSV Report**,



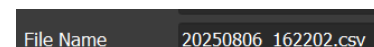
2. and the **Export Channels Data as CSV Report** Window pops up,



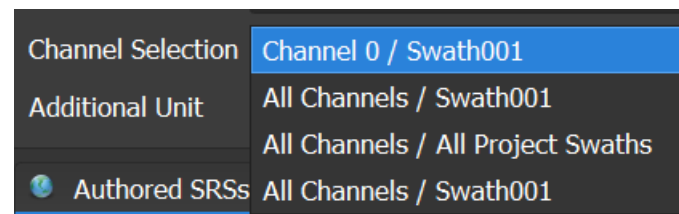
3. define the folder location where to save the export in the **Folder** box,



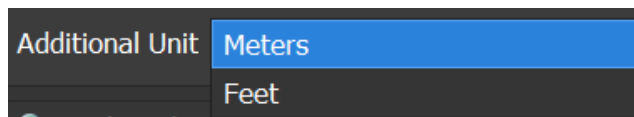
4. define the **File Name**,



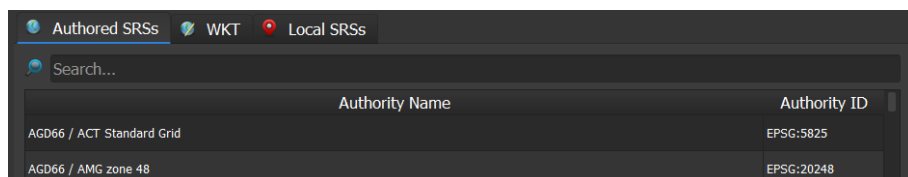
5. Choose the Channel(s) to export,



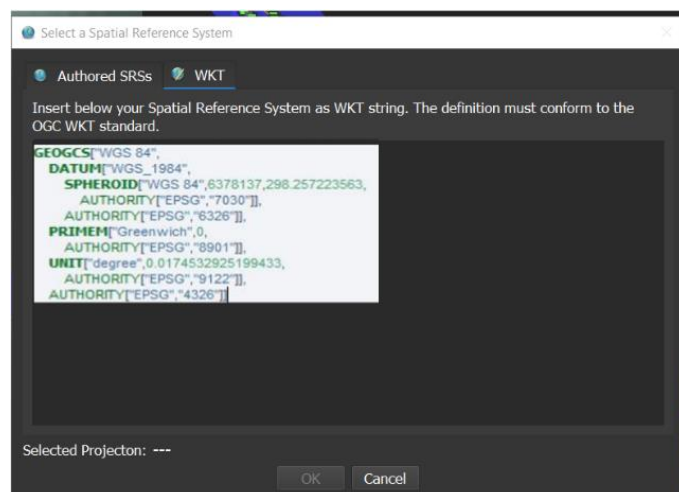
- Select if the export is in **Metric** or **Imperial**, from the **Additional Unit** drop-down,



- Select the correct **Authored Spatial Reference System** (SRS) from the available library list,



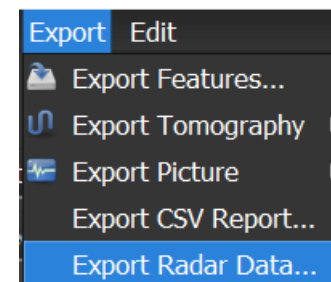
- Alternatively, enter the customized parameters of the Coordinate System into the **WKT (Well Known Text)** space,



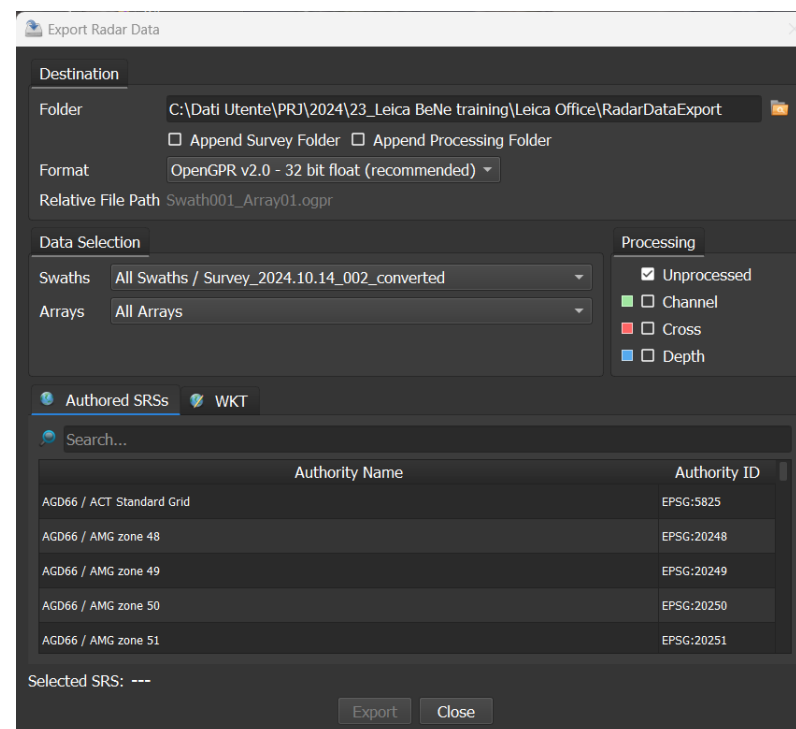
- Eventually, press **Export** to finalize the procedure.

5.6.5 How to export data in **OpenGPR** format and **SEG-Y** format

- Press the **Export** menu at the top left corner, and select **Export Radar Data**,



- The **Export Radar data** Window pops up,



3. define the folder location where to save the export in the **Folder** box,

Destination

Folder C:\Dati Utente\PRJ\2024\23_Leica BeNe training\Leica Office\RadarLayoutExport

4. select whether to append **Survey Folder**, and **Processing Folder** from the specific tick boxes,

☐ Append Survey Folder ☐ Append Processing Folder

5. Select the format of the export

Format OpenGPR v2.0 - 32 bit float (recommended)

Relative File Path OpenGPR v2.0 - 16 bit integer

Data Selection OpenGPR v1.0 - 16 bit integer (legacy)

SEG-Y

6. Select Swaths and Array to export,

Data Selection

Swaths All Swaths / Survey_2024.10.14_002_converted

Arrays All Arrays

7. Select whether to export Raw data, B Scan, T Scan, C Scan, from the Specific tick boxes

Processing

☒ Unprocessed

☐ Channel

☐ Cross

☐ Depth

8. Select the correct **Authored Spatial Reference System** (SRS) from the available library list,

Authored SRSs

Search...

Authority Name	Authority ID
AGD66 / ACT Standard Grid	EPSG:5825
AGD66 / AMG zone 48	EPSG:20248

9. Alternatively, enter the customized parameters of the Coordinate System into the **WKT (Well Known Text)** space,

Select a Spatial Reference System

Authored SRSs WKT

Insert below your Spatial Reference System as WKT string. The definition must conform to the OGC WKT standard.

```
GEOGCS["WGS 84",
  DATUM["WGS_1984",
    SPHEROID["WGS 84",6378137,298.257223563,
      AUTHORITY["EPSG","7030"]],
    PRIMEM["Greenwich",0,
      AUTHORITY["EPSG","8901"]],
    UNIT["degree",0.0174532925199433,
      AUTHORITY["EPSG","9122"]],
    AUTHORITY["EPSG","4326"]]
```

Selected Projection: ---

OK Cancel

10. Eventually, press **Export** to finalize the procedure.

Export

6 IDS GEORADAR SUPPORT

6.1 myWorld

myWorld platform of Leica Geosystems offers a wide range of services, information, and training material. With direct access to *myWorld*, the User can access all relevant services. The availability of services depends on the instrument model.

SERVICE	DESCRIPTION
<i>myProducts</i>	Add all IDS GeoRadar products that you and your company own and explore your world of Leica Geosystems. View detailed information on your products and update your products with the latest software and keep up to date with the latest documentation.
<i>myService</i>	View the current service status and full-service history of your products in IDS GeoRadar service centres including detailed information on the services performed on your system.
<i>mySupport</i>	Create new support requests for your products that will be answered by your local Support Team. View the complete history of your support requests and view detailed information on each request in case you want to refer to previous support requests.
<i>myLearning</i>	Welcome to the home of Leica Geosystems online learning! There are numerous online courses – available to all customers with products that have valid CCPs (Customer Care Packages).
<i>myTrustedServices</i>	Add your subscriptions and manage users for Leica Geosystems Trusted Services, the secure software services, that assist you to optimise your workflow and increase your efficiency.
<i>mySmartNet</i>	HxGN SmartNet is the GNSS correction service built on the world's largest reference station network, enabling GNSS-capable devices to quickly determine precise positions in the range of one-to-two-centimetre accuracy. The service is provided 24/7 by a highly available infrastructure and professional support team with more than 10 years of experience reliably delivering the service.
<i>myDownloads</i>	Downloads of software, manuals, tools, training material and news for IDS GeoRadar products.

APPENDIX A DISCLAIMER

1. General

- i. The present Disclaimer applies to all products designed, produced and distributed by IDS GeoRadar S.r.l., its Subsidiaries, Affiliated and authorized Distributors (the “Products”). IDS GeoRadar S.r.l. reserves full ownership and intellectual property rights of any “Information” contained in this Disclaimer including Trade Marks and Graphics. No part of this Disclaimer may be used or reproduced in any forms without the prior written agreement of IDS GeoRadar S.r.l.
- ii. In the event that any provision of this Disclaimer may be invalid, unlawful or incapable of being enforced by a rule of law, all other provisions shall, nonetheless, remain in full force and effect. Failure to either enforce or exercise any right, privilege, or legal remedy at any time, any provision contained in this Disclaimer, shall not be deemed a waiver of such provisions or right, remedy, or privilege.
- iii. This Disclaimer shall be interpreted, governed, construed and enforced in accordance with the laws of Italy. User/Buyer hereby consents to the exclusive jurisdiction of Pisa.

2. Initial Precautions for Setting-up and Use of the Products.

- i. The User/Buyer, for setting-up and using the Products, shall consult the official documentation provided by IDS GeoRadar S.r.l. for the Products (“Reference Documentation”) and carefully ascertain the compliance with national laws and requirements, which may limit or even forbid their use.
- ii. For Products which shall operate by circulation in Public Areas/Roads, with or without moving traffic, Buyer/User shall verify the approval of local authority and/or site’s owner according to their specific procedures. IDS GeoRadar S.r.l. shall not be liable for any direct, indirect, special, incidental or consequential damages or injuries, including without limitation, lost revenues or lost profits,

resulting by un-authorized use of the Products in Public Areas/Roads.

iii. Buyer/User further warrants:

- that these Products are not being used, in the design, development, production or use of chemical, biological, nuclear ballistic weapons. Buyer/ User will defend, indemnify and hold IDS GeoRadar S.r.l. harmless against any liability (including attorney’s fees) for non-compliance with the terms of this article.
- That, if IDS GeoRadar S.r.l. requires that Buyer/User shall carry out training with reference to some Product categories, no operation or use of the Products shall be started before its designated Operator/s has got the User Certificate, as defined by IDS GeoRadar S.r.l. specific procedure which the Buyer confirms to know and accept.

iv. For Products which include specific “Operational” software with automatic data processing and analysis “Tools”, e.g. the IBIS Products and Hydra Products, User shall be aware that the results provided by these “Tools” may be not error free. User that completely relies on the outcomes provided by these Tools only, does it at his own risk.

v. In no event IDS GeoRadar S.r.l. shall be liable for special, direct, indirect, incidental, exemplary, punitive or consequential damages including, but not limited to, loss of profits or revenue, caused by the use of the Products, either separately or in combination with other products or relied upon the results provided by the above “Tools”.

3. Disclaimer for the “Use” of the Products.

- i. The User shall follow the instructions provided by IDS GeoRadar S.r.l. in its official “Reference Documentation” for the Product, in particular the User’s Technical Manual which contains all the specific steps and recommendations for a correct setting-up and use of the Product.

- ii. In no event IDS GeoRadar S.r.l. shall be liable for special, direct, indirect, incidental, exemplary, punitive or consequential damages including, but not limited to, loss of profits or revenue, caused by the missed or incomplete observance of the instructions and prescriptions for the use of the Products, either separately or in combination with other products, including but not limited to the following main aspects:
 - a. Use of IDS GeoRadar S.r.l. Products outside its limitation of use, without proper and adequate scientific/technical knowledge or without specific training.
 - b. Use of results/outcomes of the measurements performed by the Product aimed to safety aspects without using adequate control procedures and assessment by skilled personnel.
 - c. Unauthorized changes and additions to the Products.
 - d. Use of the Products connected to suspected non-working equipment or with equipment (mainly PC) having characteristics not in compliance with the required specifications of IDS GeoRadar S.r.l. not expressly authorized by IDS GeoRadar S.r.l.;
 - e. Poor or faulty operation of the electrical and telecommunication networks not directly managed by IDS GeoRadar S.r.l. or its delegates.
 - f. Poor or faulty operation Software/Hardware of the third parties connected with IDS GeoRadar S.r.l. Equipment.
 - g. Poor or faulty operation of the Products due to Software Virus which infected the Products after their delivery.

APPENDIX B TRAJECTORY EDITOR: INSIGHTS

B.1 How to...edit GNSS trajectories by thresholds

To edit the swaths trajectories of a Survey, the User must click on **Trajectory Editor** to switch into the **trajectory editor** mode view. In the right panel (Fig. 65) User can edit:

1. Position Filters,
2. Trajectory Interpolation,
3. Radar Data Distribution,
4. Import external trajectories,

The trajectories adjustment made by the User, are displayed in real time on tomography view by clicking on the  icon locate in the upper side of project window (Fig. 66).

When uploading a project in IQMaps, the software automatically starts the default **Geographical reference module** by calculating the relative position of each scan, based on the information found in the files associated with the GPS.

○ GPS tab

In this tab it is possible to filter the positioning data by: Activating/deactivating the **RTK Position Only** option. This option allows to visualize (or not) only the positioning data Real Time Kinematic corrected.

Also, it is possible to discard those GPS points gathered when the number of satellites is below a given threshold, by selecting the

Min. satellites option. In this option box, it is possible to define the number of satellites that represents the threshold of acceptance for that survey.

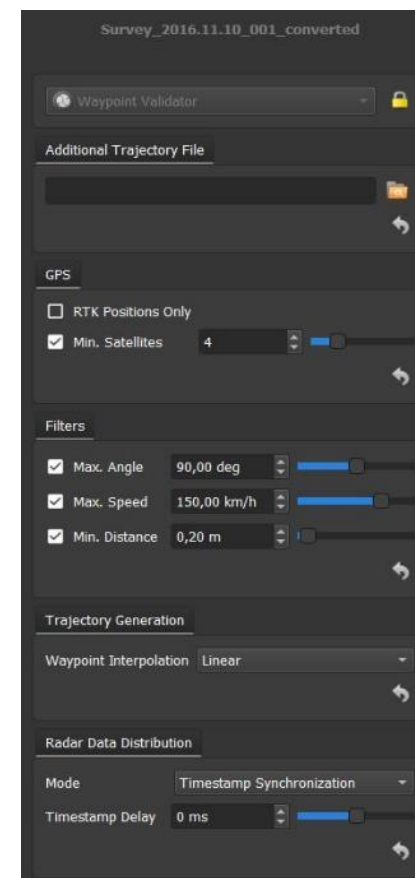


Fig. 65 – GPS Trajectories Editor Window



Fig. 66 – Trajectory preview command

- **Filters tab:** In this tab it is possible to filter the positioning data by:
- Setting the maximum angle of interpolation between each GPS point (**Max. Angle**). In case of loss of GPS coverage, the right set of this parameter avoids drift on the swath trajectories.
- Setting the **Max. Speed** drift of the GPS points; (it is advisable to set the maximum acquisition speed of the GPR system). In case of loss of GPS coverage, the right set of this parameter will avoid excessive GPS points drift.
- Selecting the **Min. Position Distance** between each GPS point, the software automatically excludes the trajectories interpolation of the positioning data points whose relative distance is smaller than the one set.

B.2 How to...edit the trajectory generation

Trajectory Generation tab: Georadar data is associated to a specific position, by performing these steps:

1. computation of the entire length of the curve by interpolating the valid positioning points.
2. computation of differential along the interpolating curve as the ratio between the total length and the number of tracks placed.
3. Positioning the single track by calculating its position along the x-axis arc length to the coordinate equal to $n \cdot dl$ where n is the index of track.

After the above, the module allows the User to interpolate the trajectory of the positioning data. The available interpolation methods are:

- **Linear interpolation:** allows for the interpolation calculation along straight paths.
- **Spline interpolation:** is the most "smoothing" of the interpolation methods (continuity of the first and second derivative in the junction points) and allows to better follow the

curves. Sometimes, the result may be oscillating in correspondence of abrupt transitions.

- **Mixed interpolation:** it follows the points "faster" and does not generate oscillations. It guarantees the continuity of the first derivative alone and it is less smooth. Essentially, it applies spline interpolation to the intermediate points and linear interpolation at the beginning and end of the trajectory.

If there are Georadar tracks acquired before the first measurement point position and after the last point, the software module performs an extrapolation placing these radar tracks on the path tangent to the first and the last point measure.

B.3 How to...edit radar data distribution

Radar Data Distribution tab: it is possible to change the way to associate the position data with the GPR data, according to the three possible approaches:

1. **Timestamp:** the position data is associated to the radar data according to a temporal proximity criterion. The GPR timestamp can be UTC (Universal Time Coordinate) (in the case of synchronization via PPS of GPS); or the timestamp GPR is the local clock of the Georadar DAD, in which case it proceeds by making an association between the two-time scales associating the first radar track on the first position data is received.
2. **Trajectories interpolation:** the radar data is associated to a specific position, by performing the following steps:
 1. calculation of the entire length of the curve interpolating the valid points.
 2. calculation of the differential along the interpolating curve as the ratio between the total length and the number of tracks placed.

- Position the single track calculating its position along the x-axis arc length to the coordinate equal to $n * dl$ where n is the index of track.

NOTE: **Trajectory Interpolation** mode is used when the timestamp is not considered reliable and cannot estimate a latency constant value that can correct the temporal association. This mode is considered reliable if the odometer is well calibrated and if the points of the beginning and the end of scan are successfully collected.

- Trajectories Extrapolation:** if there are gaps between the encoder wheel measurement and GPS measurement, it can be useful to select the trajectories extrapolation. Note that, if the final points of the GPS are not displayed, the software will draw a straight line (in this case, this function should be used only if the final path is effectively straight).

B.4 How to...add an additional Trajectory file

In the **Additional Trajectory File** tab, it is possible import a new trajectory file (*.nmea or *.txt) for the selected survey by clicking on  icon (red square Fig. 67). Automatically a window will appear allowing to import the required trajectory file. Anytime the User may go back to original trajectory file by selecting  icon (blue square Fig. 67); by pressing again the  icon, will be uploaded the imported trajectory file to current survey.

On **Radar Data Tab** will be reported which Survey adopt an external trajectory file (red square Fig. 68).

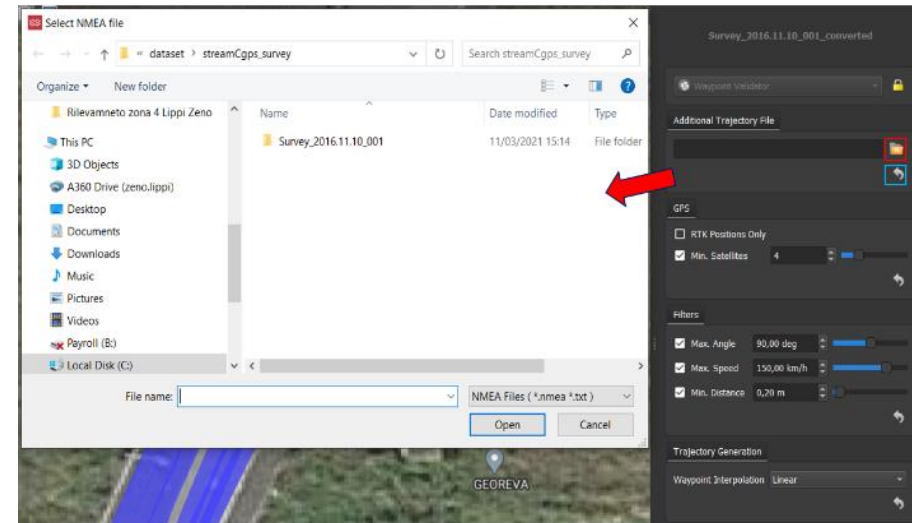


Fig. 67 – Importing an external GPS Trajectories



NOTE: Trajectory files must respect GGA format to be imported.

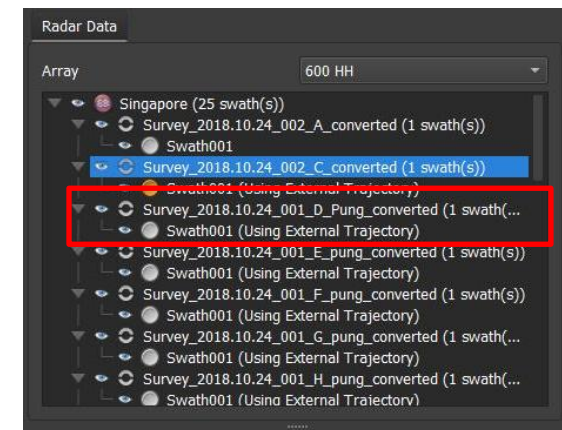


Fig. 68 – External Trajectories feedback

B.5 How to...edit TPS trajectories by thresholds

In those cases when Georadar data are collected by using a Total Station (robotic) as a positioning system, the User can perform the editing of trajectories, by selecting the **Trajectory Editor** tab.

In the right side of the main window a settings area will appear (Fig. 69).

In this area User can modify:

1. Total Station
 2. Filters (same as par. B.1)
 3. Trajectory (same as par. B.2)
 4. Radar Data Distribution (same as par. B.3)
- **Total Station:** In this tab it is possible to modify a few parameters related to TPS data, by doing the following:
 - Enter the **Origin** coordinate of TPS in order to geo-reference the acquired data.
 - Enter the **Altitude** of the TPS base.
 - Modify the Swath **Orientation** depending on the heading of the TPS base with respect to the rover (the optical prism).

If Georadar data were acquired with a **Geo-referenced TPS**, the User needs to choose the geo-reference module on **Spatial Reference System** window (Fig. 70) displayed by selecting **Spatial Reference System** option from

Georeferencing Mode tab and clicking on **No SRS Selected** icon (Fig. 71). The User can also decide to apply a customized projection by copying in the blank space of the **WKT** tab of the **Spatial Reference**

System window, the required projections (in WKT format) as displayed in Fig. 72.



NOTE: The User may import positioning data acquired with Metric or Imperial reference systems without having to perform any conversion.

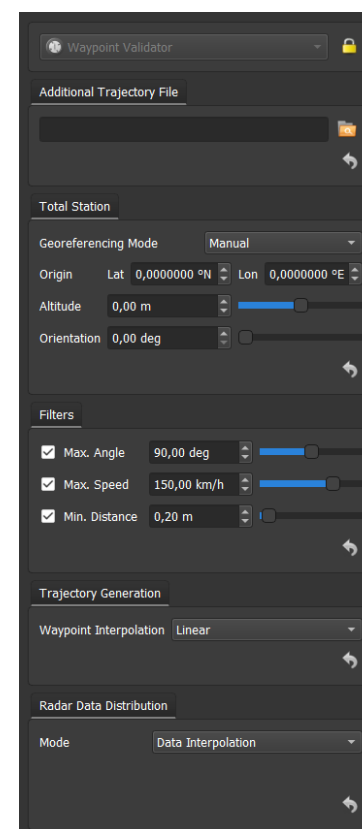


Fig. 69 – TPS Trajectories Editor Window

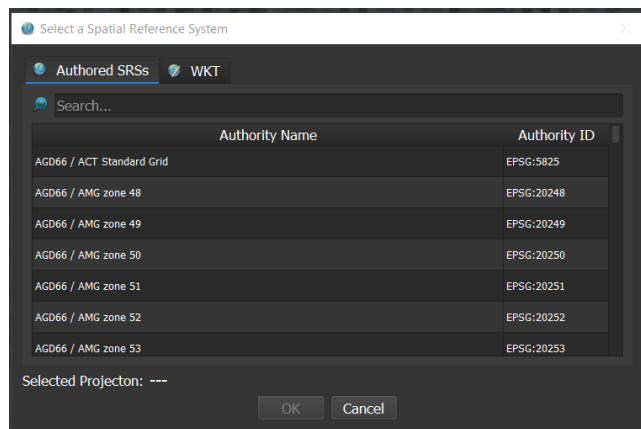


Fig. 70 – TPS Spatial Reference System Window

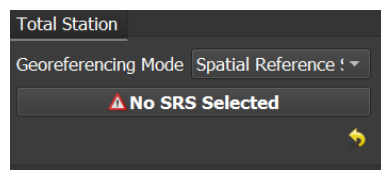


Fig. 71 –Spatial Reference System tab

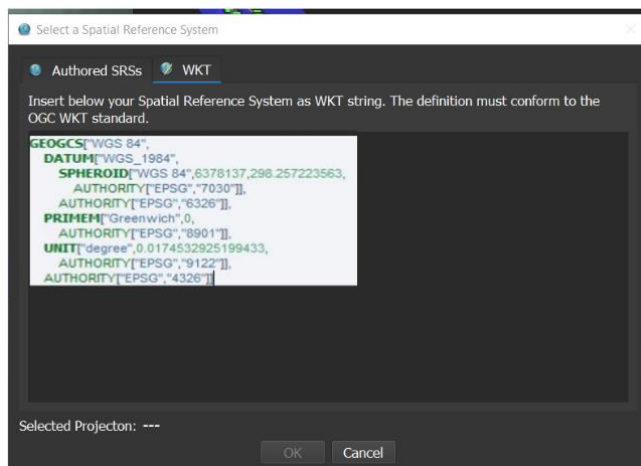


Fig. 72–Custom projection tab

B.6 How to...edit GRID trajectories by thresholds

In those cases when the Georadar data are collected by using a georeferenced GRID as positioning system, User can perform the trajectories

editing by selecting the **Trajectory Editor** tab.

In the right side of the main window a settings tab will appear (Fig. 73) allowing to:

1. Manually modify the swaths waypoints through **Waypoint validator** tab,
2. Modify Swaths trajectories through **Trajectory Generation** tab,

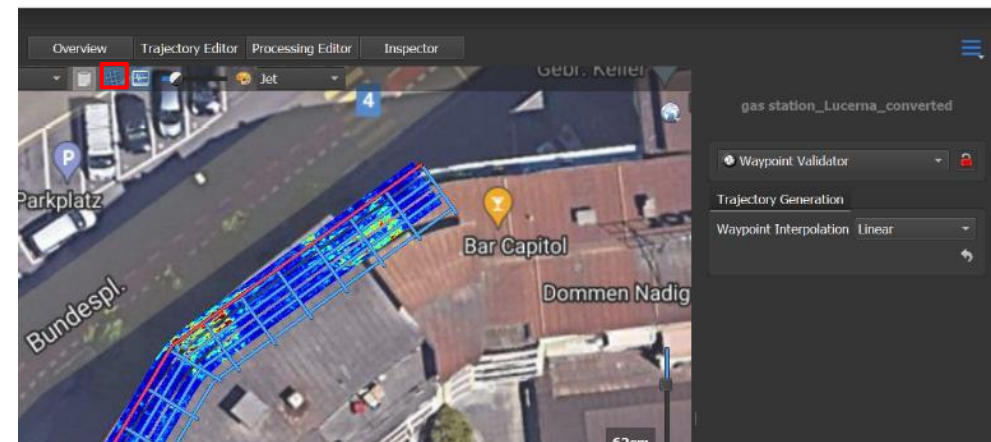


Fig. 73 – GRID Trajectories Editor Window

Additionally, it is possible turn on/off the GRID view by clicking on button (red square Fig. 73).



B.7 How to...manually position not georeferenced GRID acquisition

The User may decide to locate the acquired GRID, directly on IQMaps by selecting the **Trajectory Editor** tab. In the right side of the main window a settings tab will appear (Fig. 74) allowing the User to:

- Enter the **Origin Coordinate** of acquired GRID to geo-reference the acquired data.
- Enter the **Altitude** of the GRID origin point.
- Modify the GRID **Orientation** basing to Y axis direction (Fig. 74).

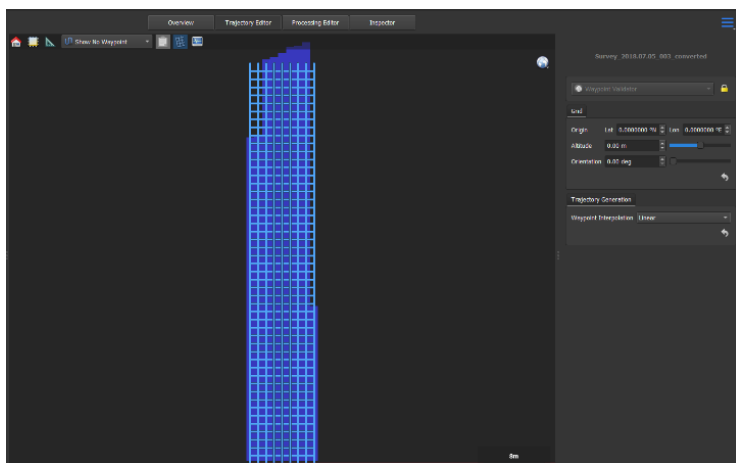


Fig. 74 – GRID repositing

B.8 How to...manually edit GNSS trajectories

The user may decide to visualize and modify each individual waypoint that created the swath trajectory. In trajectory editor, both in thresholds and manual editing, it is possible to visualize the waypoints.

It is possible to enable the waypoint visualization for the current selected swath or for the selected survey or for whole within the project by using the Waypoints drop down menu (Fig. 75).

It is moreover possible to enable the waypoints legend by clicking on  command (Fig. 75). The waypoints Legend shows the discarded Markers, and each colour is associated to the filter which removed the specific marker (Fig. 76).

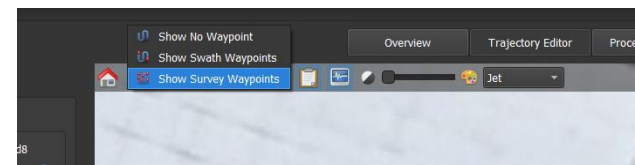


Fig. 75–Waypoint drop down menu

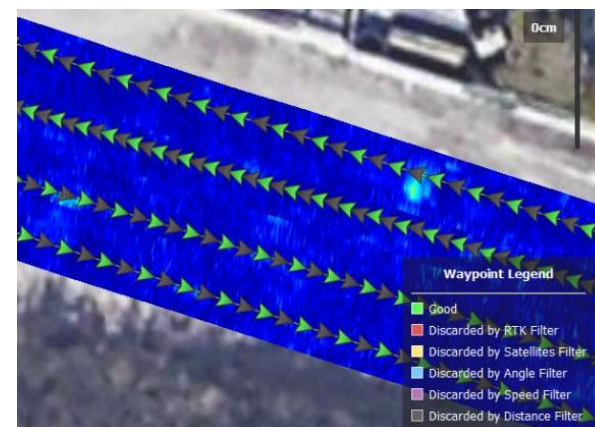


Fig. 76 –Waypoint Legend



NOTE: The entered coordinates must be reported according to WGS84 reference system.



NOTICE: Switching by the Waypoint Editor to the Waypoint Validator, the User loses all the changing made in waypoint editor.

To switch between the **Waypoint Validator** and **Waypoint Editor** shall be clicked the lock button  and selected  **Waypoint Editor** command from the Trajectory Editor window (Fig. 77).

When the User switch to the manual editor, the waypoints position is defined by the threshold settings. To lock the switch between the two trajectory editing modes the lock button  need to be pressed again.

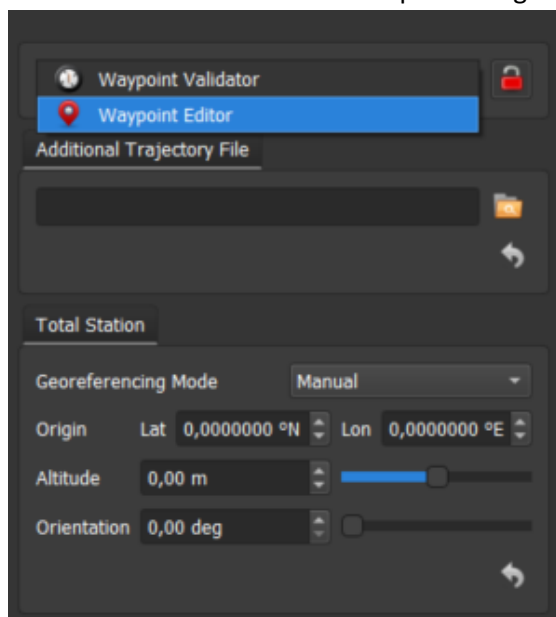


Fig. 77 –Waypoint Editor tab

It is possible to select one or more waypoints. Multiple waypoints can be selected by exploiting the following commands:

- **CTRL+ Left mouse click** → In this way the User may select any waypoint visualized.

- **Shift+ Left mouse click** → In this way the User may select two waypoints and those between them.

On the right side of the software main windows are reported value of the Lat, Lon and Altitude related to last waypoint selected (Fig. 78).

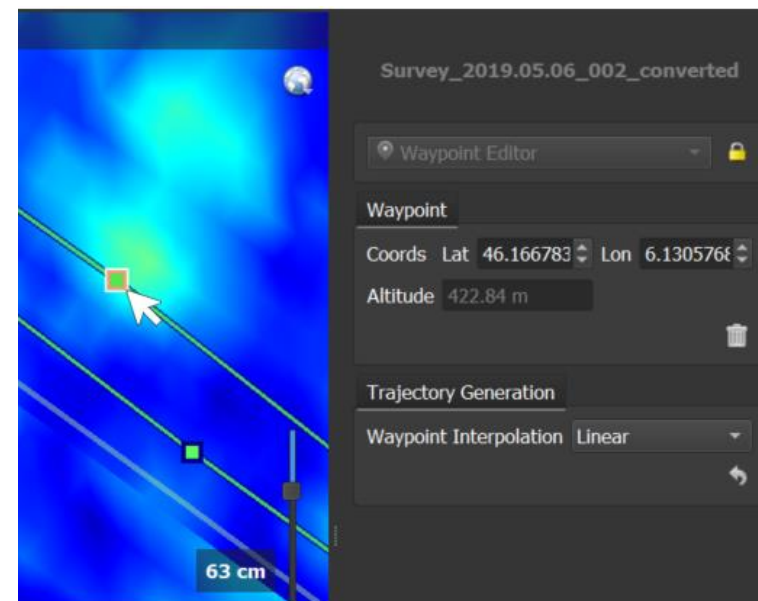
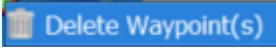


Fig. 78 –Waypoints Information window

For the selected waypoint is possible to:

- **Delete a waypoint** by right clicking on it and click  **Delete Waypoint(s)** command. Otherwise, it is possible to click on the delete button  shown on the trajectory window (Fig. 79).
- **Move a waypoint** by dragging it directly on the map or C Scan, otherwise by inserting a new value of latitude and longitude on the waypoint editor windows dedicated field (Fig. 78).

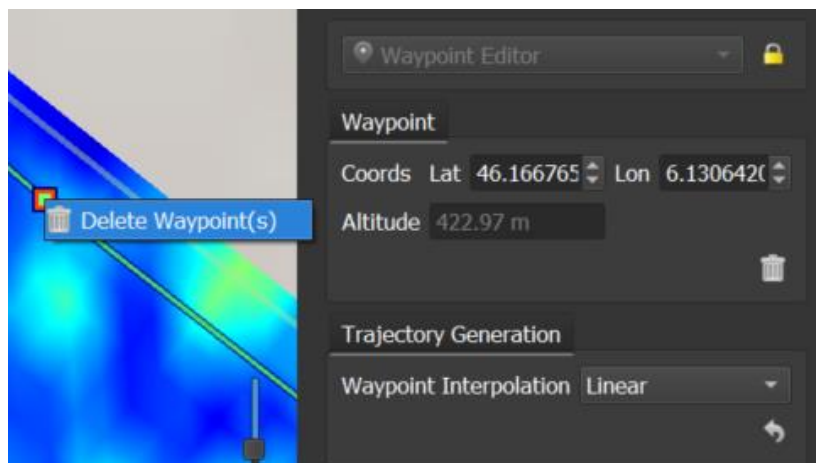


Fig. 79 – Delete Waypoint commands

- **Modify the Waypoints interpolation mode** from the Trajectory Generation Tab (Fig. 79) by choosing among options **Linear**, **Spline** and **Mixed**.
- **Add a new Waypoints** by holding CTRL button and clicking on the connection line between two consecutives. Moreover, by keeping pressed CTRL and moving the mouse pointer on trajectory line, a preview of the waypoint is shown.

B.9 How to...manually import APS trajectories (Stream UP only)

If the **Acquisition Project** file (**file.uproj**) importing is not performed, the User may manually associate the APS trajectory file to the relative Survey, by following next steps:

1. Import the required Survey files,
2. Select the **Trajectory Source Manager** button  (Fig. 80);

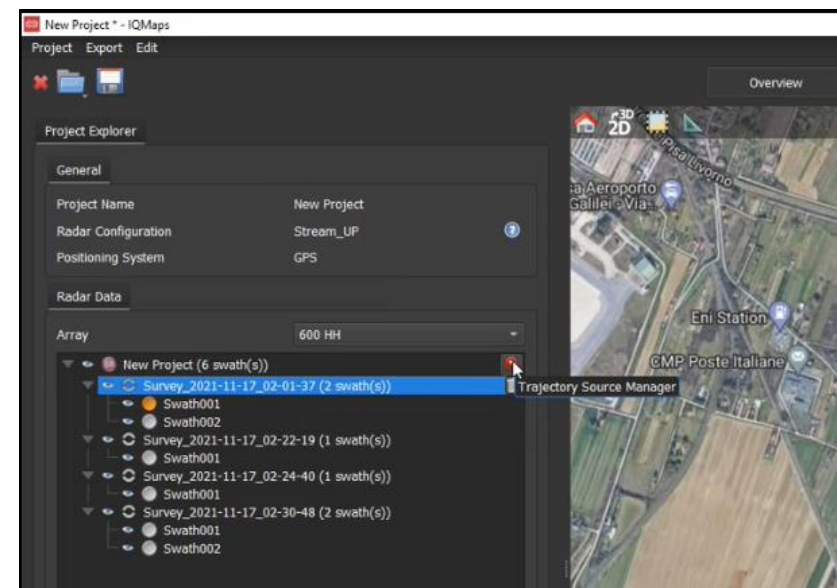


Fig. 80 – Trajectory Source Manager selection

3. Click on **Track Name** Tab associate to the APS Survey and select **<Open File...>** option from drop down menu (Fig. 81);
4. Browse from the saving folder, the **TrjData.gps** file associated to selected APS Survey and press **Open** (Fig. 82);



NOTE: Only Project surveys acquired with APS, require TrjData.gps importing from Trajectory Source Manager window.

5. Each APS Survey is associated with a **TrjData** folder which contains a **TrjData.gps** file (Fig. 83).

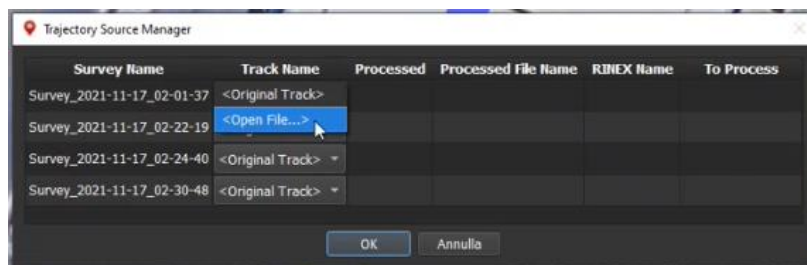


Fig. 81 – Tracks selection

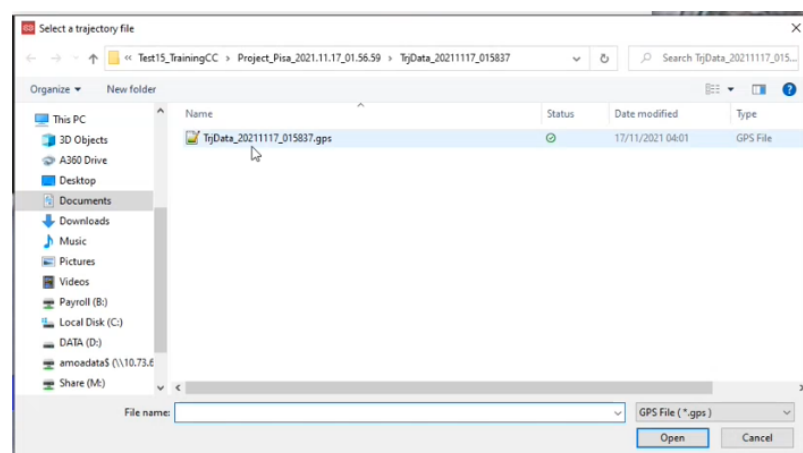


Fig. 82 – TrjData import

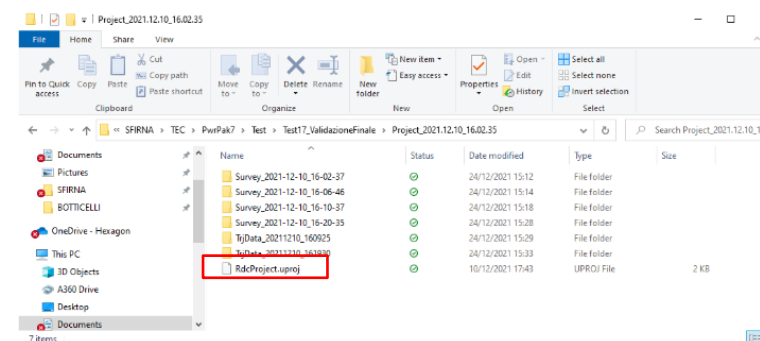


Fig. 83 – TrjData folders structure

To associate the right **TrjData.gps** file to each Survey it is advisable to check the **project file** (red square Fig. 83) by opening it with Notepad application. The **project file** report, for each Survey (blue square Fig. 84), the associated **TrjData** (green square Fig. 84) allowing to select the right one during importing phase.

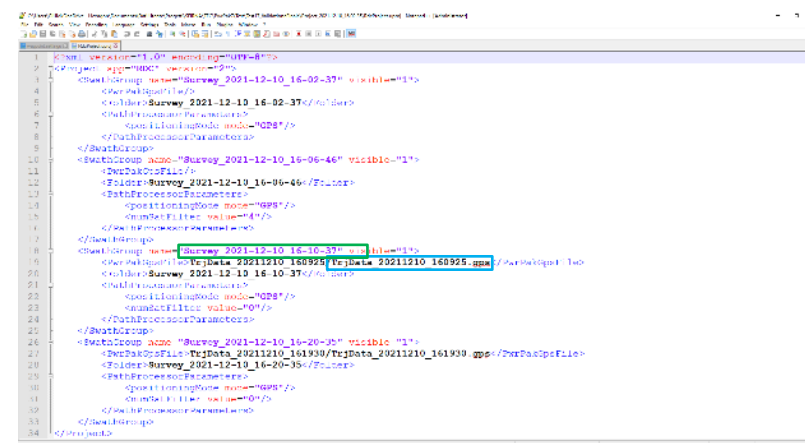


Fig. 84 – TrjData folders structure

B.10 How to...process APS trajectories (Stream UP only)

The **TrjData.gps** file is the trajectory output acquired in Realtime by Powerpack (GPS+INS) which can be corrected through radio base stations information. This information is reported on **correction files (RINEX format)** that can be downloaded from different services platforms of base radio stations.

The **correction file** must be referred to the same time interval when Surveys were acquired on field. If Project Surveys has been acquired in different days, the User must download a correction file for each acquisition day.

To apply the **correction file** to a Survey trajectory, the User should follow the steps below:

1. Click on "**select file**" option (green square Fig. 85) from **RINEX Name** tab (red square Fig. 85),

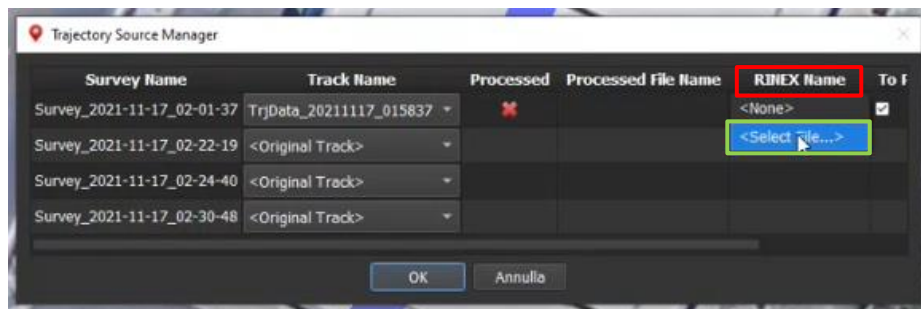
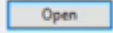


Fig. 85 – Correction files importing

2. Select the **correction file** from the current saving path and press  (Fig. 86),

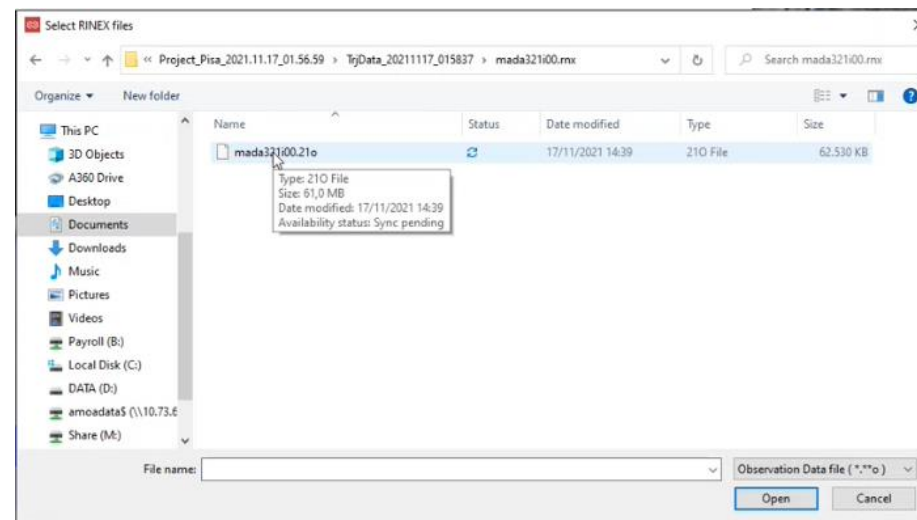
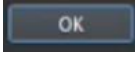


Fig. 86 – Correction file selection

3. Press  (red square Fig. 87) to start the trajectories correction process; a loading bar (Fig. 88) will update the User about processing status;

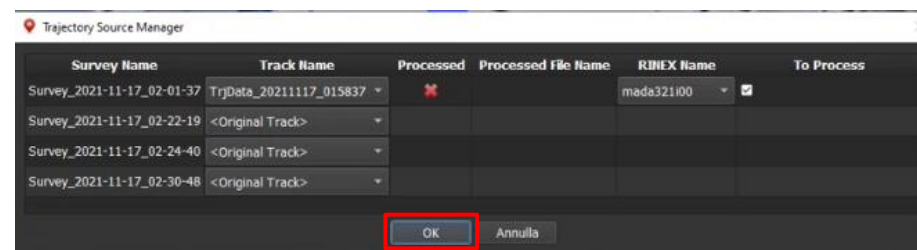


Fig. 87 – Trajectory correction start

4. When processing ends, a new trajectory file (**GPGGA.cvs**) will automatically replace the original one (**TrjData.gps**; Fig. 89) by correcting possible trajectories drifts (Fig. 90).

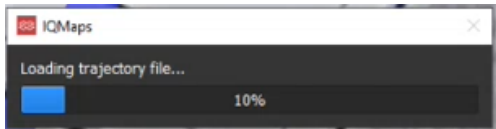


Fig. 88 – Loading bar of Trajectories processing

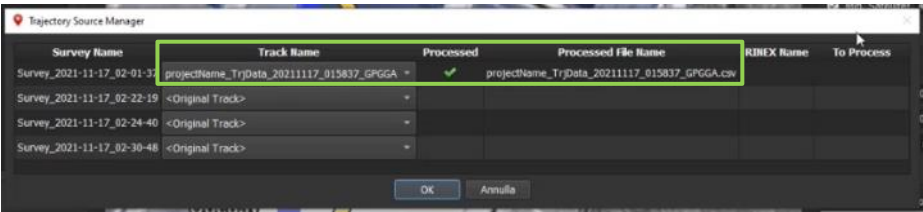


Fig. 89 – Correction file selection

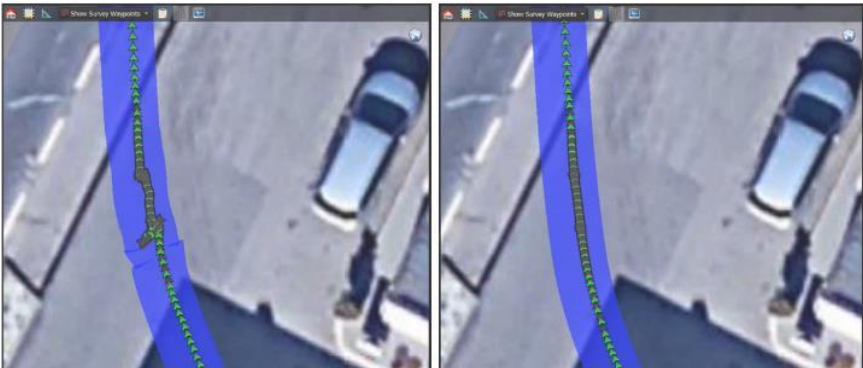


Fig. 90 – Correction file selection

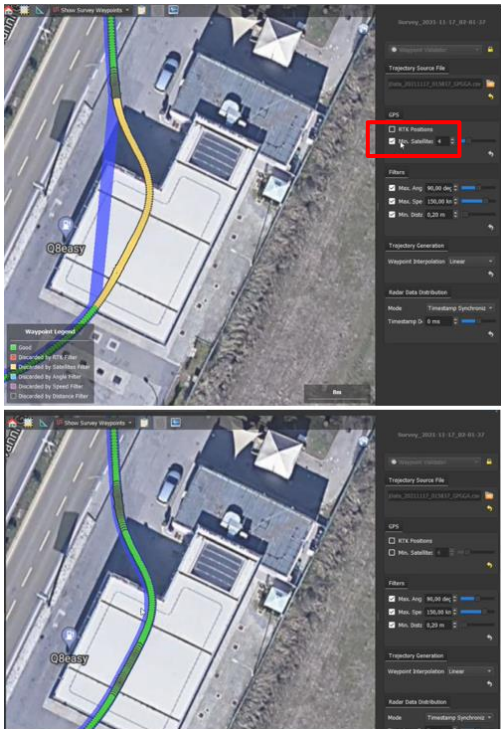


Fig. 91 – Satellites filter removing



NOTE: IQMaps will automatically copy inside the TrjData folders, a copy of the correction files.



NOTE: The User must remove the Satellite filter from Trajectory tab after APS processing (red square Fig. 91). This is because APS processing allows to resolve trajectory issues due to missing/bad satellites coverage (Fig. 91).

It is possible re-apply the original trajectory file (not processed) to the selected Survey from:

- The **Trajectory Source Manager** window by clicking on  button (1 of Fig. 91) and then selecting  option from **Track Name** tab (2 of Fig. 91);

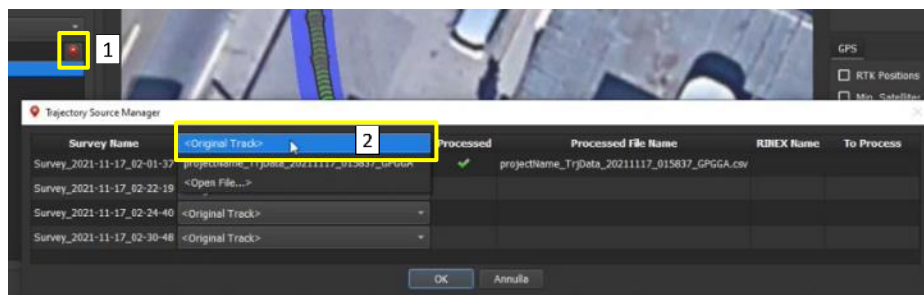


Fig. 92 – Original trajectory file restore: Option1

- The **Trajectory Source File tab** by clicking on **Trajectory Editor** Option (1 of Fig. 93) and then selecting  button from **Trajectory Source file** Tab (2 of Fig. 93).

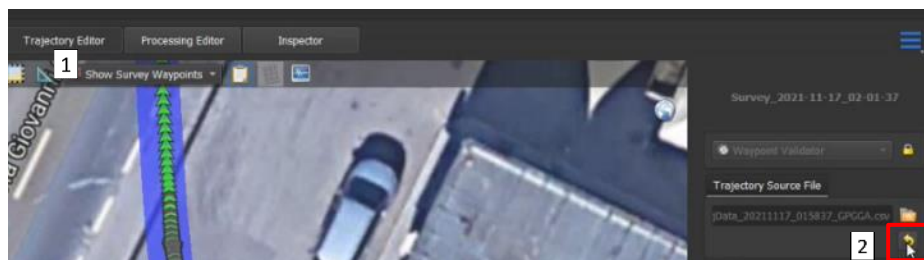


Fig. 93 – Original trajectory file restore: Option2

From the **Trajectory Source File tab**, it is furthermore possible to import back the processed trajectory file (**GPFGA.cvs**) by clicking on  button (1 of Fig. 94) and pressing OPEN option (2 of Fig. 94).

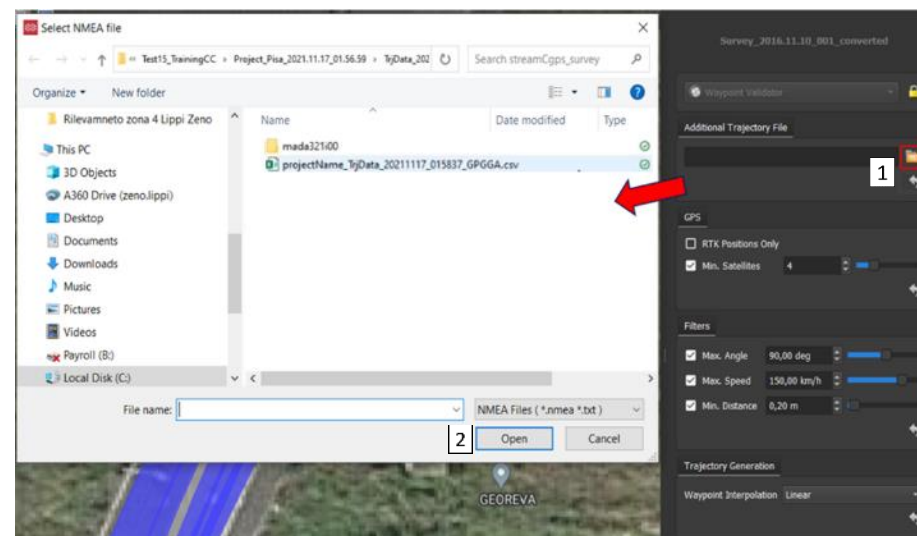


Fig. 94 – Processed trajectory file importing

B.11 How to...download RINEX data from HxGN SmartNet

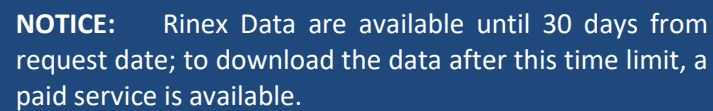
To download RINEX data, User should access to the personal [HxGN SmartNet](#) account through their own login credentials (Fig. 95).

Once connected to the home page, to access the RINEX data page it is necessary to click on  tab from top left side of HxGN platform (1 of Fig. 96) and select  option (2 of Fig. 96). From displayed window (Fig. 97) the User can get information about:

- ✓ The status and the list of HxGN SmartNet Reference Stations,
- ✓ Rinex data coverage according to green, orange, blue and red symbols,
- ✓ Data availability.



1. Insert a **project name** (facultative) on dedicated tab Fig. 97);
2. Select the closest reference **Base Station** to the Survey location. This can be done directly on the map or typing the Base Station name on searching tab (Fig. 97),



3. Select the **starting date** and **time** (local time) from dedicated icon (1 of Fig. 98), and then the duration of the request through the time bar (2 of Fig. 98);

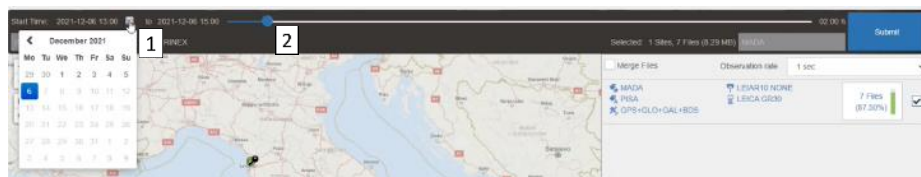


Fig. 98 – Rinx data time interval selection

4. Define the **sampling frequency** (1 of Fig. 99) and flag the "**Merge Files**" options (2 of Fig. 99). Complete the process by pressing "**Submit**" button (3 of Fig. 99),

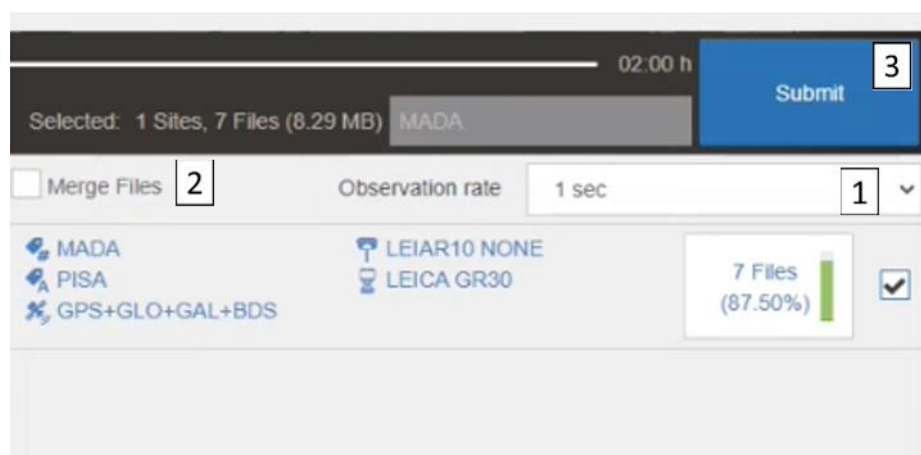


Fig. 99 – Sampling output of Rinx data

5. Select "**Confirm**" option from **Rinx Quota Window** (Fig. 100),
6. Click on "**Download**" button from **Result window** (Fig. 101); a loading progress bar will indicate the downloading status,

7. The downloaded **data** will be **zipped** into the **download folder** of User PC (Fig. 102).



NOTICE: If Merge File option is not selected, by default RINEX files of 15 minutes each are created.

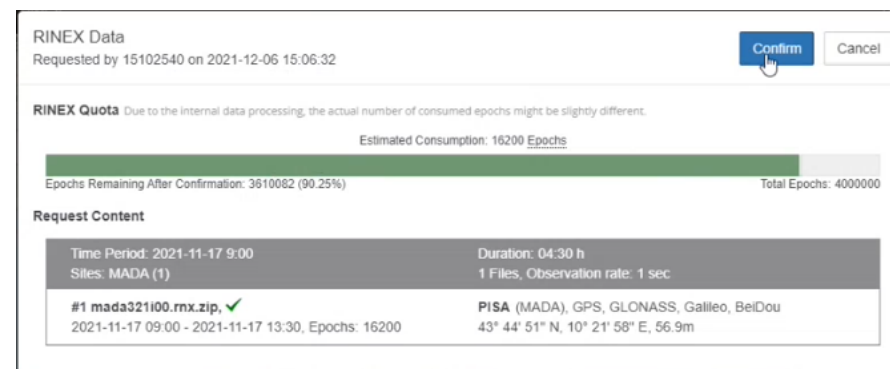


Fig. 100 – Rinx Quota WIn dow



Fig. 101 – Result window

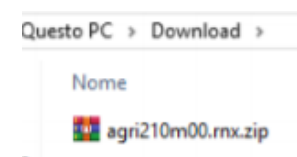


Fig. 102 – Rinx Data

B.12 How to...add new local Transformations

It is possible to insert a **.gsb** file to modify the projection parameters. This operation can be done from the Local Transformation GRID page (Fig. 103), displayed by clicking on **Local Transformations...** command from the top right drop down menu (Fig. 104).

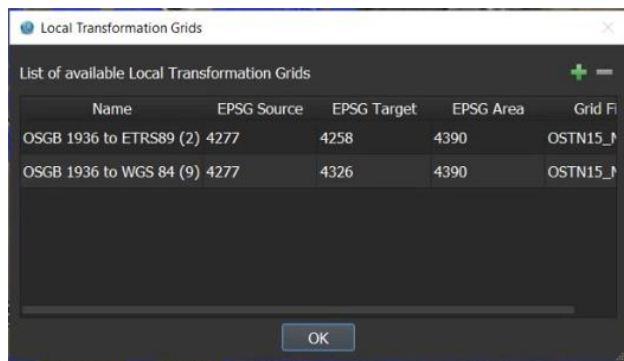


Fig. 103 – Local Transformation Grid Page

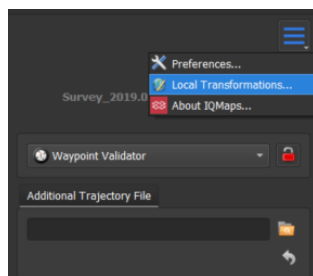


Fig. 104 – Local Transformations command

In the "Local Transformation GRID" page is possible to add a new local transformation by clicking on **+** command (Fig. 103) and importing a *.gsb file from the dedicated saving path (Fig. 105). Once selected the *.gsb file the User should insert three different parameters:

- EPSG Source
- EPSG Target
- EPSG Area

The User has the possibility also to remove one or more Transformations. Once *.gsb file is inserted, the corresponding local transformation will use this new value.

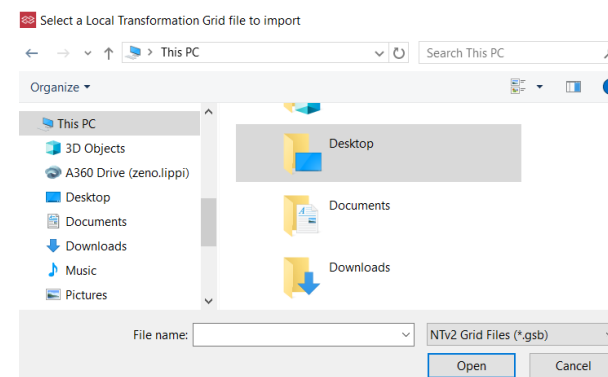


Fig. 105 – Importing a .gsb file



NOTE: The new .gsb file needs to be imported just once and it will be maintained for any future software running. When updating software versions do not update the geo folder when asked

APPENDIX C PROCESSING EDITOR: INSIGHT

C.1 General on Standard Processing

Every time that a dataset is uploaded on IQMaps, and a Project is created, IQMaps automatically applies, to the raw GPR data, a pre-defined set of “filters” whose order and parameters are customized for “that” hardware. This is defined inside “Standard Macro” files located inside the IQMaps installation folder.

In other words, for every IDS Georadar system (Stream UP, Stream DP, Chaser XR, DS4000, DS2000, Opera XR, and so on...) there is one or more Standard Macros (depending on Number of channels, dipole orientation, and antenna frequency, mainly), that is automatically applied to the raw dataset, once imported on IQMaps.

The complete list of available/applicable filters for IQMaps is illustrated in Fig. 106.

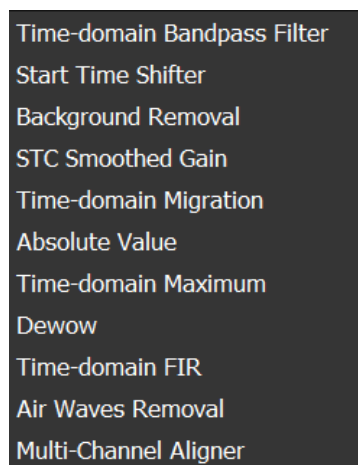


Fig. 106 – List of the IQMaps filters

Additionally, for each IDS GPR system (in Fig. 107, Fig. 108, and Fig. 109 there are examples of the standard Macro applied for Stream DP, including parameters), there is a Macro for the **B Scan** (the Channel(S)), one for the **T Scan** (cut) and One for the **C Scan** (Tomography view).

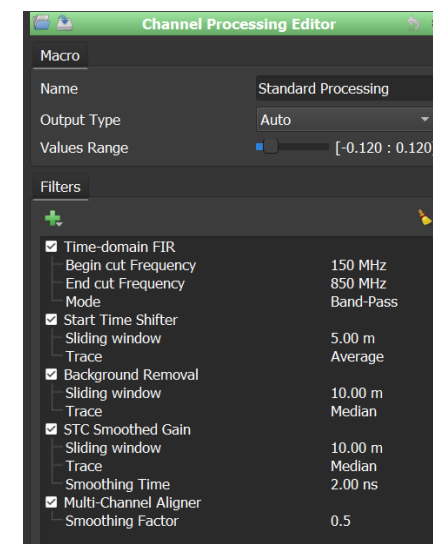


Fig. 107 Example of Standard Processing macro for B Scan of Stream DP

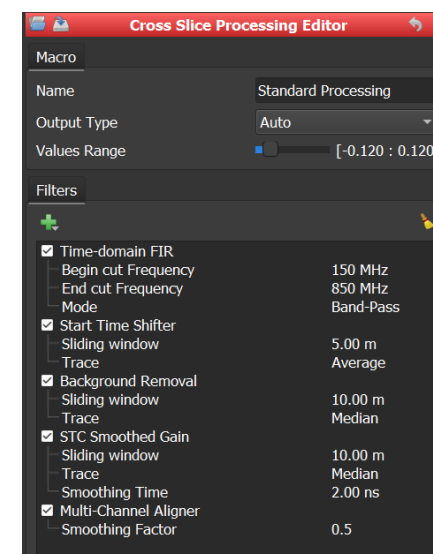


Fig. 108 Example of Standard Processing macro for T Scan of Stream DP

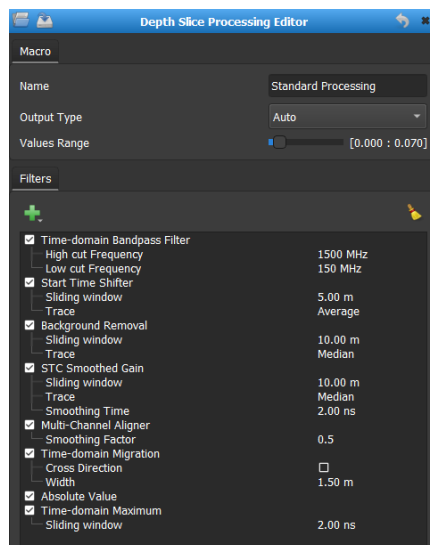


Fig. 109 Example of Standard Processing macro for C Scan of Stream DP

Filters applied, their order, and their parameters, are implemented based on “standard condition” that takes into consideration «common» soils, continuous, homogeneous, having «average» conditions of compaction and humidity. Moreover, the software would represent processed data “assuming” that the EM waves generated by the GPR (that are emitted downwards in the half-space), would propagate at an average velocity of 10 cm/nsec (Fig. 110).

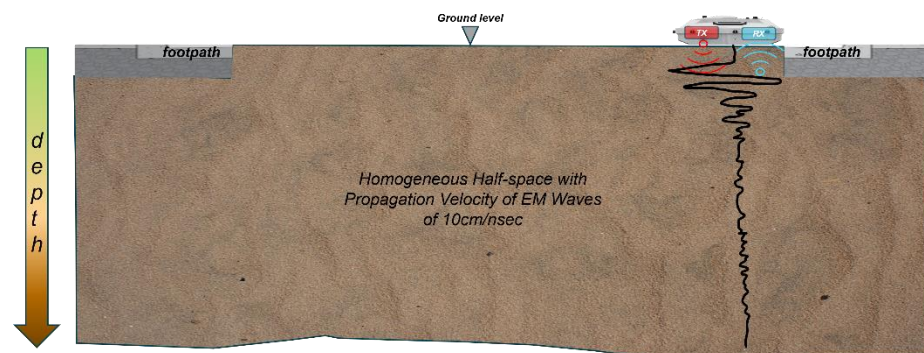


Fig. 110 Sketch of EM waves propagation

The above conditions are seldom met and, sometimes, adjustments to the filters and parameters included in the standard macros, may be necessary.

C.2 How to Modify Filters

To modify filters and their parameters with awareness, it is necessary to have some knowledge of what each filter “does” to the native data. Hence, a basic explanation of the filters available in IQMaps, is illustrated hereinafter.

C.2.1 Time-domain Bandpass Filter

It is a vertical bandpass filter in the time domain. This is a way of removing unwanted low and high frequency signal from the band, with the aim of producing clearer and more interpretable GPR maps. In Fig. 109, for example, this filter is characterized by **High-Cut Filter** set to 1500MHz and a **Low-Cut Filter** of 150MHz: this means that signal lower than 150 MHz and Higher than 1500MHz, shall be removed from the raw data.

C.2.2 Start Time Shifter

This is a processing step with which the software identifies the interface air-terrain and removes it from the radar scan. In this way, first arrival (i.e.: the first reflection of the GPR signal) moves to the zero time that should correspond to the actual position of the surface below the GPR antenna. In IQMaps, this is calculated over a horizontal moving window whose default value is 5 meters. This means that, along a moving window of 5 meters, as the systems is moved on the surface, IQMaps calculates either “Average” or the “Median” value of the signal First Brake amplitude and applies it to the data in that window. Both parameters, “Sliding Window” and “Trace”, can be modified by the User.

C.2.3 Background removal

«Clutter» is the name commonly given to an unwanted effect likely to be present in the Georadar data, due to a combination of factors, including but not necessarily limited to:

- GPR systems noise (intrinsic of the internal electronics of the system),
- Underground heterogeneity (depends on soil conditions and types),
- Surface roughness (undulations),
- External multiple reflections,
- Strong reflections of EM waves from top layers.

These would result in horizontal banding that is known as «ringing», and it may be mis-interpreted as natural layering (subsoils horizon, geological strata...) or man-made layering (asphalt thickness, concrete base and sub-base...Fig. 111). The Background Removal filter is an arithmetic process that sums all the amplitudes of reflection recorded at the same time along a profile and divides it by the number of traces found along the “Sliding window”. Either the Average or the Median value of these amplitude is then subtracted from the data set. Both parameters, “Sliding Window” width and “Trace”, can be modified by the User. Acting on the width of the sliding window, can help reducing the presence of unwanted horizontal banding from the signal. However, care must be taken in this process not to remove real linear events that may corresponds to real layering of the subsoil.

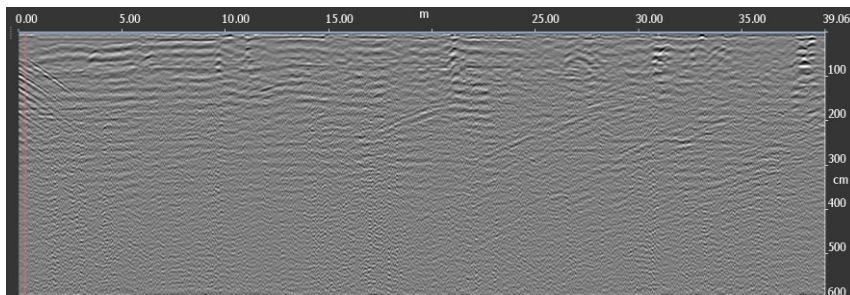


Fig. 111 "Ringing" effect

C.2.4 STC_smoothed Gain

This is a sort of amplitude “multiplier” of the signal amplitude, resulting in an equalization of the signal for deep reflection events that, because of the attenuation due to depth, may not be detected. The gain is applied on a horizontal “Sliding Window”, taking into consideration either the “Average” or the “Median” (Trace) amplitude value, and considering a Smoothing Time that, by default, is set to 2.00 ns. Users can modify all the three parameters: “Sliding Window”, “Trace” and “Smoothing Time” and verify the effect of the radar data.

C.2.5 Time Domain Migration

Migration is the mathematical technique that, using EM waves propagation velocities (default value is 10 cm/nsec), moves reflection events to the correct position, collapses diffraction hyperbolas “tails”, and increase spatial resolution of the signal. Migration applies by default only to the C Scan (tomography view) and it is characterized by two parameters: **Cross Direction** (can be either selected or un-selected) and **Width**. If **Cross Direction** is selected, then the Migration is performed across parallel channels for the “width” indicated. If this is not flagged than, Migration (hence, the tomography view) is applied along each channel (B Scan).

C.2.6 Absolute Value

Allows to visualize the absolute value of the signal after migration (without sign positive or negative) along the whole C Scan

C.2.7 Time-domain Maximum

Allows to visualize the Maximum amplitude value in a sliding window in time (depth) that can be modified by the User. Default value is 2 nsec

C.2.8 Dewow

The *Dewow* filter removes the low-frequency components (the so-called “wow”) and acts within the time domain. From the mathematical point

of view, this is a running mean value of the signal amplitude that is calculated for each value of each trace. This running mean is then subtracted from the central point of the GPR wave.

C.2.9 Time Domain FIR

This filter can be used in four Modes:

- **High Pass** -> Frequency values Higher than the High Pass threshold value are cancelled out.
- **Low Pass** -> Frequency values lower than the Low Pass value threshold are cancelled out.
- **Band Pass** -> (Default) Signal with Frequency included within the High and Low values are kept, the rest is cancelled out
- **Stop Band** -> Frequency values inside the Low and the High Pass value are cancelled out.

C.2.10 Air Waves Removal

Abnormally large and hyperbolic-shaped signature(s) may be present in a B-Scan (Fig. 112). This should not be interpreted (associated) to actual features of the underground (like layering, top of cavities, buried tunnels, large diameter pipes...), but rather seen as a complex multipath of the EM waves, due to elements above the ground (e.g.: lamp posts, trees, walls, air cabling, fences...). A combination of conditions and factors may help in identifying Air Waves in a B-Scan, including:

- Presence of large objects over the GPR (i.e.: the Operator) including Trees, Lamp posts, cables, walls, fences...
- Subsoil whose native conditions have very good GPR signal propagation,
- High performance GPR systems with broad dynamic range,

- Exceptionally high velocity of the EM-wave when attempting hyperbola fitting,

The Air Waves Removal is a F-K domain filter that acts to mitigate the effect of the airwaves. The **Roll-off** parameter (default value is 0.85) can vary from 0 to 1, where 0 indicates a minimum impact of the filter with respect to the signal and 1 is the maximum impact of it.

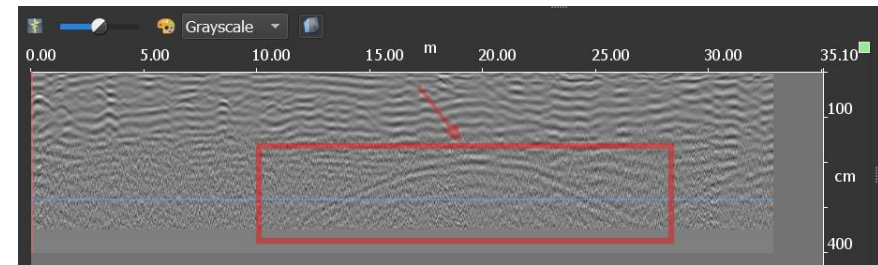


Fig. 112 "Air Waves" effect

C.2.11 Multi-Channel Aligner

This filter is specifically designed for massive array system where the effect of parallel channels may affect the quality of the C Scan.

For the User to edit existing list of filters (or the ones being added) the following tools apply:

- To edit an existing filter parameter, double click on the value and fill in with the new value (for example **1500 MHz**); alternatively, using the arrows icon on the right side of the parameter.
- To Show or hide one or more existing filters use the checkbox icon beside each filter.
- Cancel one or more filters moving the cursor on a specific filter and clicking on the icon.

All the above actions are reversible by clicking on the upper icon .

APPENDIX D OPENGPR DATA FORMAT: INSIGHTS

A file in the OpenGPR format should have the extension *.ogpr .

D.1 Conventions and Nomenclature

We call **Radar Data Volume** a three-dimensional array of “homogeneous” **Samples** acquired by a GPR system (Fig. 113). Let's say that the sizes of the array along the three main directions are **K**, **M** and **N**.

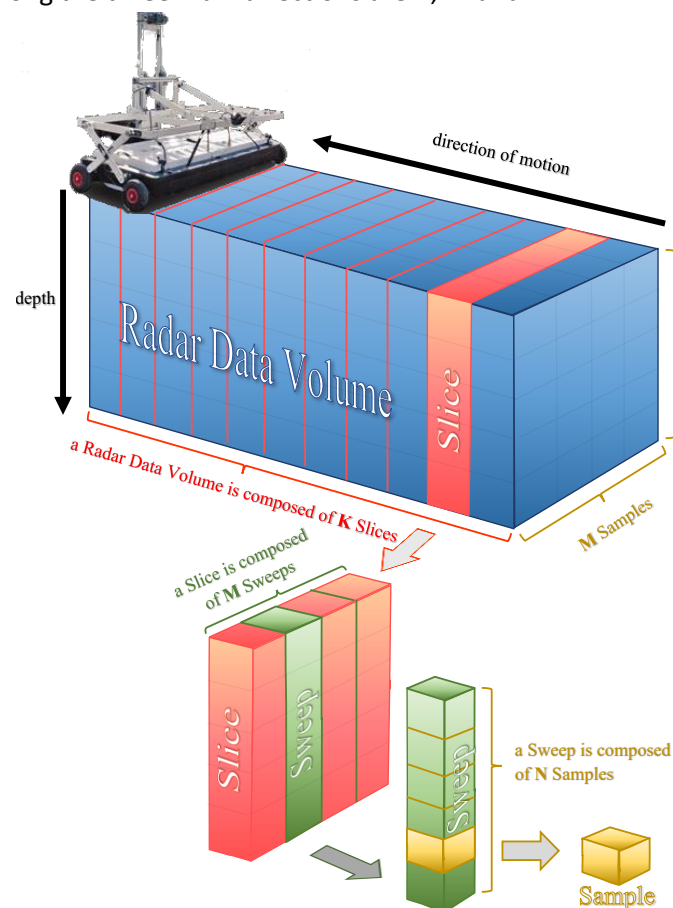


Fig. 113 Radar Data Volume and its decomposition

We call **Slice** any two-dimensional sub-array of the Radar Data Volume that is a section orthogonal to the direction of the radar motion during the acquisition. From this point of view, a Radar Data Volume is composed of K Slices. The number K of Slices usually depends on some properties of the system, on its movement and on the duration of the acquisition.

Given a Slice, we call **Sweep** any of its one-dimensional column array. A Sweep is orthogonal to the surface on which the radar moves. A Sweep is also known as A-scan. From this point of view, a Slice is composed of M Sweeps. A Sweep is usually generated by a single antenna. In a multi-channel GPR system the number M of sweeps is related to the number of channels/antennas of the system.

Each Sweep is composed of N radar Samples. The number N of Samples per Sweep depends on the properties of the GPR system.

It is assumed that all Samples of the Radar Data Volume are “homogeneous”, i.e. that they have been acquired by the same GPR system, with the same characteristics for each antenna, and that the system configuration has remained unchanged throughout acquisition or that the homogeneity condition of the Samples is guaranteed in another alternative way.

On the contrary, for multi-antenna systems where the antennas have different physical characteristics or are configured differently (multi array GPR system), or for acquisitions performed with the same system but with different settings or in other circumstances that do not guarantee the homogeneity of the radar Samples, the acquired data must be divided into two or more homogeneous Radar Data Volumes (three-dimensional arrays). In this case, each Radar Data Volume must be managed independently.

D.2 Format Structure

The content of each OpenGPR file is partly textual and partly binary and is made of the following **Parts**, each of them is broken down into sub-parts (Fig. 114):

1. **Preamble:** in text format. It is fixed length (47 bytes). And it is composed of:
 - 1.1. **Magic Number:** a fixed size text line (5 bytes) equal to “ogpr” + **LF (Line Feed)**).
 - 1.2. **md5:** a fixed size text line (32 bytes + LF) containing the md5 in hexadecimal string format (see below for more details).
 - 1.3. **JSON Header Size:** a fixed size text line (8 bytes + LF) containing the size in bytes of the JSON Header (see next Part description). The content of this line is right aligned and padded by ‘0’ (for example, a size of 1518 bytes will result in the text line “00001518\n”).
2. **JSON Header:** a variable size header in JSON format, made of one or more lines that describe the content of the file and (in particular) the content of the following Part. The JSON Header size must be less than 100 000 000 bytes (approximately 100 MB).
3. **Data Blocks:** zero or more data blocks (in binary format). The binary format of each data block depends on its type.
4. **Epilogue:** in text format. It is fixed length (33 bytes). And it is composed of:
 - 4.1. a LF character.
 - 4.2. **md5:** it is the same md5 of 1.2 (32 bytes but no new line termination).

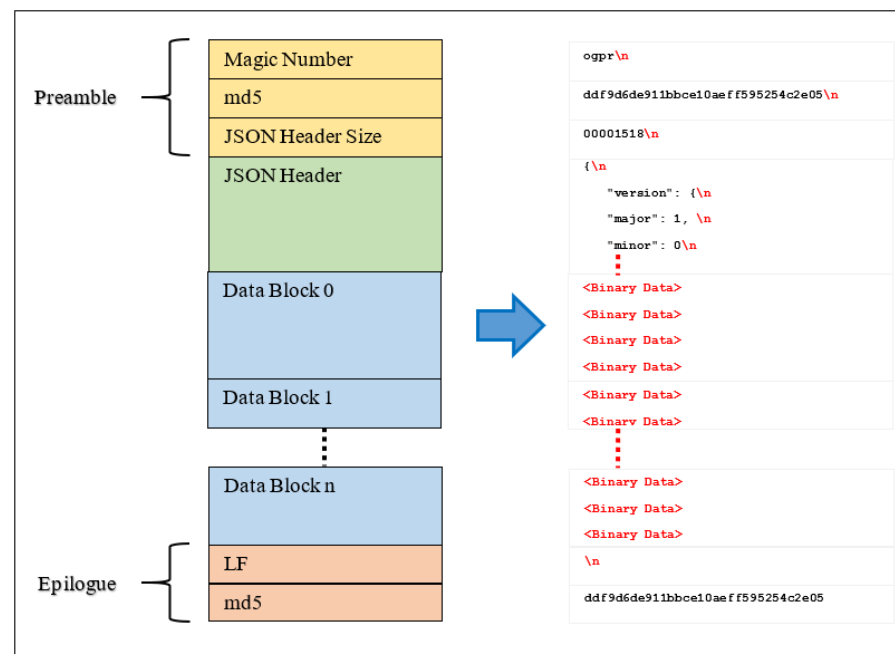


Fig. 114 - OpenGPR file structure

Each text line is terminated by a LF character ('\n' for C programmers) except for the last md5 (4.2). Each Part in text format is UTF-8 encoded.

The md5 is computed considering each byte that follows the first md5 (1.2) and that precedes the last md5 (4.2). Note the LF that follows the md5 described at 1.2 is included in the md5 computation.

Note: only part 1 and part 4 are mandatory. The JSON Header can be empty (zero bytes), in this case also the Part at point 3 (Data Blocs) must be empty. This configuration represents a valid “empty” OpenGPR file.

Section 4 contains an example of an OpenGPR file.

D.3 JSON Header

The JSON Header describes the content of the entire file and is fundamental to read and interpret the content of the Data Blocks. The variable size in bytes is written in textual format in the third line of the file (JSON Header Size).

In the following sections, the syntactic elements that make up the JSON Header will be identified following the naming convention of www.json.org and are written in *italic*.

D.3.1 JSON Header Main Element

The main *JSON element* of the JSON Header is a *JSON object* that has the key-value *members* described in the following table.

Key-Value	Mandatory	Description
"version": <i>object</i>	yes	Described in Section D.3.3
"mainDescriptor": <i>object</i>	yes	Described in Section D.3.4
"dataBlockDescriptors": <i>array</i>	yes	Described in Section D.4

D.3.2 The JSON Object "version"

The *object* "version" declares the version of the OpenGPR format to which the file conforms. It can have the key-value *members* described in the following table.

Key-Value	Mandatory	Description
"major": <i>integer</i>	yes	The major version number.
"minor": <i>integer</i>	yes	The minor version number.
"experimental": <i>true</i>	no	This member is used during the development of new versions of the OpenGPR format and should not appear in officially distributed OpenGPR files. If the member "experimental" is present, it is a sign that the structure and content of the file is provisional and not officially supported.

Currently the version of OpenGPR is 2.0. So, the "major" *value* is 2 and the "minor" *value* is 0.

To summarize, at the moment, the *member* "version" has the following form:

```
"version": {
  "major": 2,
  "minor": 0
}
```

D.3.3 The JSON Object "mainDescriptor"

The *object* "mainDescriptor" contains information about the whole file or that is essential for reading more than one Data Block.

It can have the key-value *members* described in the following table.

Key-Value	Mandatory	Description
"samplesCount": <i>integer</i>	yes	Is the number of Samples in a Sweep (as described in Section Error! Reference source not found.).
"channelsCount": <i>integer</i>	yes	Is the number of Seewps in a Slice .
"slicesCount": <i>integer</i>	yes	Is the number of Slices in the 3D Radar Data Volume
"metadata": <i>object</i>	no	Currently there are no constraints on the content of this object. It can contain any auxiliary information that is not needed to read the rest of the file, for example: system descriptive information (other than that already contained in the rest of the file), date and time of acquisition, vendor specific information, user or software specific annotations.

An example of the *object* "mainDescriptor":

```
"mainDescriptor": {
  "samplesCount": 512,
  "channelsCount": 9,
  "slicesCount": 70,
  "metadata": {
    "swathName": "Swath019",
    "swathId": "bc07edd5c9729c771c135ae6bae1d3a5",
  }
}
```

```

    "arrayId": 2
  }
}

```

D.3.4 The JSON Array “dataBlockDescriptors”

The array “dataBlockDescriptors” is an array of JSON objects, one for each Data Block contained in the file. For convenience we call each of these objects Data Block Descriptor.

Each Data Block Descriptor contains the information needed to read and interpret the binary data in the corresponding Data Block.

Each Data Block Descriptor has the key-value members described in the following table.

Key-Value	Mandatory	Description
"byteSize": integer	yes	Is the size in byte of the corresponding Data Block.
"byteOffset": integer	yes	Is the offset of the first byte of the corresponding Data Block. It is in bytes measured from the beginning of the file.
"type": string	yes	It specifies the type of the corresponding Data Block and indirectly indicate how the data layout within the Data Block. There are currently only two Data Block types identified by the strings “Radar Volume” and “Sample Geolocations”
"name": string	yes	It is a unique name/identifier of the Data Block. There are no constraints on the content of the string, but two different Data Blocks of the same type in the same file cannot have the same name.
"metadata": object	no	It has the same purpose and specifications as the analogous member of the “mainDescriptor” object.

Each Data Block Descriptor can have additional (mandatory or optional) members depending on its type. Data Block types are described in the following sections.

D.4 Data Blocks

Data blocks are in binary form. They correspond to the Data Block Descriptors in the JSON Header in number and order.

All binary data described in this document, unless otherwise stated, follow the convention of Intel-compatible x86-64 family processors, i.e.:

- Multibyte binary data are little-endian
- Floating point binary data follow the IEEE 754 standard
- Integer data follow the two’s complement encoding

In this first version of the OpenGPR format, only two types of Data Blocks are defined: “**Radar Volume**” and “**Sample Geolocations**”. They are described in the next two sections.

D.4.1 “Radar Volume” Data Block Type

A Data Block of type **Radar Volume** stores a 3D array of radar Samples. It is possible to store all Radar Sample Values in one of the following numerical representations:

- 16-bit signed integer (2 bytes per Radar Sample)
- 32-bit floating-point (4 bytes per Radar Sample)

The numerical representation used in the Data Block is declared in the respective Data Block Descriptor (*member* “valueType”).

Each Sample value can be converted to the corresponding value in volts using the following equation:

$$\text{valueInVolt} = (\text{sampleValue} / 32768.0) * 10.0$$

All Samples of the Radar Data Volume are stored consecutively inside the Data Block according to the following sorting rules.

- 5. First are stored all Samples of the first slice. Then follow all Samples of the second Slice and so on. The first Slice should be the one that the GPR system acquired first (if this statement makes sense for the GPR system that has acquired the Radar Data Volume).
- 6. For each Slice, the Samples of the first Sweep are stored first. Then follow the Samples of the second Sweep and so on. For an observer pushing the GPR system and looking at it along the direction of motion, Sweeps should be ordered from left to right. The first Sweep should be the leftmost one (if this statement makes sense for the GPR system that has acquired the Radar Data Volume).
- 7. For each Sweep, Samples are stored from the shallowest to the deepest.

The following figure summarizes the data order.

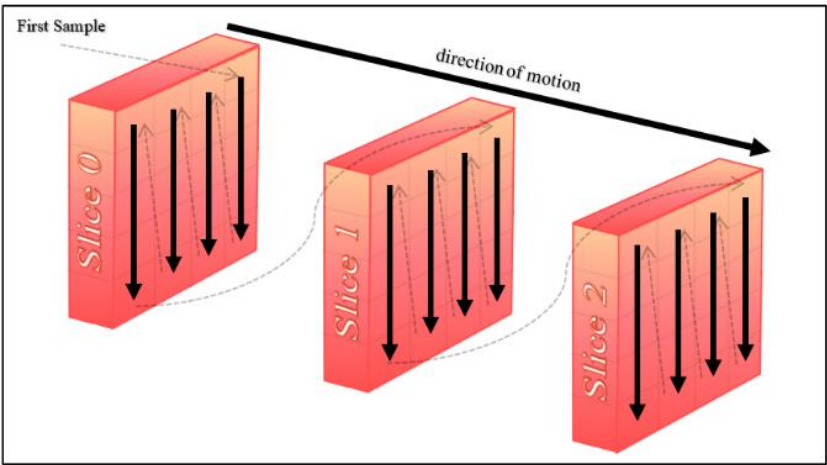


Fig. 115 - Order of Samples in a Data Block of type Radar Volume.

The number of Samples per Sweep, the number of Sweeps per Slice and the number of Slices in the Data Volume are those of the mainDescriptor members. So, the total number of bytes in the Data Block of type Radar Volume is:

$$\text{samplesCount} * \text{channelsCount} * \text{slicesCount} * \text{BytePerSample}$$

where BytePerSample is either 2 or 4 bytes depending on the numerical representation (int16 or float).

The Data Block Descriptor of type Radar Volume has the following key-value member (in addition to those that are common to all types of Data Blocks described in Section D.3.4).

Key-Value	Mandatory	Description
"radar": <i>object</i>	yes	It collects the properties of the GPR system related to the acquisition of radar Samples.
"valueType": <i>string</i>	no	It specifies the numeric representation of each Radar Sample Value in the Data Block. It can be "int16" or "float". This member is not mandatory, if it is missing, its value is assumed to be "int16".

The previous "radar" object has the key-value members described in the following table.

Key-Value	Mandatory	Description
"samplingStep_m": <i>number</i>	yes	It is the sampling step in meters along the acquisition direction (i.e.: the metric distance between two consecutive Slices).
"samplingTime_ns": <i>number</i>	yes	It is the sampling time in nanoseconds between two consecutive Samples of the same Sweep.
"propagationVelocity_mPerSec": <i>number</i>	yes	it is the average propagation velocity of the electromagnetic signal through the ground in meters/seconds that is supposed to be during the acquisition.
"frequency_MHz": <i>number</i>	yes	It is the frequency of the electromagnetic signal generated by the GPR system.

"polarization": <i>string</i>	yes	It describes the polarization of the electromagnetic signal generated by the GPR system. At the moment it can be equal to one of these values: "horizontal" , "vertical" .
-------------------------------	-----	--

An example of the Data Block Descriptor of type Radar Volume:

```
{
  "type": "Radar Volume",
  "name": "Swath019 Array02 Radar Data Volume",
  "byteSize": 1290240,
  "byteOffset": 909,
  "radar": {
    "samplingStep_m": 0.0399,
    "samplingTime_ns": 0.125,
    "propagationVelocity_mPerSec": 100000000.0,
    "fequency_MHz": 600,
    "polarization": "horizontal"
  },
  "valueType": "float",
  "metadata": {
    "processing": null
  }
}
```

D.4.2 “Sample Geolocations” Data Block Type

A Data Block of type **Sample Geolocations** stores the Sample positions of a Radar Data Volume with respect to a specified **spatial reference system**.

This Data Block is logically made of a sequence of **Slice Blocks**, one for each Slice. Each Slice Block consists of:

- 1. a 64-bit integer identifying the corresponding Slice, followed by
- 2. a sequence of **Sweep Blocks** (one for each Sweep in a Slice)

Each Sweep Block consists of two **Coord Blocks**: the first one contains the (x, y, depth, elevation) coordinates of the first Sweep Sample (min. depth), the

second contains the (x, y, depth, elevation) coordinates of the last Sweep Sample (max. depth). The position of the intermediate Samples can be determined by linear interpolation or by a more sophisticated method that takes into account the characteristics of the ground.

It is to be noted that the second Coord Block is relevant only when considering an “excavation” direction that is not perpendicular to the ground (this may be useful in the future, e.g. in order to correct tilt from GPS).

Each Coord Block is a tuple of four coordinates (x, y, depth, elevation), each coordinate being represented by a double precision floating-point number. Coordinates x and y are relative to the geographical system specified in the corresponding Data Block Descriptor (described below). The depth coordinate is expressed in linear meters from the ground level (negative numbers below the ground). The elevation coordinate is expressed in linear meters. If elevation information is not available, it should be set to zero for the shallowest (min. depth) point and should be set accordingly for the deepest point.

Note that the order of the Coord Blocks resembles that of the Fig. 115.

The Data Block Descriptor of type Sample Geolocations has the following key-value *member* (in addition to those that are common to all types of Data Blocks described in Section D.3.4)

Key-Value	Mandatory	Description
"srs": <i>object</i>	yes	It specifies which spatial reference system is used.

The previous “srs” object has the key-value *members* described in the following table.

Key-Value	Mandatory	Description
"type": <i>string</i>	yes	It declares the type of spatial reference system specification. At the moment it can be equal to one of these values: “WKT”, “EPSG”, “Local”, “Scaled Pseudo-Mercator”.
"value": <i>element</i>	yes	It is of variable type, its type and its value depend on the value of the member “type”. See the next table for more details.

The following table describes the type and the constraints on the value of the member “value” depending on the value of the member “type”.

Value of member “type”	Type of member “value”	Value of member “value”
"WKT"	<i>string</i>	A spatial reference system in valid WKT form. See: https://en.wikipedia.org/wiki/Well-known_text_representation_of_geometry https://en.wikipedia.org/wiki/Spatial_reference_system#Identifier
"EPSG"	<i>integer</i>	An EPSG code that specifies a spatial reference system. https://en.wikipedia.org/wiki/EPSSG_Geodetic_Parameter_Dataset
"Local"	<i>null type</i>	null
"Scaled Pseudo-	<i>number</i>	It is a positive real number, let's call it Scale Factor . This spatial reference system is defined by the following

Mercator"	equations: Scaled Pseudo-Mercator X Coord = Pseudo-Mercator X Coord / Scale Factor Scaled Pseudo-Mercator Y Coord = Pseudo-Mercator Y Coord / Scale Factor Scaled Pseudo-Mercator Z Coord = Pseudo-Mercator Z Coord Where Pseudo-Mercator Coords are relative to the EPSG:3857 spatial reference system.
-----------	---

Some examples of the Data Block descriptor of Sample Geolocations:

```
{
  "type": "Sample Geolocations",
  "name": "Swath019 Array02 Sample Geographic Locations",
  "byteSize": 30800,
  "byteOffset": 646029,
  "srs": {
    "type": "EPSG",
    "value": "PROJCS[\"WGS 84 / Pseudo-Mercator\",GEOGCS[\"WGS 84\",DATUM[\"WGS_1984\",SPHEROID[\"WGS 84\",6378137,298.257223563,AUTHORITY[\"EPSG\",\"7030\"]],AUTHORITY[\"EPSG\",\"6326\"]],PRIMEM[\"Greenwich\",0,AUTHORITY[\"EPSG\",\"8901\"]],UNIT[\"degree\",0.0174532925199433,AUTHORITY[\"EPSG\",\"9122\"]],AUTHORITY[\"EPSG\",\"4326\"]],PROJECTION[\"Mercator_1SP\"],PARAMETER[\"central_meridian\",0],PARAMETER[\"scale_factor\",1],PARAMETER[\"false_easting\",0],PARAMETER[\"false_northing\",0],UNIT[\"metre\",1,AUTHORITY[\"EPSG\",\"9001\"]],AXIS[\"Easting\",EAST],AXIS[\"Northing\",NORTH],EXTENSION[\"PROJ4\",\"+proj=merc +a=6378137 +b=6378137 +lat_ts=0 +lon_0=0 +x_0=0 +y_0=0 +k=1 +units=m +nadgrids=@null +wktext +no_defs\"],AUTHORITY[\"EPSG\",\"3857\"]]"
  }
}
```


<pre>{ "type": "Sample Geolocations", "name": "Swath019 Array02 Sample Geographic Locations", "byteSize": 30800, "byteOffset": 646029, "srs": { "type": "EPSG", "value": 32632 } }</pre>
<pre>{ "type": "Sample Geolocations", "name": "Swath019 Array02 Sample Geographic Locations", "byteSize": 30800, "byteOffset": 646029, "srs": { "type": "Local", "value": null } }</pre>
<pre>{ "type": "Sample Geolocations", "name": "Swath019 Array02 Sample Geographic Locations", "byteSize": 30800, "byteOffset": 646029, "srs": { "type": "Scaled Pseudo-Mercator", "value": 0.75 } }</pre>

D.4.3 OpenGPR File Example

The following table shows an OpenGPR file as it would look if opened with a text editor, the binary part has been removed.

<pre>ogpr 6f1c8f5c98f6114572648518376321cb 00000862 { "version": { "major": 2, "minor": 0 }, "mainDescriptor": { "samplesCount": 512, "channelsCount": 9, "slicesCount": 70, "metadata": { "swathName": "Swath019", "swathId": "bc07edd5c9729c771c135ae6bae1d3a5", "arrayId": 2 } }, "dataBlockDescriptors": [{ "type": "Radar Volume", "name": "Swath019 Array02 Radar Data Volume", "byteSize": 1290240, "byteOffset": 909, "radar": { "samplingStep_m": 0.0399, "samplingTime_ns": 0.125, "propagationVelocity_mPerSec": 1000000000.0, "frequency_MHz": 600, "polarization": "horizontal" }, "valueType": "float", "metadata": { "processing": null } }] }</pre>

```
        "type": "Sample Geolocations",
        "name": "Swath019 Array02 Sample
Geographic Locations",
        "byteSize": 30800,
        "byteOffset": 646029,
        "srs": {
            "type": "EPSG",
            "value": 3857
        }
    ]
}
*** Binary Data Blocks ***
6f1c8f5c98f6114572648518376321cb
```

APPENDIX E IDS GCAD PLUGIN: INSIGHTS

From IQMaps it is possible to export data (scans and targets) into the AutoCAD environment (see par. 5.6.1). The *ids_gcad_xyz* application is an additional tool that the User may use as an AutoCAD plugin (refer to par. 4.5 for the installation).

Once AutoCAD application is launched, the *ids_gcad_xyz* toolbar will automatically pop up (Fig. 116) and the user should follow the steps below to open and work on the export of the IQMaps project in the AutoCAD environment:

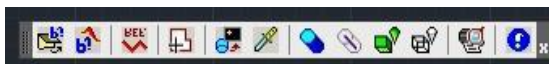


Fig. 116 – ids_gcad_xyz toolbar

1. Click on  command and select the folder where the project is contained (not the raw data). The User must click on the  button (Fig. 117).

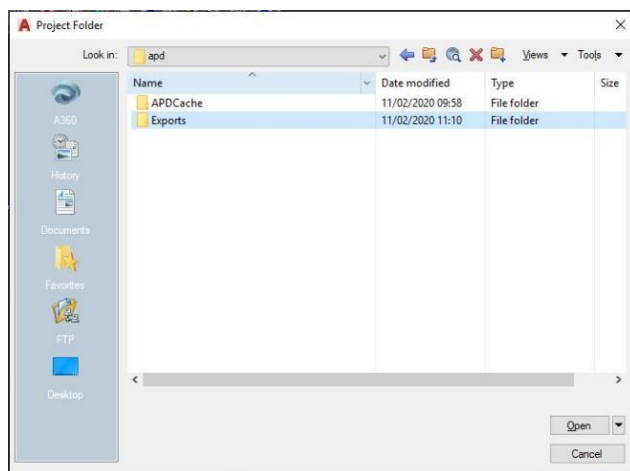


Fig. 117 – Project export path

2. Click on  command to load the selected project. The data will be automatically represented on the drawing and geo-referenced with respect to the present GPS/TPS system of the hardware used.
3. The **Current Project** command () allows to manage more than one project on the same CAD design. The procedure is:
 - Repeat step 1 & 2 for each new project that should be imported in CAD (remember that IQMaps SW creates a different ASCII file for each project inside its directory). The user will choose the current project among the list shown in the window in (Fig. 118).

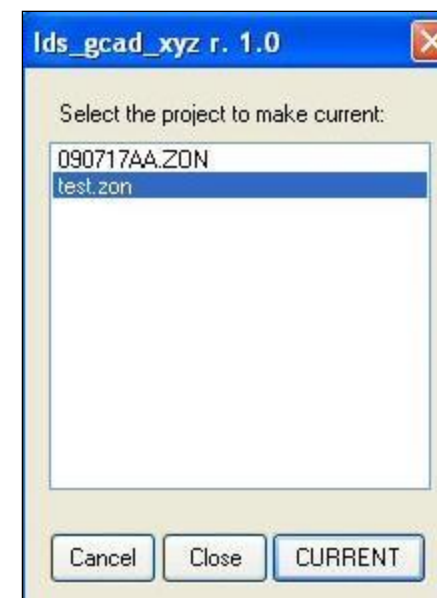


Fig. 118 – Multiple Project management

4. Once the project has been selected, the User must click on **Objects creation** command () from the Ids_gcad_xyz toolbar. This command opens the **New object window** (Fig. 119).

5. Point  allow to insert in the AutoCAD file, particularly interesting cartographic points surveyed in the countryside for ex. fixed points. After having defined the point's name in the **Object name** field (Fig. 119), select Point to open the window in Fig. 120.

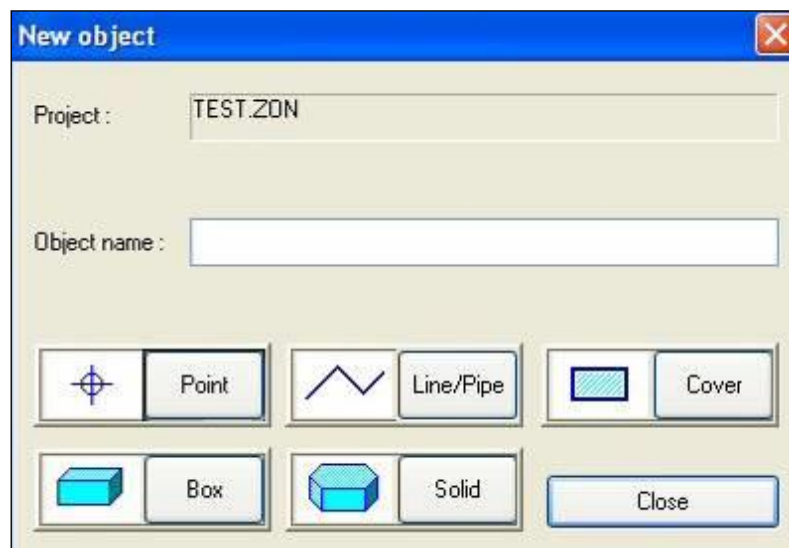


Fig. 119 – New object window

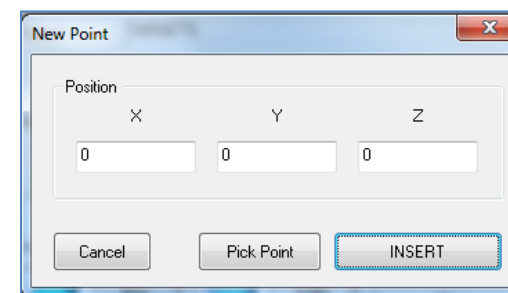


Fig. 120 – Insert X, Y and Z coordinate of New Points

6. The user may enter the Points in the AutoCAD file by activating:

- Pick Point
- INSERT

Pick Point allows selecting the desired points directly on the zone of the plan view, after the cursor has automatically become a pointer.

INSERT allows the user to insert the points in the AutoCAD file after having inserted the X, Y and Z coordinates in the corresponding fields of Fig. 120. The position of the inserted points in the AutoCAD file refers to the CRL previously designed.



7. The line/Pipe button allows to insert in the selected zone, particularly interesting lines taken in the countryside, for example, street borders, fencing, trenches, pipes, etc. After given the line a name in Object name (see Fig. 119) and having pressed Line (see Fig. 121) you may select buttons:

- Select Object
- Pick Points

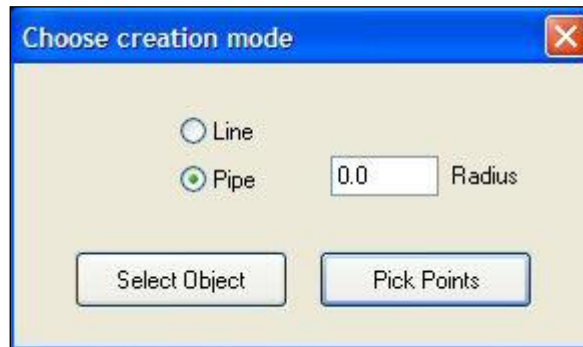
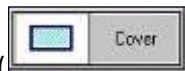


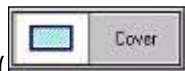
Fig. 121 - Choose creation mode window

Select Object allows the User to select and assign a pre-existing 3D line in the AutoCAD file and insert it in the DataBase.

Pick Points allows User to manually insert a poly-line in the AutoCAD file and insert it in the Database. It is possible to choose to represent a Pipe or a Line and assign a radius in meters to the selected object.

The User can draw a Line or a Pipe in AutoCAD workspace by pressing left mouse button to insert each point and then pressing the right mouse button to stop the process. Once a Line/Pipe is created it can be rotated or modified using AutoCAD command.



The Cover command () activates the window showed in Fig. 122 where the User may insert the infrastructures, found in the Symbol list, in the AutoCAD file. The procedure used to insert the infrastructures is:

- In the **Symbol** field the User can select the desired infrastructure and insert its X, Y, and Z coordinates. The User must also define the base and height of the symbol scale and

set the symbols angle with respect to the X axis (Fig. 122); in particular:

- **Base scale** command changes the dimension of the cover that will be insert in the Main window along the reference line;
- **Height scale** command strain the cover perpendicular to the base line;
- **Angle from X axis**: angle that the cover will have compared to X axis.

- By clicking on  command (Fig. 122) the User can select the colour that will be associated to the symbol from the window show in Fig. 123.

- By clicking on  command (Fig. 122) the desired symbol will be loaded in the AutoCAD file. The item will be automatically loaded in the desired position, following the established coordinates and the previously defined CRL.

- The User can decide to manually position the Cover by clicking on  command and, with the left mouse button, to select the point on the Main Window where to place the cover.

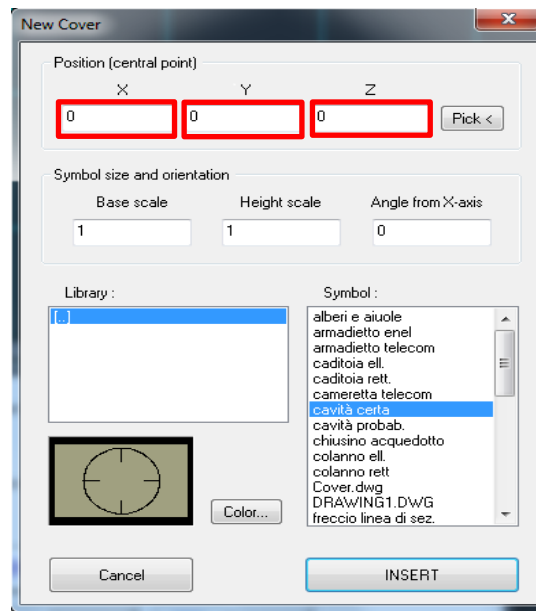


Fig. 122 - New Cover window



Fig. 123 – Symbol colour selection window



The **Box** command () allow the User to insert in the AutoCAD file 3D elements representing existing, buried infrastructures (Fig. 119).

By clicking the **Box** command, the window shown in Fig. 124 will appears; here the User needs to put the coordinates of the opposite vertexes along the structure's major diagonal line and also the angle with respect to the T axis. Fig. 125 shows the pattern of a Box insertion in the field.



Click  button to insert the first corner of the box (**P1**) and Repeat the same process with the opposite corner (**P2**).

Z coordinates value is “0” both first and opposite corner; before selecting **Insert** command, modify this value in order to create a parallelepiped structure; if **Z** value is zero, software creates on XY Plane a rectangle.

The User, on the **Angle from X-axis** space (Fig. 124), must set the Box angle respect to X – Axis and click on **INSERT** button in order to finalize the Box insertion.

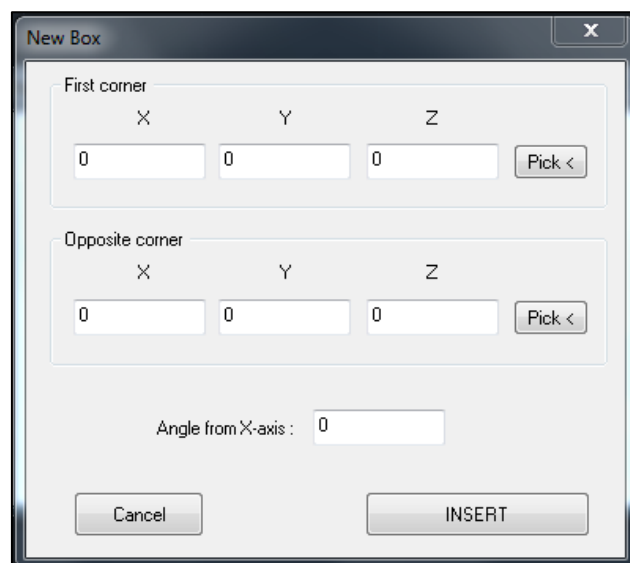


Fig. 124 - Insert first (P1) and opposite corner (P2) to create a parallelepiped structure

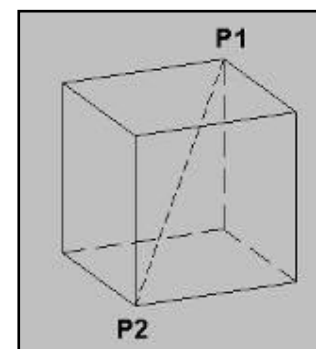


Fig. 125 – Box insertion pattern



The **Solid** command () allows the User to insert in the AutoCad file an item having a polygonal structure (Fig. 119). By clicking on the **Solid** command, a window automatically will show up allowing the User to insert in the **Top depth field**, the depth in meters of the shallowest part, and the width in the **Thickness** field of the element to insert (Fig. 126). By activating **PICK POINTS** button, the cursor will become a pointer and the User may design the object's 3D polygonal structure that will be inserted in the AutoCAD file.

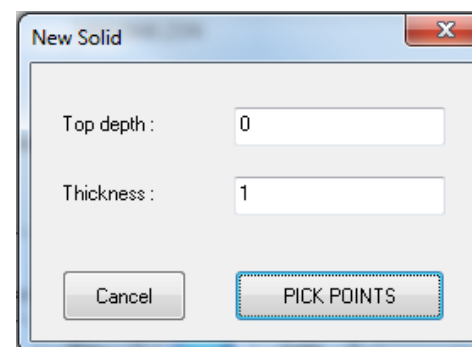
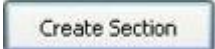


Fig. 126 –New Solid window

8. by clicking on  command, the User may automatically obtain schematic longitudinal, traversal, horizontal and diagonal sections of the cartography imported from the Database. Activating this button, the window in Fig. 127 will appear allowing to decide which section to perform. In order to create a section, the User must edit the following fields:

- **Section name:** insert the name of the section
- **Vertical Limits:** insert the min & max depth of the vertical section
- **Grid:** insert the X-step & Y-step of the reference grid
- **Vertical Scale Factor:** select the scale relation in the vertical direction.

After having completed the setting phase, the User must press the

 button to save the settled parameters and start to draw the section (Fig. 127) following the steps reported below:

- Select two or more points with the left mouse button on the Main Window.
- Stop the selection by pressing the right mouse button.
- Choose where place the new section in the AutoCAD's workspace; there the User can visualize the name of the section, intersected targets, grid's step, length and the depth of the section.



NOTE: The User must verify that the inserted object is a closed polygonal to finalize the process.

Fig. 127 – Section setting window

In Fig. 128 it is possible to see a section NE – SW oriented. It cuts three targets represented with different colours and Est – West oriented

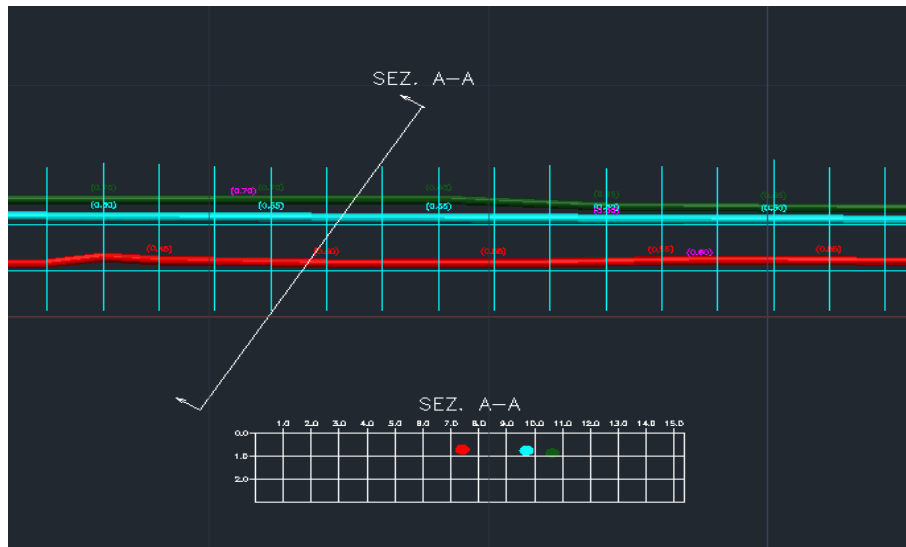


Fig. 128 - A-A Section across three horizontal targets

9. The **Dress Pipe** command () allows to “dress” the pipes, creating around the pipe axis the vectors of a cylinder. This type of viewing gives the pipe a so-called “iron view” look (Fig. 129). After activating the *Dress pipe*, the cursor becomes a small square allowing to select the pipes to “dress”.

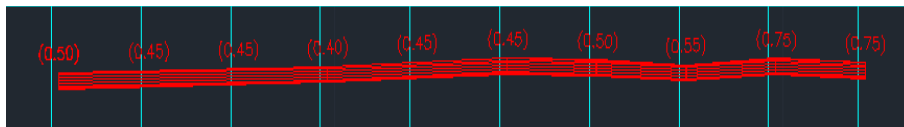


Fig. 129 - Dress Pipe

10. The **Undress Pipe** command () allows to “undress” the pipes, highlighting the pipe axis. This command allows to reduce the pipe in a poli-line (Fig. 130) and be modified using normal AutoCad tools.

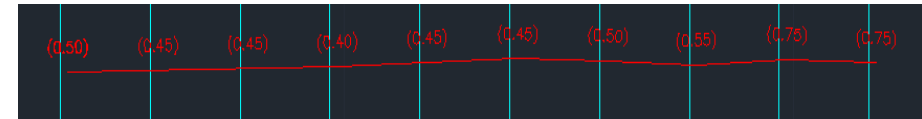


Fig. 130 - Undressed Pipe

11. The **Shade 3D Objects** command () automatically execute a “rendering” of all the 3D elements in the AutoCAD file (Fig. 131).

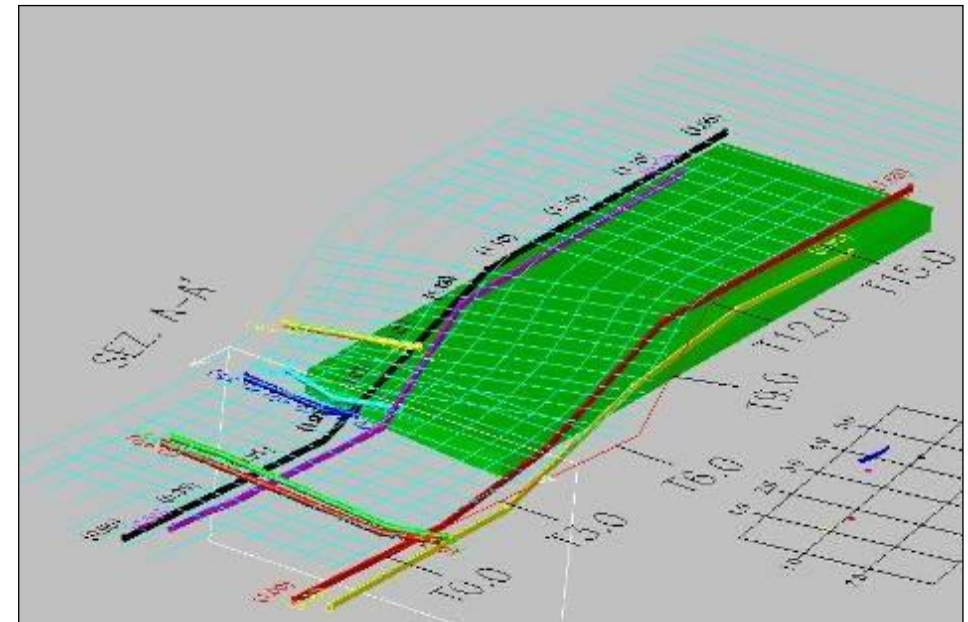


Fig. 131 – 3D View with rendering activated

12. The **Unshade 3D Objects** command () allows to view the 3D elements in the so-called “iron wire” view (Fig. 132).

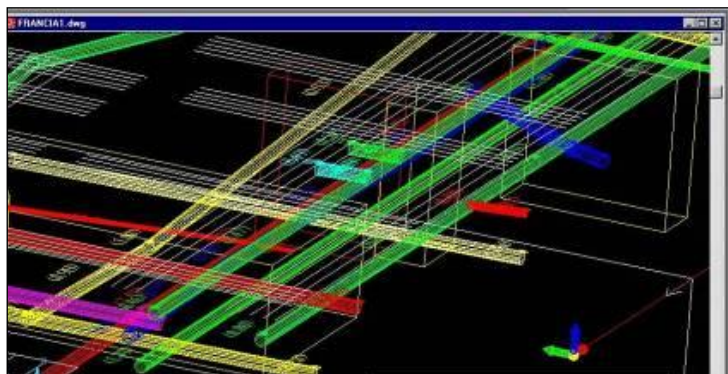


Fig. 132 – 3D View in iron wire mode

13. The **Layout Manager** command () activates the window in Fig. 133 that manages the creation of the various printing Layouts allowing to:

- **Select Paper Size** (from A0 sheet to A4 sheet)
- **Define Output Scale**
- **Insert Frame**: select one or more frames to represent in final layouts (Click one time with left mouse button to insert the first frame, move mouse to rotate it and, when the process is complete, press right mouse button. Repeat the same process to insert other frames);
- Create various printing layout selecting **Make all Layout** command.
- View printing borders in AutoCAD design by flagging **Visible frames when closing** command.

Examples of final report are shown in Fig. 134, Fig. 135 and Fig. 136.

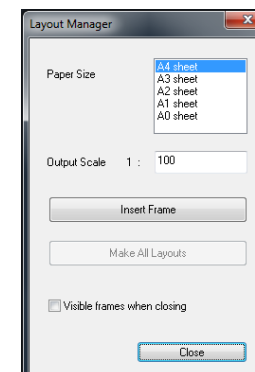


Fig. 133 - Layout Manager window

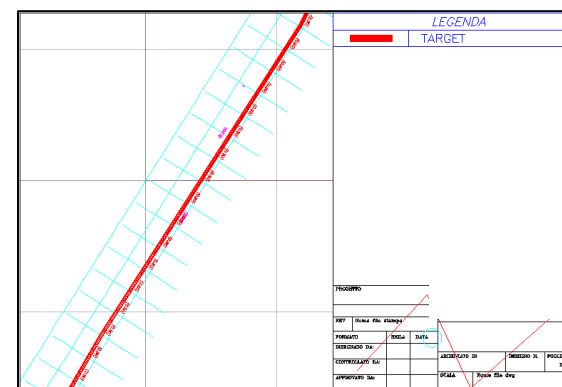


Fig. 134 - Example of Layout

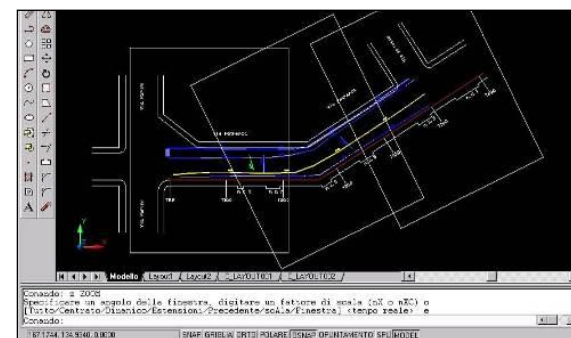


Fig. 135 – Example of area subdivision in printings

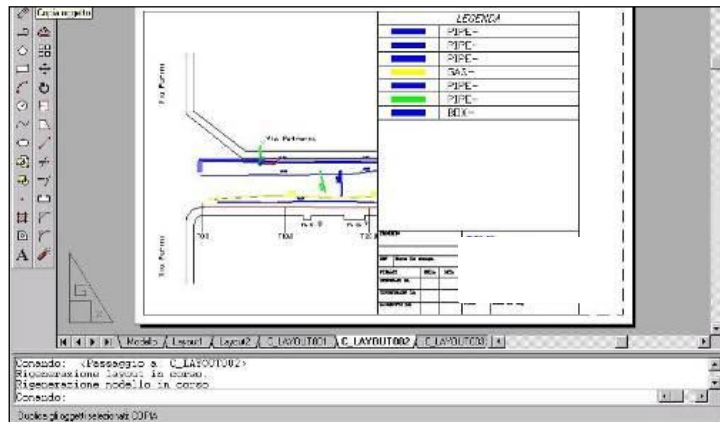


Fig. 136 – Example printing Layout

If the User wishes to cancel a layout, they must repeat the insertion by selecting **Make All Layouts** command (Fig. 133); an alert message (Fig. 137) will show up allowing, by selecting **Yes** button, the User accept creating new printing layouts which will overwrite the ones previously created.

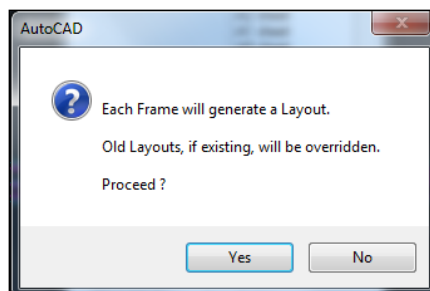


Fig. 137 – Overwrite window

14. The **Object Information** command () transforms the AutoCAD cursor into a white square; by selecting any desired item the window show in Fig. 138 will appear. This window will display all the general

information regarding the Project, Zone and about the selected object (Fig. 138); here the User can decide to edit the following fields:

- **Colour:** By clicking on  command, it is possible to choose from the AutoCAD palette, which colour to associate to the selected element.
- **Diameter of Pipe:** change pipe diameter.

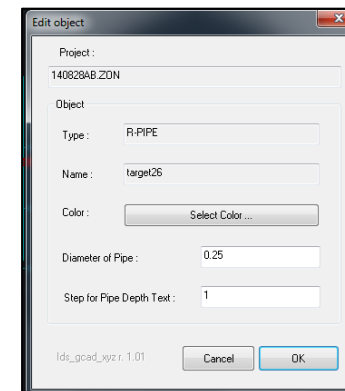


Fig. 138 - Information command window

- **Pipe Depth Text:** select the distance interval of the fields associated to each pipe. In this way the User can easily read the depth of the target in respect to reference surface (XY Plane) (Fig. 139).

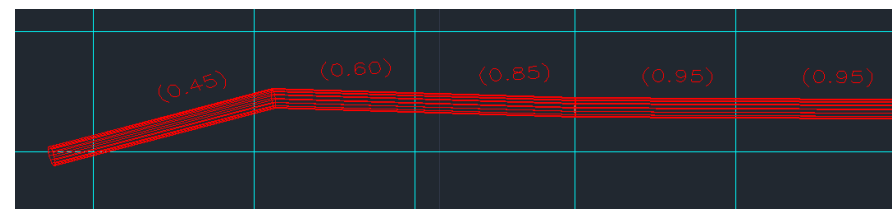


Fig. 139 - XY section of the target. Step for Pipe Depth Text = 1

APPENDIX F CLOUD DATA SHARING (CDS): INSIGHTS

Cloud Data Sharing (CDS) is a functionality relying on third-party cloud service, allowing to share data between the field PC connected to IDS GeoRadar systems and the office PC (and vice versa), via commercial Cloud paired with uMap-Logger Advanced, uNext Advanced, DXplore, IQMaps software. This functionality is meant to enhance the collaboration between field and office, optimizing the whole workflow.

The general idea behind CDS for IQMaps is that the office Team may download a survey uploaded from field and start post-processing far away from field. This workflow is shown in the simplified sketch of Fig. 140.



Fig. 140 Cloud Data Sharing scheme

As of today, the cloud that can be linked and used with IQMaps, Google Drive™ (<https://accounts.google.com/>), Microsoft OneDrive, HxGN GeoCloud Drive, Leica ConX drive.



Note

Google Drive is a SaaS solution provided by an independent third party, not provided by IDS GeoRadar, and subject to its relevant Terms and Conditions of Service.

IDS GeoRadar shall not be liable for any damages related/caused by such Cloud solutions, including but not limited to data breaches and/or losses whatsoever.

If User does not have any of the above Drive accounts, the first step is to have an active account.

As an example, the following shall report the procedure relating to Google Drive (for more information, please refer to <https://www.google.com/account/about/>), and then a Google Drive account (<https://drive.google.com/>) to be paired with IQMaps.

Login the Google Drive can be done directly on Google or through IQMaps, by the following steps, below:

1. On the IQMaps main page the Users should click on the *upload* or *download* icon of (Fig. 141 Fig. 141 link to Cloud Drive) under the “project” tab

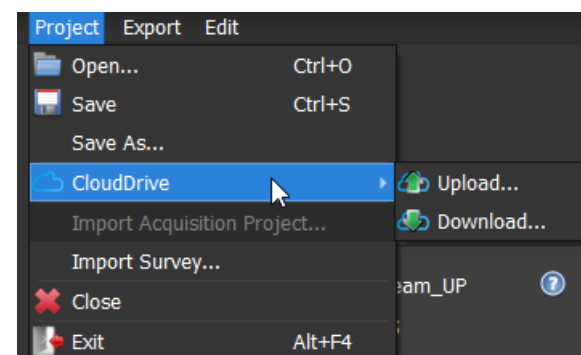


Fig. 141 link to Cloud Drive

2. The User shall be re-direct to the default internet browser to the third-party service provider Google
3. Then, follow the steps indicated by Google to finalize the procedure
4. Eventually, the message below shall appear on the browser page, indicating that the process has been successful

“Authentication successful. Please return to the Leica field software”

Once the Google account is active, and the Cloud Drive is paired with IQmaps through the above steps, Field crew and Office Team can share data by pointing to the same Cloud drive having had access with the same credentials.

The installed plugin version can be verified by the Hamburger menu on the top right corner of IQMaps (Fig. 142) after pressing the  icon

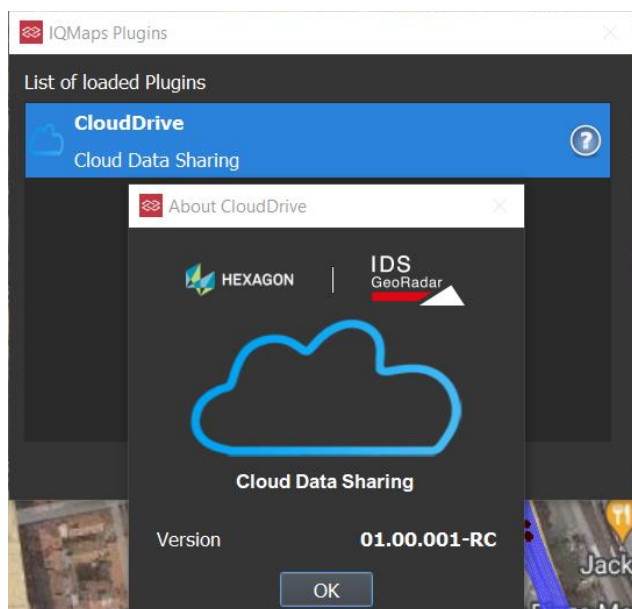


Fig. 142 Cloud Drive version

Functions that can be done by the User by the CDS function implemented in IQMaps are:

1. Upload or download to/from Cloud drive, a project

At the start of IQMaps, Users may start post-processing a Project after downloading it from CloudDrive (Fig. 141). User shall be prompted to window Fig. 143, and the data folder can then be saved locally to start working on it.

A project, may also be uploaded onto the cloud by CDS, performing the very same operation described above, by using the “upload” function.

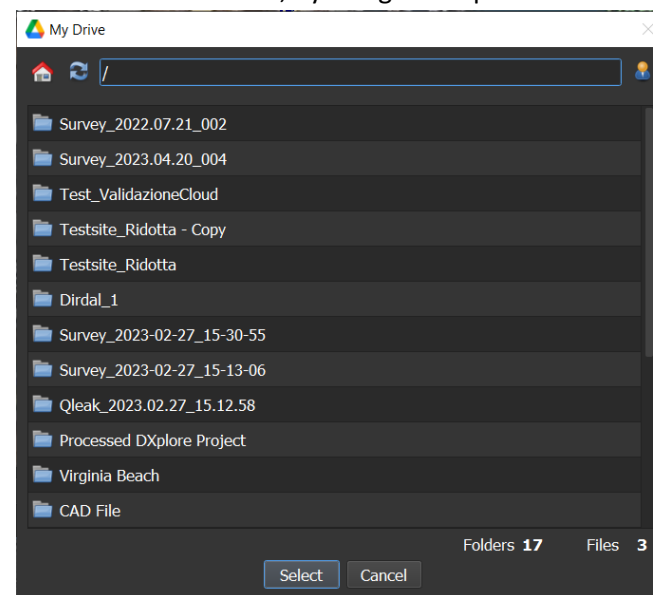


Fig. 143 upload and download files and survey data folder

2. Download from Cloud Layers

Vector layers can be added to the tomography view panel either, by selecting them from a local directory, or after download from Cloud Drive and local save, through pressing and hold the  icon.

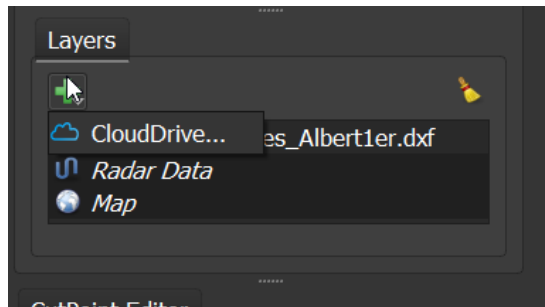
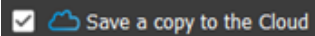


Fig. 144 Add Vector Layers

3. Upload to Cloud Drive any of the export

A copy of the whole exports may be saved on Cloud, after ticking the



icon of Fig. 145.

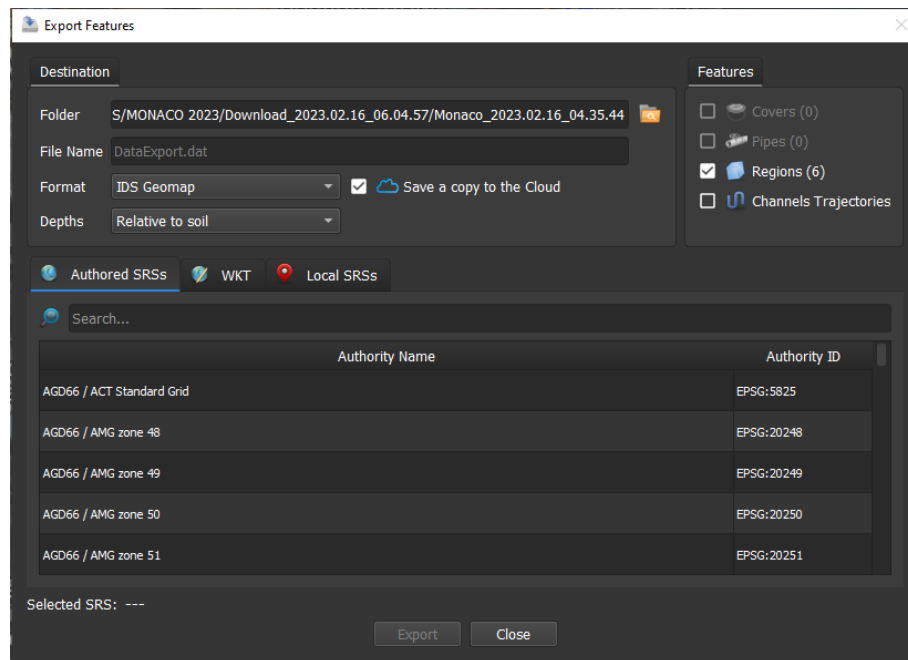


Fig. 145 Upload export on Cloud

APPENDIX G LICENSE MANAGEMENT: INSIGHTS

G.1 General

IQMaps and its plugins can only run “**on Premise**” and not “**on Cloud**”. This means that the software can only work on a physical computer, either PC or Desktop.

Furthermore, the CLM to activate the license EID must be installed on a PC that can be either:

1. on the very same computer where it is installed IQMaps or, alternatively,
2. it can be installed on a PC-server that can be reached remotely via IP protocol from another “physical” PC, provided that both PC and PC-server are in the same IP network.

The differences between the two options are:

- **IQMaps is installed and running on a PC:** in this case, only the “owner” of the PC where IQMaps and plugins are installed (including the CLM that activate licenses, of course), can use the Software.
- **IQMaps is installed on a PC and the CLM on a PC-server:** in this case, the CLM and the activation EID are installed and activated on a “remote” PC (the “Server”). The User(s) that installed IQMaps onto its(their) own PC can use the active license installed into the “Server” and “point” at the PC-server IP address from “configure License Server” option of IQMaps. The advantage in doing this is that more Users, not at the same time of course, can use the same license from different PC desktops that are in the same IP network.

To recover all information on the IQMaps license in use, enter [myWorld](#) web site, and follow the instruction on the **myProducts** page.

Here, after filling in with the 25-digit EID in use it is possible to recover all information like those illustrated in Fig. 146.

The information that can be recovered includes:

- Type of license
- Article number

- Delivery date
- Expiration date
- SW maintenance end
- Activation status
- Number of rehost
- Operative system installation
- Last rehosting date

The screenshot shows the 'myWorld' web interface. At the top, there are logos for HEXAGON and Leica Geosystems, and the text 'myWorld'. Below the logos, there is a navigation bar with 'Products & Services' and a language selector set to 'English UK'. The main content area has a breadcrumb trail: 'Home > Products & Services > My Products'. There are three tabs: 'Overview', 'Software', and 'Manuals'. The 'Overview' tab is active. Below the tabs, there is a 'Back' button and two buttons: 'Offline Activation/Return' and 'Entitlement Certificate'. The 'General' section displays a product image of a computer monitor showing a map. To the right of the image, there is a form with the following fields: 'Product' (IQMaps Sales DEMO (1 year)), 'My Reference' (a text input field with a 'Save' button), and 'Entitlement ID' (00106-77501-00030-82378-2A693). To the right of these fields, there is a table with the following data: Article No. 907489, Delivery Date 18.02.2020, and Equipment No. 10677501. The 'Software' section is also visible, showing a table with the following data: Start Date 20.02.2020, SW Maint. End 19.02.2021, Expiration Date 19.02.2021, Duration 1 Year, License Type SalesDemo, and License Model Node locked.

Fig. 146 myProducts page to recover IQMaps licenses information

G.2 How to install and run IQMaps on PC

When IQMaps and its plugins are installed on a single PC or Desktop (following the procedure described at Par. 4) it is necessary to specify where the EID has been activated through the CLM, that is that PC.

In this case, at the first start after installation and activation, the “Configure License Manager” window shall pop up (Fig. 147) asking where the EID is activated. In this case, as the CLM is installed in the same Computer, and the EID activated, the User must type “localhost” in the Address space and press “connect”.

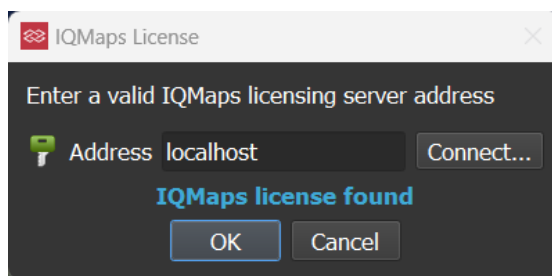


Fig. 147 Configure License Manager window

The “Configure License Manager” window can also be reached from the Hamburger menu on the top right corner of IQMaps Fig. 148.

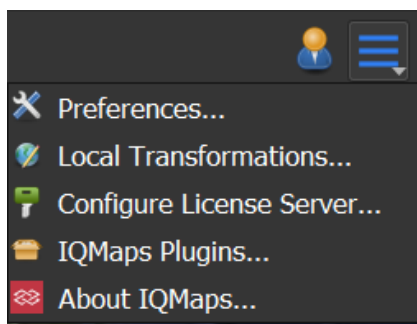


Fig. 148 Hamburger Menu at the top right corner of IQMaps

G.2.1 How to return a license and rehost on another Computer

To deactivate IQMaps from one computer and use it on another, the EID activated on CLM must be returned first and then the same EID can be activated on CLM on another computer. To do this, follow the steps below:

1. Start CLM and select the EID you want to return (for example, the third one in Fig. 149)

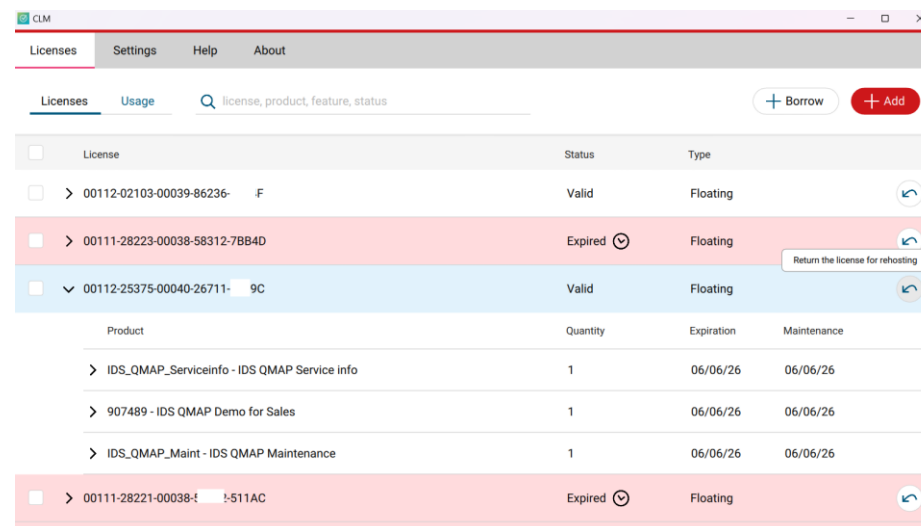


Fig. 149 the CLM interface to return a license

2. Press on the  icon at the right side,
3. The window of Fig. 150, shall pop up. Press .

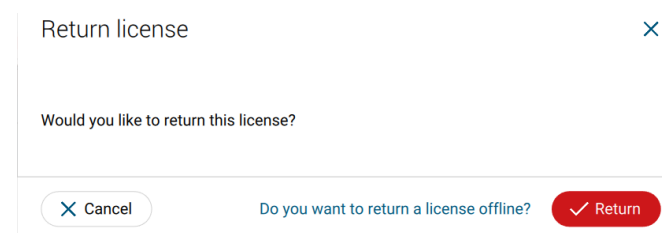


Fig. 150 Return license of IQMaps

- 4. Now, the same EID just returned can be activated (rehost) on another PC, following the procedure described at **Par. 4.1 How to activate an EID using CLM**

G.2.2 *How to work out the rehost error message*

Occasionally, the return and rehost procedure described at par. G.2.1, may not succeed and the error message of Fig. 151 may pop up.



Fig. 151 Example of rehost error message

In this case, send a message to the technical support e-mail address support.geo@idsgeoradar.com , indicating the EID number of the license that cannot be rehosted.

G.3 **How to run IQMaps with license activated on PC-Server**

When IQMaps and its plugins are installed on a PC or Desktop (following the procedure described at Par. 4) but the license EID is activated on a CLM that is installed on a PC-server, it is necessary to specify where the EID has been activated through the CLM.
In this case, at the first start after installation and activation, the “Configure License Manager” window shall pop up (Fig. 147) asking where the EID is activated. In this case, as the CLM is installed on another PC (the Server), the User must type IP address in the Address space and press “connect”.

This setup can be convenient when more than one User of the same Organization have access to the same internal network where the Server is located and the license is activated. The same license can be used by multiple computers that indicate the IP address of the PC-server. However, multiple computers (i.e.: Users) cannot use the same license at the same time

G.3.1 *How to work out the activate license error message*

Occasionally, the activation of a license may fail, and the error message of Fig. 152.

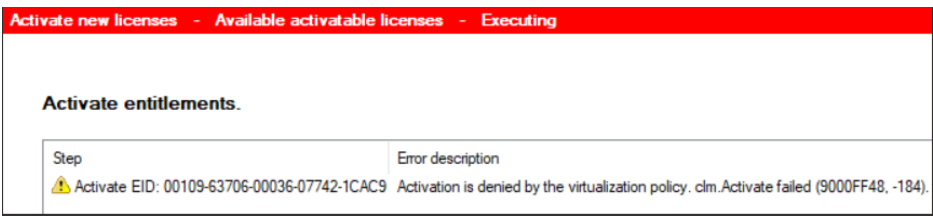


Fig. 152 Activate license error message

In this case, send a message to the technical support e-mail address support.geo@idsgeoradar.com , indicating the EID number of the license that cannot be activated.

G.4 **AIMaps-related topics**

AIMaps is a plugin of IQMaps allowing to automatically process **Stream UP**, **Stream DP**, **Stream C** data over [HxDR](#) web platform using an Artificial Intelligence generative algorithm. And, the procedure to activate the plugin and having granted the access to the platform, is described at “**4.3 How to install and activate AIMaps**”.

G.4.1 *How to create a HxDR account*

AIMaps is a plugin that uses a weblink between IQMaps and HxDR cloud platform using a User-specific account that is linked to IQMaps licenses.

- To generate a HxDR account to use AIMaps only use the e-mail address AIMaps.geo@idsgeoradar.com and follow the procedure described at par 4.3.
- If an HxDR account is generated using a direct procedure on the [HxDR.app](#) website, this shall not be linked to IQMaps causing the plugin not to work.

G.4.2 *How to avoid mismatching issues with old licenses*

When/if a new AIMaps EID is activated to replace an existing one that is expired, it is recommended to remove the old, expired EID, from CLM.

G.4.3 *How to work out data upload issues*

Occasionally, Organization may have IT restriction to internet connection with specific web sites or categories of web site.

If the upload of the data on the HxDR platform fails, try the following:

1. Verify that the website [HxDR.app](#) can be reached from the browser,
2. Verify that the HxDR.app is not blocked on your computer.

APPENDIX H SHORTCUTS

- **Ctrl+O**: to Open a Project
- **Ctrl+S**: to Save a Project
- **Alt+f4**: to exit a Project
- **Ctrl+z**: to undo
- **Ctrl+Y**: to redo
- **Alt+left** click mouse: select or deselect area
- **Ctrl+left** click mouse: to add points on trajectory editor waypoints
- **Ctrl+right** click mouse: to add a cut on the Processing Editor tab or Inspector tab

APPENDIX I CONTACTS

The IDS GeoRadar Support team is available to answer any question about:

- general enquiries,
- request of advertising material,
- technical issues,
- As well as listening to your suggestions too.

Please, feel free to contact us at:



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